

Slot-Chain: A Preconfirmation Protocol

Built on Per-Slot Signature Chains

v2 Design Specification

Draft v1.51 · August 2026

Abstract

This document specifies the successor design (v2) of the Taiko preconfirmation protocol. L2 time is divided into one-second slots; a lookahead schedule, published two epochs in advance, assigns each slot to a unique builder. The builder produces a block and signs it, and because each signature covers the parent hash, the not-yet-landed blocks form a *signature chain*. Submitting such a chain to L1 together with a validity proof is called *landing*: an aggregator elected by a perpetual auction performs it in the normal case, but anyone may perform it when the aggregator is absent, because a block's authority comes from its builder's signature rather than from whoever lands it. Finality is given by the proof system together with a deterministic *settlement window*: among the candidates that have landed and been proven, the heaviest one is finalized when the window closes at a predetermined L1 height. We give the trust model, the liveness accounting, the forced-inclusion floor, the slashing rules and the parameter geometry, and we state the accepted residual risks and the remaining open items explicitly rather than eliding them.

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Status of this document. This is the successor design to the Taiko preconfirmation protocol, superseding the earlier v15 design line (perpetual auction plus epoch-level determination; its full text and review record are preserved in the git history, and the conclusions worth retaining are given in legacy-summary.md). The design originates in the owner's instruction of 2026-08-25: adopt a sub-epoch failover granularity in which rotating builders produce and sign one block per slot, the signatures are strung into a chain, and the result is verified by the proof system. This document is the English edition; the Chinese edition is the normative source (see the note at the head of this file).

How to read this document. The main text (§0–§11 together with Appendices A–C) is the specification proper and can simply be read in order; a reader in a hurry to grasp the trunk can build a full picture from the §0 one-page summary and the §8 liveness accounting table. The many parenthetical notes in the main text of the form "[1]" or "[2]" are audit traces of the design process — they record which round of adversarial review prompted a given rule — and a first reading may skip them outright without any loss of understanding; return to them when the provenance of a particular rule needs to be traced. The complete per-version change history (from v1.31 onward, including the findings and fixes of every review round) is collected in Appendix D and is no longer placed ahead of the main text.

0 One-page summary

In one sentence: L2 has one slot per second, and the lookahead — which states "who holds the right to produce a block in this second" — is published two epochs in advance. The builder whose turn it is produces a block and signs it, and the signature covers the parent block hash; the blocks, which are not yet on chain, are thereby strung into a signature chain. The act of packing this chain onto L1 together with a validity proof is called landing; in normal operation the aggregator produced by the auction is responsible for it, but when the aggregator goes offline anyone may do it — because a block's authority comes from the builder's signature, not from the lander.

The complete path from block production to finality:

This structure solves three things at once:

1. **Second-scale liveness.** A builder going offline costs only its own slot (about 1 second), and the next builder simply continues producing blocks on top of the last visible chain head. There is no window in which "one participant goes offline and the chain stops for tens of minutes".
2. **No L2 consensus is required.** "Who should produce the block in this slot" is decided by the lookahead, and "whether this block is legal" is decided by pure-function rules inside the proving circuit (verify the signature, verify the lookahead assignment, verify the link) — anyone reading L1 computes the same answer, with no voting and no arbiter.
3. **Landing carries no privilege, and the aggregator is not a single point of failure.** The signature chain carries its own authority; landing is merely transport. If the designated aggregator times out without landing, anyone may pick the signature chain up, land it, and collect a reward out of the aggregator's bond.

What is kept and what is removed (relative to v15):

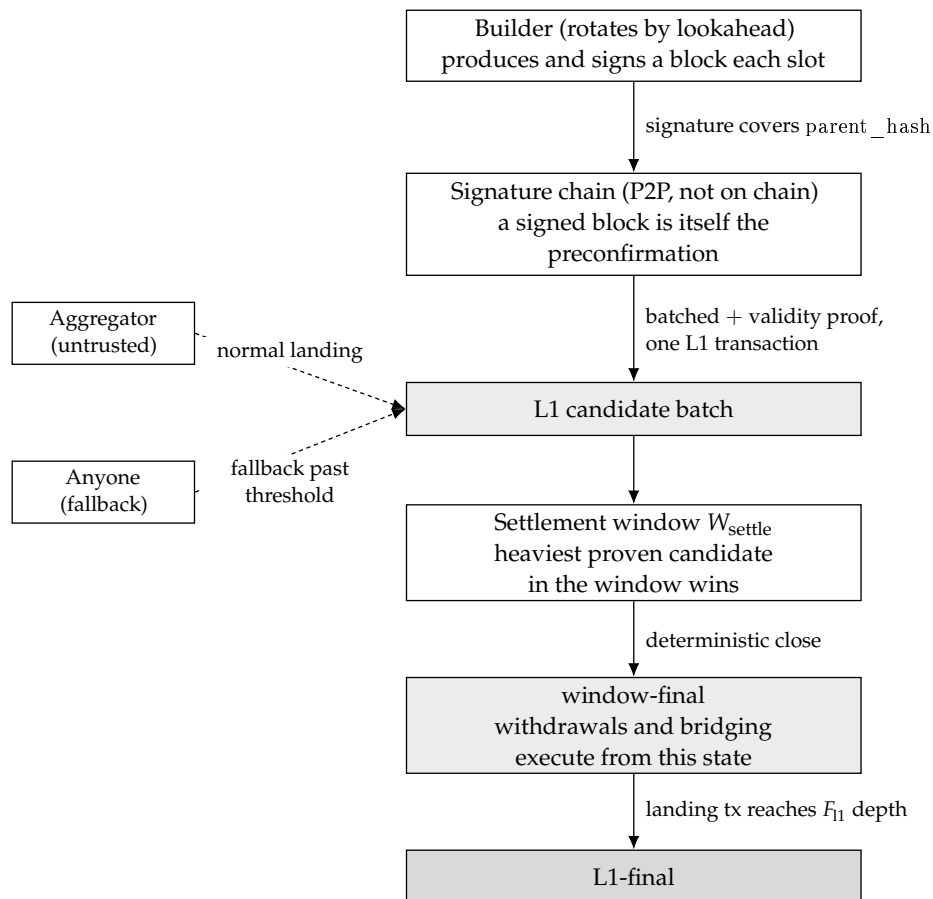


Figure 1: The complete path from block production to finality: builder signatures produce a signature chain that never touches the chain, anyone may land it on L1 together with a validity proof to become a candidate, and once the settlement window closes it becomes window-final and then L1-final. Landing carries no privilege, so the aggregator and the fallback lander are interchangeable.

Kept	Removed
The perpetual auction skeleton for the aggregator seat	Epoch boundary commitments (EBC) and the once-per-epoch irreversible determination
The per-block signature chain and the two-tier parent rule (new here, not inherited from v15)	The default derivation rule (§6.8 default derivation)
—	The forced-inclusion queue — removed wholesale in v1.51 by the owner's decision (simplification), together with forced-only blocks, P_forced, the dual queues and C_force/C_bridge/F_delay/H_force
The "party at fault pays, permissionless substitution" model (fault-paid permissionless fallback)	Total Anarchy mode (§8) — landing itself can already be backstopped permissionlessly, so it is no longer needed
The tiered bond / slashing concept	K_empty, availability certificates (AC), the H_cancel cancellation cascade, and the rest of the machinery supporting the epoch determination
The per-block anchor approach to L1→L2 bridging	Handover and seat-switching arbitration mechanisms (replaced by circuit rules)

Timeline illustration (slot = 1 second, epoch = 384 slots) — going offline leaves only a gap, and the chain does not stop:

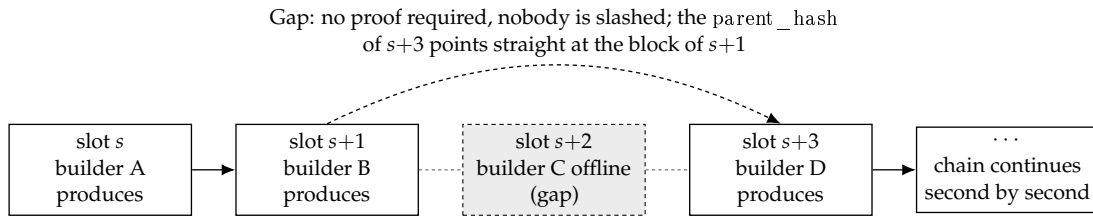


Figure 2: A single builder going offline leaves nothing but a one-slot gap in the timeline: the next builder takes the last visible block as its parent and carries on, so the chain never stalls.

The relationship between landing and fallback (details in §6): anyone may take a contiguous segment of the signature chain plus a validity proof and land it on L1 in a single transaction, making it a candidate; the heaviest proven candidate within the settlement window W_{settle} is final at the window close (§5.6). When the aggregator times out without landing, anyone may land, and the reward is paid out of the aggregator's bond (§6.3).

1 Design goals and non-goals

Goals:

- **[G1] Second-scale continuity of ordering.** If any single participant (a builder or the aggregator) goes offline, the interruption of ordering service perceptible to users lasts at most a few slots (seconds), not a few epochs (minutes to hours).
- **[G2] No L2 consensus.** The lookahead, legality and canonicity are all pure functions of L1 data and circuit rules; L2 nodes neither vote nor arbitrate among themselves. This inherits the spirit of v15's I1: derivation is a deterministic function of L1 data.
- **[G3] Landing carries no privilege.** Moving an already-signed chain onto L1 requires no identity whatsoever; the designated aggregator is merely an efficiency arrangement (avoiding gas competition and duplicated proving), not an arrangement of power.

- **[G4] Proof plus a deterministic window is finality** ^[3]. A landed batch carries a validity proof, which L1 verifies on the spot, and the batch becomes a candidate; at the close of the settlement window the heaviest proven candidate is finalized — there is still no optimistic period, no fraud proof and no arbitration, and there is no intermediate "proposed but unproven" state, so the entire body of machinery built around such an intermediate state (sealing deadlines, cancellation cascades, and most of the timing exemptions) is unnecessary.
- **[G5] (withdrawn) An inclusion floor is no longer a goal of this design.** In v1.51 the owner decided to remove forced inclusion wholesale, for simplicity. There is therefore **no** entry path to L1 that does not depend on builders: whether a transaction gets included depends on whether the scheduled builder is willing to include it. If the entire set colludes to censor an address, the protocol offers no remedy whatsoever, and that address's funds have no exit path independent of the builders. This is an explicitly accepted cost; see the all-cartel row of §8 and the trust model below. One thing closes along with it: the blocking-level nonce-preemption gap of r27-1 (DeepSeek round 6 Critical 1). With no forced path, the gap has nothing to apply to, and item 12 of §11 is retired with it.
- **[G6] A low barrier to entry.** The builder bond is priced at the order of magnitude of "the extractable value of a single slot", far below the full collateral for one seat in v15.

Trust model (normative)^[4]:

- **Builders: honest majority.** We assume that within each lookahead window (on the order of an epoch), most of the scheduled builders honestly follow the protocol (producing blocks on time, broadcasting block headers and block bodies, and extending the highest available chain head).
- **Aggregators: untrusted.** An aggregator may misbehave arbitrarily — delaying, truncating, choosing among forks, colluding with a minority of builders — and the protocol must, under such conditions, still hold the bounds given in the §8 table. The scope of those bounds ^[5]: the bounds in §8 are liveness bounds (advancing finality, seat turnover); the exposure of preconfirmation safety to collusion between the aggregator and malicious builders — isolation without slashing, a single-block deep skip of depth $\leq G_max - 1$, and a minority coalition relaying a sparse branch to a depth that can reach the length of the unlanded tail ($\approx \Delta_lag + \text{fallback response}$) — is the safety residual risk explicitly priced in §5.4, and it falls outside the "bounded" promise.
- **L1: safe and live.** The standard assumption of every L2. Added to it is one assumption that this design explicitly acknowledges, the Bounded-Inclusion Assumption ^[6]: a candidate or landing transaction that pays a sufficient fee is included within $T_include,max$ L1 blocks, and the parameters satisfy $T_include,max < \min(W_settle - P_prove,max - margin, D_anchor_max - D_anchor - \Delta_lag - P_prove,max)$ (the setter relations of §11). Targeted L1 censorship exceeding that bound is out-of-model — in that case a better candidate may miss the close and a recovery block's anchor may expire — and every statement that relies on "robustness under temporary censorship" (including the recovery bound of §6.4) is uniformly conditional on this assumption. This gathers the same timing dependency, implicit at various layers from v1.28 to v1.40, into a single stated assumption; it is not newly introduced trust.

The honest majority is an economic and social assumption, not one enforced by the protocol

[7]: the draw is weighted by bond and $w_{\max}$ is no defense against Sybils (§3.2), so a single capital holder could in principle buy up the majority of the weight and push the system out of the model — of the same kind as the honest-majority assumption of any PoS chain. This design does not claim that the builder role is entirely trustless or entirely permissionless; this is the trust model explicitly accepted by the owner (2026-08-25), and what is bought in exchange is not paying the cost of protocol-level DA to cover collusion by all participants (the cost identity is given in Appendix A-3). Each promise is calibrated accordingly: the in-model bounds are given in the §8 table; out-of-model, after v1.51 removed forced inclusion, nothing is guaranteed. A note on the acceptance criterion [8]: if the acceptance criterion is "no role may be trusted, everything fully permissionless", this design neither satisfies it nor attempts to — the acceptance criterion is the trust model accepted by the owner in this section; the user-facing promises, graded into three tiers, are stated in §5.4.

How much of the assumption each mechanism actually consumes (written out, to keep the assumption from being quietly strengthened):

- Finality liveness (fallback landing always having a chain available to land) requires only ≥ 1 honest builder per window (§6.3) — weaker than the assumption, so there is margin;
- The frequency with which preconfirmations are isolated (skips, body withholding) is bounded by the share of dishonest builders — this is where "majority" is used, and it is load-bearing there: skipping is profitable for the skipper [9], not a behavior that extinguishes itself for being unprofitable;
- Collusion of the entire builder set together with the aggregator (block bodies kept private, the chain landable by no one) lies outside the model: the protocol provides no finality guarantee in that case [10]. After v1.51 removed forced inclusion, this case has **no defense in depth left**: the former guarantee that "even if the assumption is broken, forced transactions and bridge messages still operate independently on L1 data alone" no longer holds, and a user's inclusion and exit depend entirely on the honesty of the scheduled builders. The user's right to exit does not lapse with the model;
- **Preconfirmation safety for the tail additionally depends on the completeness of the lander's view** [11]: a lander that is honest but eclipsed or blinded will land the wrong tail and revoke the preconfirmations of a heavier tail that it never saw. The mitigation is "landing depth $\geq$ propagation settling" (§5.2), which compresses this risk down to "equivocation (slashable) or eclipse (the P2P layer)"; together with "the honest majority is an economic assumption" and "liveness under a Byzantine aggregator is conditional on fallback being economically viable", it belongs to the explicitly stated non-pure-protocol dependencies.

Non-goals (not addressed in this version):

- Per-transaction fair exchange: the preconfirmation receipt is block-level (§5).
- User restitution: slashing is a deterrent, not insurance.
- Resistance to deep L1 reorgs: as with any L2, a deep L1 reorg can roll back L2 state.
- A built-in prover market: the aggregator either brings its own proving capability or procures it.
- **Collusion of the entire builder set plus the aggregator**: under the trust model above this is an out-of-model situation. After v1.51 removed forced inclusion there is neither a finality bound nor any defense in depth for entry or exit.

2 Roles

- **Builder:** the set of addresses that have registered with the L1 contract and posted a bond. When the lookahead brings a builder's own slot around, it produces a block, signs it, and broadcasts it on the P2P network. The builder is the preconfirmer: the block it signs is itself the preconfirmation commitment for the transactions inside that block.
- **Aggregator:** the service provider that holds the sole seat by way of the perpetual auction. It has exactly one duty: to pack the signature chain into a batch, generate a validity proof, and land it on L1. It does not build, does not order, and holds no discretionary power over content — the only residual discretion is "which prefix to land and when to land it", and both are constrained by deadlines and by permissionless fallback (§6).
- **Fallback lander:** anyone. A fallback lander appears once the aggregator has timed out and is paid under the fault-paid model. This is not a registered role.
- **User:** submits transactions to builders in exchange for preconfirmations; when censored, goes by way of the forced-inclusion queue.
- **L1 contracts:** the builder registry, the aggregator auction, the forced-inclusion queue, the batch entry point (which verifies proofs), and the slashing entry point.
- **The proof system:** the circuit that verifies "this segment of the chain is legal and its state transition is correct". It is the arbiter of this design.

3 Time and scheduling

3.1 Slots and epochs

- **slot = 1 second.** The time unit of L2. There is at most one block per slot, and a block's timestamp is identically the starting second of its slot — the timestamp is not chosen by anyone, it is fixed by the lookahead (which eliminates the entire class of problems around the origin of timestamps found in Appendix C of v15).
- **epoch = 384 slots** (the current value is retained, aligned with L1's 32×12 seconds). In this design an epoch is merely a unit of measure (the granularity of lookahead windows and of fee settlement); it no longer carries the mechanism-level meaning of "one determination per epoch".

3.2 The lookahead

- The lookahead is a mapping $\text{lookahead}(\text{slot}) \rightarrow \text{builder address}$ that gives, for every future slot, the unique holder of the right to produce a block.
- **Computation (a pure function)**^[12]: $\text{lookahead}(\text{slot}) = \text{draw}(\text{hash}(\text{seed}(\text{window}), \text{slot}), \text{registry_snapshot}(\text{window}))$ — the seed is taken once per window, not once per slot:
 - **Window partitioning**^[13]: a window is a fixed aligned partition: $W(\text{slot}) = \text{floor}(\text{slot} / W_size)$, $W_size = 768$ (two L2 epochs); the starting slot of window W is $W \times W_size$, and "the L1 slot corresponding to that start" is converted by the genesis mapping $L1_slot(s) = \text{GENESIS_L1} + \text{floor}(s / 12)$ (an L2 slot is 1 second, an L1 slot is 12 seconds). The lookahead assignment of any given slot has the same value for every observer at every moment.

- **A unique snapshot height:** the snapshot height of window W is $H_snap(W) =$ the L1 slot corresponding to the start of $W - D_snap$, where $D_snap =$ the lookahead horizon H_look (2 L1 epochs) + the finality distance F_final (2 L1 epochs) + margin (1 L1 epoch) = 5 L1 epochs ^[14]. The setter invariant: $D_snap \geq H_look + F_final + margin$. It follows that at every moment at which the lookahead is required to be already determined, $randao_source(W) = H_snap(W) \leq$ finalized L1 head $<$ start of W — there is only one seed, it holds simultaneously for all slots in the window, and the lookahead is fixed before it is used and is unaffected by L1 reorgs.
- **The seed:** the RANDAO of the L1 epoch containing $H_snap(W)$ (provable in-circuit via the EIP-4788 beacon root); the variation between slots is derived by $hash(seed, slot)$, so no independent per-slot entropy source is needed.
- **The registry snapshot:** the registry state at that same $H_snap(W)$; the entry and exit delay (§4.1) guarantees that the snapshot covers the whole window. The circuit’s lookahead check for every block in the window uses $H_snap(W)$ uniformly, independently of each block’s own anchor block ^[15].
- **The draw:** deterministic bond-weighted sampling (independent for each slot). The weight of a single address is capped at w_max (initially 20%). An honest characterization ^[16]: w_max only prevents concentration in a single address and is no defense against a Sybil that splits its bond — large capital need only split across several addresses to recapture the weight, and it costs not one cent more. The real upper bound on concentration is economic (buying weight requires genuinely locking up bond), the same trade-off as v15’s acceptance that “the highest bidder can keep on winning”. The role of w_max is operational hygiene; it is not a promise of decentralization.
- The full codification of the computation above (the constraint at each step is in the comments; for a runnable version with property tests see `lookahead-model.py`, and the sampling operator is the candidate for settling item 18(b) of §11):

```

W_SIZE = 768 # lookahead window = 768 L2 slots (fixed aligned partition, not a sliding horizon)
H_LOOK = 768 # lookahead horizon
D_SNAP_L1 = 5 * 32 # snapshot delay (L1 slots); invariant D_snap ≥ H_look + F_final + margin
W_MAX = 0.20 # cap on a single address's effective weight (operational hygiene; no defense against a bond-splitting Sybil)

def window_of(slot):
    # fixed aligned partition: the same slot maps to the same window at every moment and for every observer (r10-5)
    return slot // W_SIZE

def snapshot_height(w):
    # unique snapshot height = the L1 slot corresponding to the window start - D_snap.
    # the three terms of D_snap guarantee: before the lookahead is used, this height is already
    # L1-finalized — neither the seed nor the registry is affected by an L1 reorg (r8-1)
    return l1_slot_of(w * W_SIZE) - D_SNAP_L1

def seed(w):
    # seed = the RANDAO of the L1 epoch containing the snapshot height (provable in-circuit via the EIP-4788 beacon root).
    # taken once per window, not once per slot: all slots in the window share one entropy source (r7-1)
    return hash("seed", randao_at(snapshot_height(w)))

def effective_weights(registry):
    # effective weight = min(bond, w_max × total bond). capped once, no iterative renormalization —
    # the excess is voided rather than redistributed; a single linear pass, circuit-friendly
    total = sum(registry.values())
    return {a: min(b, W_MAX * total) for a, b in registry.items()}

```

```

def lookahead(slot):
    # lookahead(slot) = draw(hash(seed(window), slot), registry_snapshot(window)).
    # the registry snapshot and the seed are taken at the same height (r1-
    # uniqueness); the variation between slots is derived from the hash
    w = window_of(slot)
    reg = registry_at(snapshot_height(w))
    r = hash(seed(w), slot)
    return weighted_pick(reg, r)

def weighted_pick(registry, r):
    # deterministic weighted sampling (the candidate operator for settling §11-18(b)):
    # 1. addresses in a fixed order by registration index (an order intrinsic to the snapshot, preventing grind-
    # ing on the registered name);
    # 2. prefix sums of the capped weights (fixed-point integer wei, no floating point);
    # 3. x = r mod total weight; whichever prefix interval x falls into determines the address selected.
    # probability of being selected = share of effective weight; in-circuit = one prefix-sum pass + one modulo + one lookup
    eff = effective_weights(registry)
    addrs = sorted_by_registration_index(registry)
    x = r % sum(eff[a] for a in addrs)
    acc = 0
    for a in addrs:
        acc += eff[a]
        if x < acc:
            return a

```

- **Lookahead availability:** at any moment, the lookahead for at least the next $H_{\text{look}} = 768$ slots (about 12.8 minutes) is already determined and computable by anyone (the remainder of the current window plus the whole of the next window; the horizon term of D_{snap} guarantees that the next window's snapshot is already finalized). The temporal geometry of snapshots, windows and the lookahead horizon:

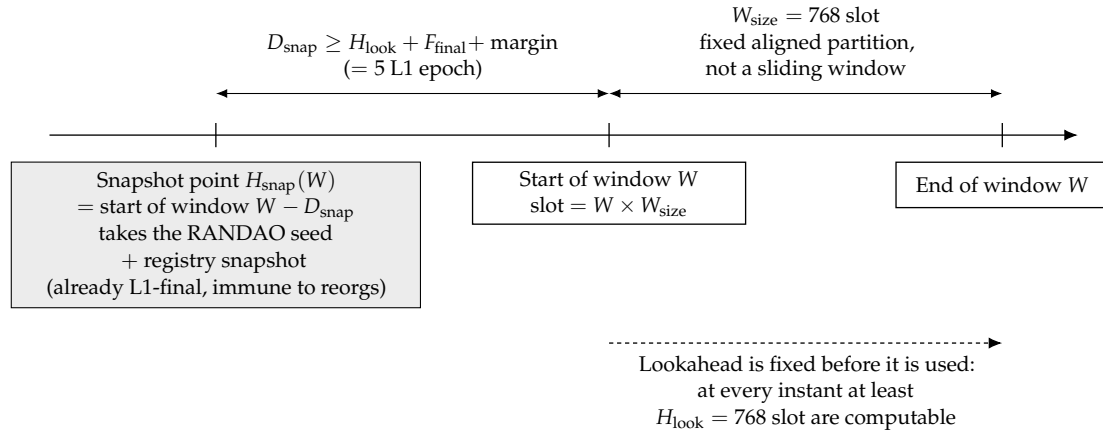


Figure 3: The time geometry of the snapshot point, the lookahead window and the lookahead horizon: entropy and the registry are frozen once and for all at D_{snap} before the window starts, so at every instant at least one full window of lookahead can be computed by anyone.

- **Why "the hash of the most recent proof" is not mixed in as entropy** ^[17]: one and the same statement can generate an unbounded number of byte-distinct valid proofs (proof generation carries randomness of its own), so a proof hash is entropy that the prover producing it can re-grind at zero cost — mixing it in would amount to handing part of the control over the lookahead to the aggregator. Pure L1 beacon randomness can be biased by L1 proposers by roughly one bit each, but it is good enough for a use such as the lookahead and introduces no new controlling party. The principle: entropy comes only from the L1

consensus layer, and the moment at which its value is taken is earlier than the lookahead window.

- **A known cost, stated openly:** a public lookahead means that "who produces the block in the next second" is public, so a targeted DoS against a builder about to produce a block is theoretically feasible. Mitigations: a slot is only 1 second long (the attack window is minuscule), builders can use sentry or private networks, and a slot that is DoS'd merely turns into a gap (§5.3). Ethereum itself runs with the same public lookahead.

4 Builders

4.1 Registry

- Maintained by an L1 contract. Registration requires posting a bond B_{builder} , which serves two purposes (one account, two red lines):
 - **Equivocation slashing allowance** L_{eq} (ETH). The pricing base is the cumulative exposure over a window, not a single slot^[18]: before a builder has been slashed for the first time and that slashing has taken effect (δ_{slash} in §4.3), it can pre-sign both forks in bulk for every slot it holds in the lookahead for the current window, so $L_{\text{eq}} \geq$ the maximum number of lookahead slots held by that builder within one exposure horizon $\times$ the worst-case extractable value per slot $\times$ a safety factor, where the exposure horizon = the remaining lookahead window + δ_{slash} + a margin for detection and submission latency^[19]. Computed with $w_{\text{max}} = 20\%$ and a window of 768 slot, the magnitude is a cumulative exposure on the order of one to two hundred slots. The legitimate way to push this number down is to shorten the slashing-effectiveness delay and the detection latency, not to pretend that the exposure is only one slot. The concrete valuation method is calibration content under item 2 of §11.
 - **Nuisance buffer:** a small amount covering the handling cost of sporadic violations.
 - **The honest upper bound of L_{eq}** ^[20]: L_{eq} is priced against an estimate of the extractable value visible to the protocol; for value external to the protocol (bribes, cross-domain positions, downstream value on the bridge) the protocol can give no upper bound — so the equivocation deterrent is a "deterrent priced against an estimate", not a guarantee against arbitrary external incentives. Any party that relies on commitments exceeding the size of the bond must price that residual risk itself (the same honesty clause as "deterrence, not compensation" in v15 §5.2; compensating users is a non-goal to begin with, §1). Evidence is idempotent: multiple pieces of equivocation evidence against the same builder trigger slashing only once (L_{eq} is confiscated in full, in one shot, and any later evidence is a no-op) — the total deterrence ceiling for a single builder is exactly L_{eq} , which is precisely why §4.1 prices it against the exposure horizon as a whole and why §4.3 uses δ_{slash} to freeze the builder's remaining slots into gaps as quickly as possible.
- **Entry/exit delay and bond retention period**^[21]: after an exit request, every slot already scheduled to the builder in the lookahead remains valid (it either produces a block or becomes a gap) — any lookahead entry that has already entered a window's snapshot $H_{\text{snap}}(W)$ is valid throughout that window without exception, and does not lapse even if that address subsequently (after the snapshot height) exits the registry (the earlier blanket

statement that "the lookahead never contains an address that is no longer registered" was ambiguous: an address that exits after the snapshot does still appear in the already-fixed lookahead, and the correct formulation is "its scheduled slots remain valid and are handled as either a block or a gap"); moreover, the actual unlocking of the bond must wait until all of that builder's slashing exposure has expired — setter invariant: **exit delay $\geq$ the distance to its last scheduled slot + δ_{slash} + a margin for detection and submission latency**, the same formula as the exposure horizon of L_{eq} above. Otherwise a builder could equivocate in bulk shortly before exiting and reclaim its bond ahead of the evidence being submitted and taking effect, zeroing out the deterrent.

- Registry capacity cap N_{max} (initially 64): decentralized enough, while keeping lookahead verification inside the circuit cheap. Applicants beyond the cap compete for seats ranked by bond (detailed rules pending, §11).

4.2 Block production and signing

- The builder whose turn it is at slot s constructs a block header:

```
header = (chainId, slot, parent_hash, anchor, final_ref(tier (ii) blocks only, zero otherwise),
txs_root, state_root_claim(optional), coinbase, gas_used, ...)
sig = the builder's signature over keccak(header)
```

Key fields:

- slot: the slot number of this block; timestamp is uniquely determined by it and does not exist separately.
- anchor: the L1 block referenced by this block (the freshness baseline and the L1→L2 consumption point, §7)^[22].
- parent_hash: the hash of the parent block header. Having the signature cover parent_hash is a central element of this design: the choice of parent (including "whom to skip") is an explicit act that the builder has signed, not a fuzzy state that can be repudiated afterwards^[23]. The link uses the hash of the parent block header rather than the parent's signature — the hash is structurally necessary (it defines the chain), whereas putting the parent's signature into the child block as well is redundant (the circuit verifies every block's signature anyway); it may serve as an evidence-packaging convention at the P2P layer, but it is not a consensus rule.
- **Signing-domain uniqueness invariant**^[24]: the canonical digest of the block header and the signature scheme must be one shared object — the direct verification by the L1 slashing contract (§4.3) and the validity verification in the circuit (§5.1) verify exactly the same bytes, the same hash, and the same curve. If BLS (via the EIP-2537 precompile) or a SNARK-friendly hash is adopted to reduce circuit cost, both sides must be switched at the same time. The choice of key system is therefore a blocking open item (item 6 of §11) that must be settled before the remaining parameters.
- The builder broadcasts (block header + signature + block body) over P2P. This signed block is itself the preconfirmation: the receipt given to the user = (block header, builder's signature, inclusion proof of the transaction) — the receipt contains the block header hash and txs_root, and is publicly forwardable evidence (used in §5.5).
- The decision procedure of the two-tier parent rule (the main statement and its exceptions

are given in the next item):

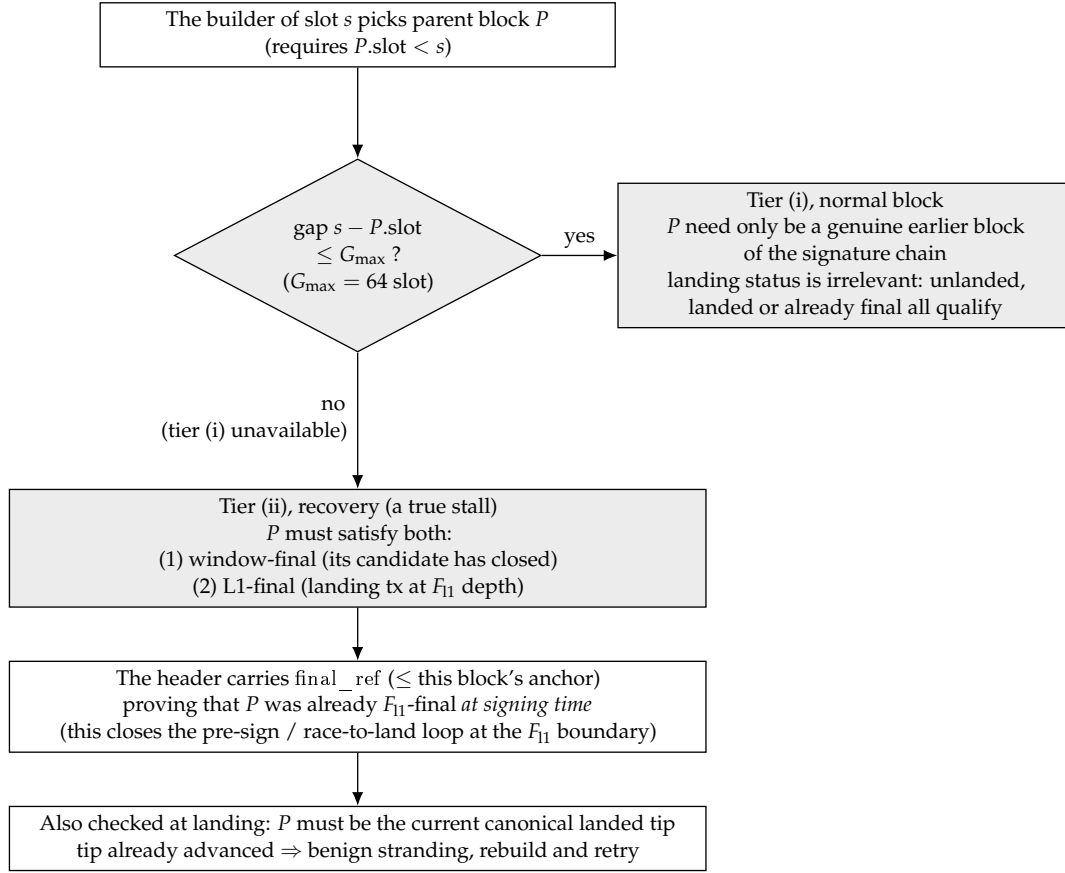


Figure 4: The decision procedure of the two-tier parent rule. The criterion is the size of the gap on the signature chain, not the landing status of the parent, so the rule evaluates identically at signing time, at proving time and at landing time; only a true stall, whose gap exceeds G_{max} , drops into the recovery tier that demands a final parent.

- **The parent rule — two tiers, with the criterion being [the size of the gap] rather than the parent's landing status** ^[25]: $parent.slot < s$, and the parent satisfies one of the following. The criterion is the gap $s - parent.slot$, evaluated over the structure of the signature chain (the gap + parent_hash) — the evaluation at signing time, at proving time and at landing time agree, and it does not depend on whether the parent "has landed / is already final" at this instant, a state that drifts with time:

- **(i) Bounded-gap tier (normal block production)**: $s - parent.slot \leq G_{max}$ (the gap cap, initially 64 slot). The parent need only be a real earlier block in the signature chain (parent_hash points at it), and its landing status is immaterial — not landed, landed but not final, or even already final are all admissible. Normal block production (including the next block after every landing) always takes this tier: after a landing, the next block attaches with a small gap to the (landed) chain head, and a gap $\leq G_{max}$ is legal, so "landed but not final" therefore constitutes no coverage hole at all ^[26]. This tier guards against deep reorgs: G_{max} caps the rollback depth of a single block, and a malicious builder that wants to orphan a deeper unlanded tail needs a relayed sparse branch (see the residual risk below).

- **(ii) Unbounded-gap tier (recovery):** $s - \text{parent.slot}$ is arbitrary — but the parent must simultaneously be [window-final] (its candidate has already closed in the §5.6 settlement window)^[27] and [L1-final] (L1-final, i.e. its landing L1 transaction has reached F_{l1} depth; terminology distinction, review r38 DeepSeek warning 3: "final" here refers specifically to L1-final = having reached F_{l1} depth on L1, as distinct from landed-final in §6.1 = the window-final finality of an L2 batch that has passed the close of the settlement window^[28] — the two-tier parent rule always takes L1-final as its criterion; for the unification of terminology throughout the document see open item 18(c) of §11). It is entered only when the gap $> G_{\max}$ (tier (i) is unavailable, i.e. a true stall). Why a large gap may be unbounded: by §5.2 a final landed block can no longer be reorged, so building on top of it cannot reorg any landed block; it is therefore safe by construction and needs no $G_{\max}$. This tier is the only mechanism for recovery (it replaces the entire old re-anchoring episode machinery, see §6.4). The point of evaluation and the F_{l1} wait^[29]: that the parent "is final and is the current canonical landing tip" is checked by the §5.6 landing rule at landing time, not at the moment of signing. Consequently, if a stall begins shortly after a landing and the current landing tip has not yet reached F_{l1} , the recovery block must wait for the current tip to become final (an extra $\leq F_{l1}$, on the order of minutes) before it can land — building on an older final head would fork off the newer tip and be rejected by §7. A steady-state catastrophic stall (where the last landing became final long ago) incurs no such wait. Determinism likewise falls back on the landing rule: a batch may only extend the current canonical landing tip (§5.6), and a recovery block built on an old tip, if the tip has since advanced, is benignly stranded and can simply be rebuilt (there is no episode or deposit that could get out of order; during a true stall the tip does not move, so the recovery block is certain to land). F_{l1} makes the predicate "final landed block" stable under shallow L1 reorgs (linked to L1 reorgs in §11).
- **A tier (ii) block header must carry an L1 reference attesting that [at signing time the parent was already F_{l1} -final] — closing the pre-signing loop at the F_{l1} boundary**^[30]: checking the parent's finality only at landing time hands the aggregator a loop: pre-sign and pre-prove a tier-(ii) block while the tip is not yet final, then race to land it in the first L1 block at which the tip has just become final; this gives one iteration every F_{l1} , the lag never crosses its threshold, and the honest content has not yet aged to Δ_{prop} and therefore cannot challenge $\rightarrow$ a sparse forced-only chain lands forever. The fix: the tier (ii) block header gains a new field final_ref = a reference to an L1 block, and the circuit / L1 entry point verifies that "this L1 block has already witnessed the parent reaching F_{l1} depth", with $\text{final_ref} \leq$ this block's anchor (available at signing time). The parent must therefore already be F_{l1} -final at signing time, and an unfinalized tip cannot be pre-signed; the race-to-land loop is broken (to land a tier (ii) block one must wait until the parent is truly final before signing, and by that time the honest content tail has had time to be proved and to land in the settlement window as a heavier candidate, which by §5.6 beats the sparse forced-only candidate outright). Normal recovery is unaffected: a recovery block is emitted only after its parent is final anyway.
- **Justification for the value $G_{\max_landed} = \infty$ (direction B)**^[31]: tier (ii) has no cap because the reorg exposure through a landed head is naturally bounded by the length of the unlanded tail, and the tail length is bounded by Δ_{lag} — a tail older than Δ_{lag} has already been landed by the fallback, is final, and can no longer be reorged. "Unbounded" therefore does not increase the worst-case reorg depth (still $\leq \Delta_{\text{lag}}$, the same bound as

the existing residual risk in §5.4), yet it lets a stall of arbitrary duration be recovered in a single hop (see §8). A finite value loses at both ends: below the proving latency it can never catch up with a long stall, and set equal to Δ_lag the tightening is an illusion (nature already gives Δ_lag) — for the full argument see Appendix D, entry v1.31.

- **"tail $\leq \Delta_lag$ " is conditional, not unconditional** ^[32]: the claim above that "the tail length is bounded by Δ_lag " holds only when permissionless fallback can land an old tail within $\approx \Delta_lag$, which is exactly the same conditionality as the one §6.3 ^[33] attaches to the liveness bound under a Byzantine aggregator. If fallback is economically infeasible (nobody is willing to be a fallback lander) or the fallback landing transaction is itself censored on L1 (already listed in §6.3/§5.2), the unlanded tail will grow without bound as the stall continues, $> \Delta_lag$; in that case $G_max_landed = \infty$ in tier (ii) lets a single malicious builder orphan this over-long tail with one block (no longer requiring the coalition lookahead density of tier (i), which grows harsher as the tail lengthens), and a colluding lander that lands it thereby orphans honest preconfirmations reaching far beyond Δ_lag . The honest conclusion: the worst-case reorg depth of tier (ii) is $\leq$ the actual unlanded tail length, and it is $\approx \Delta_lag$ if and only if the fallback lands on schedule; when the fallback stalls, the depth grows with the stall, which is the same condition and the same root cause as the possibility in §6.3 that finality stalls indefinitely (no protocol-level DA + fallback economics / L1 censorship resistance). This is not a hole newly introduced by ∞ (a relayed coalition in tier (i) can equally orphan an entire over-long tail, merely at greater expense), but ∞ lowers the builder-side cost in that regime to "one person, one block", and this must be spelled out together with the optimistic annotation of $\leq \Delta_lag$. r41 note: under the same "someone is willing to land" condition, the §5.6 settlement window turns this kind of orphaning branch into a self-defeating move (a heavier honest tail candidate simply overrides it inside the window), but when "nobody provides the fallback / the fallback is censored on L1" the window may contain only malicious candidates — the conditionality is unchanged, only the failure mode shifts from "the malicious chain becomes final" to "the malicious chain becomes final in an uncontested window". This is still stated honestly.
- **Deep-reorg residual risk** ^[34]: no protocol-level DA $\Rightarrow$ L1 cannot see the unlanded tail $\Rightarrow$ a colluding lander can always land a legal branch that excludes certain unlanded blocks, orphaning the preconfirmations they contain. The worst case depth = the unlanded tail length $\approx \Delta_lag$ + the fallback response time (once a tail is landed it is final). The builder-side cost under the two-tier rule: in tier (i) a single block reaches $\leq G_max - 1$, and going deeper requires a relayed sparse coalition; in tier (ii) a single block already suffices to offer a branch whose parent is a landed head and which orphans at most the entire unlanded tail. But both take effect only when landers collude — an honest lander follows the §5.2 fork choice, lands the heaviest tail and ignores such short branches (see the lander strategy in §5.2); the worst-case depth ($\leq \Delta_lag$) and the binding constraint that "lander collusion is required" are both unchanged, and all that changes is that tier (ii) lowers the builder-side cost of a fully orphaning branch from "a dense coalition" to "a single malicious builder". This is a residual risk explicitly accepted under the trust model of §1 (the cost identity in Appendix A-3), and it breaks only the tier-2 commitment (§5.4, which already declares that it can be revoked by a single malicious builder). G_max (tier i) remains the governance knob that suppresses the feasibility of a sparse coalition.

4.3 Equivocation slashing (verified directly on L1)

- **Definition of the fault:** the same builder signs two byte-wise different block headers for the same slot.
- **Evidence and enforcement:** anyone may submit the two (block header, signature) pairs to the L1 slashing contract; the contract verifies the two signatures + same slot + different hashes directly on chain — no zero-knowledge proof and no arbitration period are needed, and once verification passes L_{eq} is slashed, of which $\geq 80\%$ is burned and the remainder is paid to the submitter (following v15's burn-dominant principle, which prevents self-slashing arbitrage).
- **The point at which a slashing takes effect is a deterministic, slot-based rule** ^[35]: if the timestamp of the L1 block that includes the slashing transaction corresponds to L2 slot s_0 , then the slashing takes effect from slot $s_0 + \delta_{slash}$ (initially $\delta_{slash} = 64$ slot): once it is in effect, every slot already scheduled to that builder in the lookahead is treated as a gap — the circuit rule: when a batch lands, the L1 contract hands the list of (slashed party, effective slot) entries it has itself recorded to the proof verification as a verified input; any block within the batch that is "signed by a party whose slashing is already in effect, with slot $\geq$ the effective slot" is invalid. Effectiveness is delimited by slot (rather than by each block's anchor block), which guarantees that all forks agree on the judgement of "who has already been slashed", so that no anchoring race exists. Without this clause, a slashed builder with a zero bond could still produce legal blocks (§4.1 and §5.1 would then contradict each other).
- **Effectiveness must additionally be pinned to [the point of batch acceptance], closing off the backfilling of historical slots** ^[36]: the rule above decides purely by $header.slot \geq effective\ slot$, yet $header.slot$ is filled in by the signer itself — a slashed, zero-bond old key can, after the slashing has taken effect, sign a historical block with $header.slot < effective\ slot$, bypass that comparison and keep producing legal blocks (which can also be used to grief with a stale tip, §6.4 / independent review finding 5). The criterion ^[37] = the L1 landing time of the [candidate batch] the block belongs to — inherently unforgeable, replacing the commitment timestamp of v1.40 (the commitment layer was deleted along with the old §5.7) and the δ_{land} guess of v1.39 ^[38]: once a slashing has taken effect on L1, no candidate batch whose [L1 landing time $\geq$ the effective time] may contain a block from that signer — regardless of what $header.slot$ says; in candidates that had already landed (provisional or window-final) before the effective time, that signer's blocks are grandfathered in as valid. A candidate's landing time is an intrinsic, unforgeable property of the L1 transaction: a historical block backfilled after the fact can only appear in a candidate that lands after the effective time $\rightarrow$ it is necessarily rejected; a block that had already landed inside some candidate before the effective time $\rightarrow$ it is retained. The cost (stated honestly) ^[39]: that signer's in-flight blocks, signed before the effective time but not yet landed in any candidate by then, are rejected along with the rest — and the honest successor blocks built on them are stranded in cascade because their $parent_hash$ runs through such a block (when the equivocator broadcasts only one of the two forks, honest successors all build on it; when δ_{slash} is shorter than the time to prove the first batch, every candidate that contains it after the effective time is rejected). This still falls within the already-accepted residual risk of "slashable equivocation + $\leq$ the actual tail depth", but the pricing of L_{eq} and the user-facing semantics must acknowledge this successor-cascade exposure; no δ_{land} grace parameter is needed (deleted).

- This class of fault is submitted by a challenger and verified mechanically by L1: one level stronger than the preconf-vs-record variant in v15 (§7b) that relied on an L2 evidence chain — zero delay after submission, no watchtower needs to run a proof, and all that is required is that someone be willing to submit (there is a bounty, so it is paid permissionless work).

5 The signature chain: validity, canonicity, preconfirmation semantics

5.1 Single-block validity (circuit rules)

A block is valid if and only if all of the following hold (all of them circuit-verifiable):

1. The signature is valid and the signer = lookahead(slot) (the lookahead is recomputed inside the circuit from L1-anchored data). There is no exception to this rule (after v1.51 removed forced inclusion, the P_forced forced-only-block exemption went with it — the right to produce a block belongs to the scheduled builder alone; the cost is that under a full cartel there is no longer an any-key escape path for block production, see §8 and §1). This replaces the old "three conditions" formulation that was already deleted before v1.31^[40];
2. parent.slot < slot, and the parent block satisfies one of the two tiers of the §4.2 rule^[41]: (i) when the gap satisfies $\text{slot} - \text{parent.slot} \leq G_{\text{max}}$, the parent may be any real earlier block in the signature chain (landing status is irrelevant); (ii) the gap has no upper bound, but the parent must be an L1-final landed block. parent_hash points to that valid parent. A tier (ii) block additionally requires a valid final_ref^[42]: final_ref witnesses that the parent has reached F_l1-final and that final_ref ≤ this block's anchor (§4.2); that the parent "is the current canonical tip" is still checked by §5.6 at landing time (every candidate is verified against the frozen window baseline; the two predicates are kept separate: at signing time finality is proved by final_ref, at landing time canonicity is verified by the entry point);
3. slot does not run past the range declared by the batch that lands it; the timestamp rule is automatically satisfied (= slot);
4. The execution of the transactions in the block is valid (ordinary state-transition verification); the anchor tx satisfies the freshness rule (§7) and satisfies the §7 causal-ordering invariant $\text{anchor.L1_timestamp} \leq \text{L2_timestamp}(\text{slot})$ ^[43]; and inbound L1→L2 messages are consumed per §7 up to the [unified maximum prefix] — the longest prefix of the FIFO order that simultaneously satisfies "count ≤ C_anchor" and "cumulative declared gas ≤ the L1→L2 message gas share"^[44]; taking that longest prefix is valid, and only taking less than it is invalid (this rules out the indefinite censorship of "using a fresh anchor while processing zero inbound messages")^[45].

5.2 Canonicity and fork choice

- **The portion already landed on L1**^[46]: the canonical chain = the chain obtained by stringing together the window-final batches at the close of each settlement window (§5.6). Conflicting candidates within a window are superseded by the §5.2 total order, and the close settles the matter for life; only after the close is a candidate "permanently out".
- **The coordination convention for the unlanded portion (the P2P tail) — the ordering criterion is given uniformly by the "best-chain total order" of the next bullet**^[47]: honest builders/nodes build on the chain head that is best under the total order of the next bullet

(the old text's "most blocks + highest slot + hash" has been folded into levels 2/3/4 of the total order and is no longer listed separately, so that it cannot conflict with the total order). Positioning of its effect^[48]: this is an off-chain P2P coordination convention, not a circuit rule or an L1 entry-point rule — it constrains only the convergence of honest P2P nodes, and cannot constrain the fork choice of a malicious aggregator / fallback lander / private relay / L1 proposer; the real constraint on malicious landers is given by the §5.6 settlement-window competition (a worse candidate is mechanically overridden by a heavier one), and the bounding of the residual deep-skip exposure is still found in §5.4.

- **The best-chain total order — shared by three places: fork choice, landing strategy, and the §5.6 window candidate comparison^[49].** This bullet supersedes the v1.35 "recovery first / frozen dead tail yields" rule — the independent review (rounds 1/2) pointed out that that rule would order an honest lander to discard a fully visible, landable, healthy long tail, which is wrong: "cannot be extended over P2P" does not mean "should be discarded". Given a fork point F, the candidate chains extending F are compared, in the following order: The object of comparison is a scalar triple, not a pairwise structural comparison^[50]. Each candidate chain extending F independently computes the triple $\text{key} = (\text{count}, \text{tip_slot}, \text{tip_hash})$:
 1. **count**: the number of valid blocks starting from F;
 2. **tip_slot**: the slot of the tip;
 3. **tip_hash**: the block-header hash of the tip (the smaller one wins; this is the last level and settles the tie-break)^[51]. Comparison = the lexicographic order of the triple (count descending, tip_slot descending, tip_hash ascending) — a lexicographic order over scalars is naturally reflexive, total and transitive, so §5.6's "strictly heavier" relation and the winner at the close are thereby well-defined and independent of submission order. (The formalized property test is listed as item 18 of §11.) **lane retires in v1.51**: r42 had put **lane** (the class of the first block: a discretionary block = content lane, larger; a forced-only block = forced-only lane, smaller) in the leading position of the key, so that content chains would outrank forced-only chains and the whitewash of r40 finding 3 would be resisted. With forced inclusion removed there are no forced-only blocks, all candidates are of the content class, **lane** is constant and draws no distinction at all, so the component is dropped. The non-transitive cycle ($A > B > C > A$) that r42 fixed does not come back: its root cause was that **lane** had been a pairwise criterion, whereas the three remaining components were always scalars belonging to the candidate itself. **A "frozen long tail" is no longer discarded**: a long tail whose head lags the wall clock by more than G_{max} and can no longer be extended over P2P (no tier-(i) child, and, not being final, no tier-(ii) child either) is still the best chain among the candidates — the lander should land it first (honoring its preconfirmations) and only then resume producing blocks from its final tip (§6.4); landing it does not require extending it over P2P, only proving and posting it. The previously feared "deadlock between recovery and fork choice" thereby disappears: first land the best chain (frozen ones included) → then recover in one hop from its final tip.
- The final arbiter is still L1: disagreements are resolved once and for all by the §5.6 total order at the close of the settlement window^[52]. Equivocators are slashed (§4.3); blocks that forked innocently because of a network partition become orphans, and the transactions in them return to the mempool to be repacked.

- **The lander's tail-selection strategy**^[53]: when a lander faces several candidate tails, it applies the best-chain total order above to its own view and lands the best one. This no longer needs to be a "trusted obligation" — landing a worse chain gets it directly overridden by a heavier candidate inside the §5.6 window and its cost is spent for nothing, so landing the best chain is the only non-futile strategy; the protocol does not need to judge what a lander "ought to land" (which is exactly what v1.39/v1.40 proved cannot be decided mechanically), it only needs to let the heaviest proven candidate win. Δ_prop is retained as an off-chain strategy reference for the completeness of the lander's view(land the frontier where propagation has fully settled)^[54], and is no longer any kind of on-chain liability parameter. The handling of "there is a better tail but the lander did not see it"^[55]: some lander did not see the heavier tail and landed a worse candidate? — that is fine: anyone who does see it lands the heavier tail together with its proof inside the same §5.6 window and directly overrides it. The incompleteness of a single lander's view is no longer a safety dependency^[56]; overall it is still required that at least one participant inside the window has a complete view and is willing to land — normal P2P propagation takes seconds and the pending frontier has long since spread everywhere, so this condition is naturally satisfied under the §6.3 fallback feasibility; network-wide eclipse-grade blinding is delegated to the §11 P2P spec.
- **The visibility assumption and the "economic only, not enforced" characterization**^[57]: this design assumes that, under normal P2P propagation, the aggregator (and any lander) can in most cases see the currently longest available signature chain — but the protocol cannot, and does not attempt to, force a lander to land the longest chain. The reason is structural: visibility of the chain can itself be taken away — the simplest example is a builder at the tail of the chain withholding the block it has just signed (§5.5 withhold-body / withhold-signature); the aggregator cannot obtain it and therefore cannot include it, and that is not a fault. Hence "land the longest visible chain" is a rational choice under economic incentives, not a slashable obligation: a lander that lands a shorter chain is not penalized, and the consequences run through only two economic channels — (i) being superseded by a heavier candidate inside the §5.6 window, having paid gas and proving fees for nothing; (ii) under the base-fee revenue-sharing rule below (§5.6/§6.5), the shorter the chain landed, the smaller the base on which the share is computed. v1.39/v1.40's attempt to turn "ought to have landed but did not" into an adjudicable fault, and the refutation of that attempt, is precisely the cautionary example for this characterization.
- **A node's local chain-selection rule = the total order above(equivocation forks included)**^[58]: the criterion an L2 node uses to dynamically maintain its "local best signature chain" is exactly the triple total order above; no second set of rules is needed. Equivocation is heavily slashed under §4.3, but one cannot assume that it does not happen — when it does, two signed blocks appear at the same slot and the chain splits in two; the two forks are still two individually valid candidate chains, and the triple yields a total order as usual (differing block counts are settled by count, equal block counts are arbitrated by tip_slot/tip_hash), so the convergence of node views is not interrupted by equivocation; slashing the equivocator (§4.3) is orthogonal to chain selection. Why the fee total does not enter the total order(a design trade-off)^[59]: it was at one point considered to make "the sum of all base fees on the chain" one of the ordering inputs. The conclusion is that it cannot enter the canonicity criterion: fees are a quantity an attacker can manufacture by paying itself — a malicious builder that stuffs the block in its own slot with high-fee self-transfers can make a shorter chain beat an honest longer chain on the fee dimension, which amounts to

buying out the honest majority's chain-selection outcome with capital (and if part of the fees flows back to the lander, the colluder's cost of manufacturing those fees is even partly refunded). The total order therefore keeps the number of genuinely signed blocks as its primary key (forging one requires stealing a builder's private key and cannot be manufactured); the fee total enters only the reward (the §6.5 base-fee sharing) and not the ordering — incentive alignment relies on money, canonicity determination relies on signatures.

- **Honest bounding**^[60]: before landing, an equivocating builder + one colluder willing to land its fork (the aggregator, or any lander inside the fallback window, or even a bribed L1 proposer) can make the "other" tail canonical and isolate the preconfirmations contained in it. This attack (i) requires one equivocation that L1 can slash directly (the cost = L_{eq} , priced by the exposure accumulated over the window, §4.1); (ii) has a depth bound = the length of the unlanded tail. The tail length is not "hard" capped by Δ_{lag} ^[61]: $lag > \Delta_{lag}$ only opens the authorization for fallback landing, it does not force any batch to be accepted; while an aggregator that withholds landing drags things out, the tail keeps growing throughout the fallback response time. The honest depth bound = Δ_{lag} + the fallback response time (one round of proving
 - landing $\approx 10\text{--}15\text{ min} \approx 2\text{--}3\text{ epochs}$), and it can be stretched further by sustained L1-level censorship of the fallback transaction — the latter being the most expensive censorship class, already priced in §6.3. This is a soft upper bound plus an explicitly stated response-time assumption, not a hard cap. Therefore the safety of a preconfirmation before landing is deterrence plus bounded depth, not absolute — the user-facing semantics of §5.4 are worded accordingly. Keeping the tail short (§6.3) is the principal means of shrinking the payoff of this class of attack.
- **Canonicity is still determined by L1 alone (G2) — but L1 does not adjudicate "which chain ought to be landed"; the settlement window selects it mechanically**: off-chain fork choice gives honest nodes convergence (the total order above), while "the best chain wins" is delivered mechanically by the §5.6 settlement window — L1 does not adjudicate "who ought to land what", it only lets the heaviest proven candidate inside the window become final. This design is therefore not "L1 enforces the longest chain" (that would require protocol-level DA in order to see the unlanded tail) but "L1 picks the heaviest among the already-landed, already-proven candidates" — no DA, no ex post adjudication; it is the mechanical middle point chosen by the owner under the §1 constraints^[62].

5.3 Gaps

- No valid block appears at slot s (the builder went offline, was DoSed, or its block body did not propagate), the chain is left empty at s , and the builder at $s+1$ builds on an earlier chain head. Gaps are handled as a normal case, not as an anomaly: no proof of "why it is missing" is required, and nobody is slashed for a gap (producing a block is an opportunity, not an obligation — inheriting the principle of v15 §9.3, but now applying to every slot).
- The cost of a gap is absorbed by incentives: a skipped slot earns no fees, and users who hold preconfirmations settle accounts with that builder's reputation. Nuisance-level chronic absence is handled by the registry's liveness rules (an optional item, §11).

5.4 Deletion vs. gaps: what the signature chain fixes and what remains (stated explicitly)

The parent-block-hash chaining fix raised by the owner has precisely the following effect:

- **Fixed: the aggregator's selective deletion.** Because a child block's signature covers `parent_hash`, pulling a block out of the middle of the chain breaks the chain for every successor and the proof necessarily fails. Over a signature chain, the aggregator retains only the power to take a prefix: land $[..m]$ and leave $[m+1..k]$ outside the chain. (The deep-reorg path in which "a malicious builder starts a short alternative branch from an old point without equivocating, for the aggregator to fork-choose" — a single block is capped by $G_{\max}$, but a relayed sparse branch can reach the entire unlanded tail)^[63] And the suffix that is left behind still attaches to the new L1 chain head, so the next batch (anyone's) can land it just the same — truncation is delay, not destruction. Together with the landing deadline (§6.3), a malicious truncation by the aggregator buys little.
- **What remains: the skip by an adjacent builder — and it is profitable^[64].** Builder $s+1$ builds on $s-1$ and claims never to have seen the block at s — which cannot be falsified inside the protocol ("I did not receive it" is unprovable). v1.4 previously extrapolated "the actor gains nothing" onto skipping, which was wrong: the skipper can repack the high-priority-fee transactions and the MEV from the skipped block s into its own block and still collect the entire revenue of its own slot — a skip is a profitable, non-slashable revocation of preconfirmations that a single untrusted builder can carry out on its own. Four points of honest bounding: (1) the choice is fixed by a signature and publicly visible (`parent_hash` in black and white, plus the signed block at s circulating over P2P), so attribution has hard evidence; (2) the one-shot payoff of an adjacent skip is bounded by the fee/MEV transfer of one slot; **Deep skip / sparse private branch^[65]:** with a single block, one malicious builder can, in a single deep skip, affect at most $G_{\max} - 1$ unlanded blocks; but a minority coalition that relays several blocks along a private branch (each hop keeping the parent distance $\leq G_{\max}$, every block valid and with no equivocation)^[66] can affect the entire unlanded honest tail. The coordination convention as revised in §5.2 makes the honest network not follow such a fork, so it must be landed on L1 ahead of the others in order to take effect. The crux of the race^[67]: if the colluding lander is only a fallback lander, this really is a race against the honest batch already in flight (the window $\approx$ the normal landing cadence, and the fallback right does not open at all before $\text{lag} > \Delta_{\text{lag}}$); if the colluder is the incumbent aggregator, then during normal operation it is the only routine lander and there is no race to speak of. This combination is a security residual risk that this design prices explicitly, and its depth bound splits into three tiers according to the form of the attack^[68]:
 - **Tier (i), single-block deep skip (unlanded parent):** depth $\leq G_{\max} - 1$ (a tail of roughly 1 minute), one malicious slot;
 - **Tier (i), relayed sparse branch:** depth $\leq$ the unlanded tail $\approx \Delta_{\text{lag}}$ + the fallback response, at the cost that the coalition must hold one scheduled slot in every $G_{\max}$ window of the tail (the density requirement becomes harsher as $G_{\max}$ is tuned smaller);
 - **Tier (ii), single block via a landed head^[69]:** one malicious builder produces a block whose parent is a final landed head, and in a single step can supply a branch that isolates up to the entire unlanded tail, with no coalition density required. **Precondition^[70]:** for that branch to be landable on L1 by a colluding lander, its final parent block must at the same time be the current canonical landed tip (§5.6 only lets candidates that extend

the current tip land) — that is, the current tip must itself already be final. In steady state the tip is usually already final, so the precondition usually holds and the threat is not weakened; the point here is only to state the residual risk accurately (when the tip is not final, this single-block attack must first wait for the tip to become final, which has the same origin as the F_{l1} wait for recovery in §4.2). **The depth upper bound $\leq$ the length of the unlanded tail**, which lowers the builder-side cost of a fully isolating branch from "a dense coalition" to "one person, one block"; the worst-case depth and the binding constraint that "a lander must collude" both stay unchanged — this is the price accepted for $G_{\max_landed} = \infty$, in exchange for one-hop recovery from a stall of arbitrary length (§6.4/§8). " $\approx \Delta_{lag}$ " is a conditional upper bound^[71]: the tail length is $\approx \Delta_{lag}$ only if permissionless fallback lands on schedule (the same condition as the §6.3 liveness bound under a Byzantine aggregator); when fallback stalls, or the landing transaction is censored by L1, the tail grows without bound along with the stall, and a single tier-(ii) block can isolate a tail far longer than Δ_{lag} — see that conditional argument in §4.2 for details. This is not unique to ∞ (a tier-(i) relay coalition can likewise isolate an over-long tail, merely at greater cost), but ∞ lowers the cost in that regime to one person and one block, and this must be stated honestly alongside it. The frequency of both tiers is $\leq$ the number of scheduled slots the coalition obtains, everything is attributable throughout (the parent choice is signed and on record, and the aggregator's truncated landing is public on L1), the only mechanical deterrents are seat value and reputation, and the protocol does not slash builders; but from r41 on, for these isolating branches actually to be finalized they must in addition beat a heavier honest candidate inside the §5.6 settlement window — as soon as somebody lands the honest tail together with its proof into the window, the isolating branch is self-defeating; the residual risk narrows to "nobody lands the honest tail inside the window" (the fallback feasibility condition, §6.3). Complete structural elimination still requires protocol-level DA (Appendix A-3), which is not adopted, per §1. $G_{\max}$ is thereby a governance knob: tuning it up strengthens gap tolerance (§6.4), tuning it down raises the coalition's scheduling density required for a relayed sparse branch. The direct repacking payoff is still bounded by the capacity of the attacking block itself; the frequency of both kinds of skip is bounded by the fraction of dishonest builders — the "honest majority" of the §1 trust model is load-bearing exactly here (this is not conservative redundancy); (3) the harm to users is layered: pure "inclusion"-class commitments are usually honored in the next slot via repacking (the skipper has exactly the incentive to collect the fees of those transactions), while what is genuinely broken are the "ordering / exact state"-class commitments; (4) the handling = public attribution + the registry's liveness/reputation rules (item 1 of §11) + users penalizing repeat offenders with their flow. This is a deterrence-grade residual risk, which this design accepts and states explicitly; structural elimination requires protocol-level DA / availability proofs (for the cost identity see Appendix A-3).

- **Corollary — preconfirmation semantics(the version written for user documentation)**^[72]: there are four ways in which the preconfirmation of a signed block can be broken:
 1. The builder that signed it equivocates and a colluding party lands the other fork — the cost is being slashed directly by L1 for L_{eq} , priced by the exposure accumulated over the window (§4.1); the isolable depth is $\leq \Delta_{lag}$ + the fallback response time(a soft upper bound)^[73];

2. Some later builder skips it — profitable for the skipper (fee/MEV transfer) and not slashable^[74]; publicly visible and fixed by a signature; an adjacent skip affects a single slot, a single-block deep skip affects at most $G_{\max} - 1$ unlanded blocks, and a minority coalition's relayed sparse branch can reach the entire unlanded tail^[75]; it generally has to win the landing race^[76] — but no race is needed when the incumbent aggregator colludes^[77]; the frequency is bounded by the honest-majority assumption; pure inclusion-class commitments are usually honored via repacking (§5.4 above);
3. The builder that signed it withholds the block body (§5.5) — the block can never be landed, the preconfirmation lapses as a matter of course, and the handling is likewise deterrence-grade;
4. **A deep L1 reorg** — a residual risk shared by every L2. The correct statement is this: no single participant can annul a batch of preconfirmations "at no cost and silently" — every path either pays a slashing that L1 can verify directly, or leaves a public record of behavior with a signature on file; but "collusion that can afford the cost" can do it within a bounded depth. This is still stronger than the single seat of v15 (where the silence of one seat annulled the entire epoch's service with nobody able to substitute), but it is not an absolute guarantee and should not be advertised to users as an absolute guarantee. **The product must distinguish three tiers of commitment^[78]**: (1) inclusion intent — the transaction will get on chain, and this usually survives skips/withheld bodies (via repacking); (2) ordering/state preconfirmation — in this design it can be profitably revoked by a single malicious builder in the next slot^[79], and is backed only by frequency bounds and public attribution, with no slashing; (3) landed finality on L1 — absolute (up to the L1 reorg depth). User-facing wording must be honestly labeled by tier, and tier 2 must not be advertised as a strong safety commitment.

5.5 Signing the header without publishing the body (withhold-body) — a known non-slashable misbehavior^[80]

- **The behavior**: the builder broadcasts (block header + signature) but withholds the block body. The consequences: later builders cannot compute its state and can only skip it; the aggregator cannot obtain the data and can never land it; the preconfirmations of users holding receipts lapse.
- **Structural self-healing^[81]**: a builder can only build on a parent block whose body it holds and whose state it can execute out — this is a physical fact of execution, and need not be "required" as a rule. A suffix whose bodies are withheld will therefore be automatically bypassed as soon as the first non-colluding builder's turn comes (it has no choice but to attach to the last available block), producing an available fork that anyone can land. For a withheld-body chain to hold up the finality of the whole chain, every scheduled builder after it would have to keep colluding to share the bodies privately — that is the full-cartel case, governed by the backstop clause of §7.
- **The characterization, stated explicitly**: this is not a slashable fault — "it has the body but does not publish it" is unprovable inside the protocol (the same undecidability as "skipping" has at its root). This design does not pretend to solve it; the handling is:
 1. **Preconfirmations before landing explicitly carry a DA assumption**: user documentation must state that "a preconfirmation before landing relies on best-effort availability of the block body on the P2P network, with no protocol-level DA guarantee";

2. The receipt format (§4.2) leaves the victim user holding (block header + signature + inclusion proof) — public evidence of the withholding behavior, for use by the reputation system and by off-chain accountability;
 3. The withholder itself gains nothing (its block earns no fees and is bound to be skipped), so this is purely spiteful behavior, and deterrence rests on reputation and the registry's liveness rules (item 1 of §11).
- The honest conclusion of this design is therefore^[82]: **the slashable faults are the two classes that L1 can adjudicate mechanically — equivocation (§4.3) and landing-timeout strikes (§6.3)**; a lander landing a worse chain is not a "slashable fault" but "self-defeating" (inside the §5.6 window it is overridden by a heavier proven candidate and its cost is spent for nothing — there is neither a need nor an ability to adjudicate "what ought to be landed" mechanically)^[83]; skipping and withholding bodies at the builder level remain non-slashable. The wording of §9 and §10 is updated accordingly. An honest inventory of the weak-enforcement class^[84]: withholding a body usually yields no direct protocol revenue; skipping (deep skipping included) can capture fees/MEV and can revoke ordering/state-class preconfirmations, and is not slashable — the only handling is frequency bounds (honest majority) and public attribution.

5.6 Settlement windows and heaviest-proven-batch finality (settlement-window finality) — mechanical, DA-free tail protection^[85]

*This section supersedes in its entirety the "challenge slashing" of v1.39 (old §5.6) and the "chain-head commitment accumulator" of v1.40 (old §5.7). Independent review rounds 3 and 4 proved the fatal weakness common to those two layers: for L1 to adjudicate mechanically that "the lander skipped a better chain", L1 must be able to prove that that chain existed at the time, that its block bodies were available, and that its execution was valid — without full DA plus a challenge/deposit/generational-ledger state machine this simply cannot be done, and doing it would be episode-level complexity flowing back in. On this basis the owner decided to change the root of the design (option C): no longer to adjudicate after the fact that a chain "should have been landed but was not", but instead to relax finality from "the first batch to be proven is final" to "the [heaviest proven batch] inside the settlement window wins and is final". A "better chain" therefore need not be recognized by L1 after the fact — it lands on its own, carrying its proof, and wins outright inside the window. Comparison takes place only among candidates that have all landed and all passed validity proving, hence: - **No block-body availability problem**^[86]: a chain that cannot produce its block bodies can never become a candidate at all, and nobody is punished on its account; - **No commitment-ledger/poisoning problem**^[87]: there is no cross-window commitment state whatsoever; every window starts afresh from the current final head; - **No challenge/deposit/slashing state machine**: there is no L_land/L_chal/L_commit, no evidence submission, no rebuttal, no response period — landing a worse chain is not a "punishable fault" but a self-defeating futility (inside the window it is simply overwritten by a heavier candidate, and the gas and proving fees are wasted). Finality is still delivered by the proving system plus a deterministic window — only batches carrying a validity proof may compete, and window close is a deterministic L1 height; the price is that finality waits one additional W_settle (see §6.2; the seconds-scale preconfirmation experience is unchanged).*

The window lifecycle in a single diagram (in one-to-one correspondence with the pseudocode below and with the executable model of Appendix C):

- **Candidates and the settlement window — including the [baseline freeze + candidate versioning + close commit] state machine**^[88]: let the current window-final head be F. Opening the window immediately freezes the baseline tuple base = (F, F.stateRoot, F_consumed, m_consumed); every candidate inside the window is verified against the same

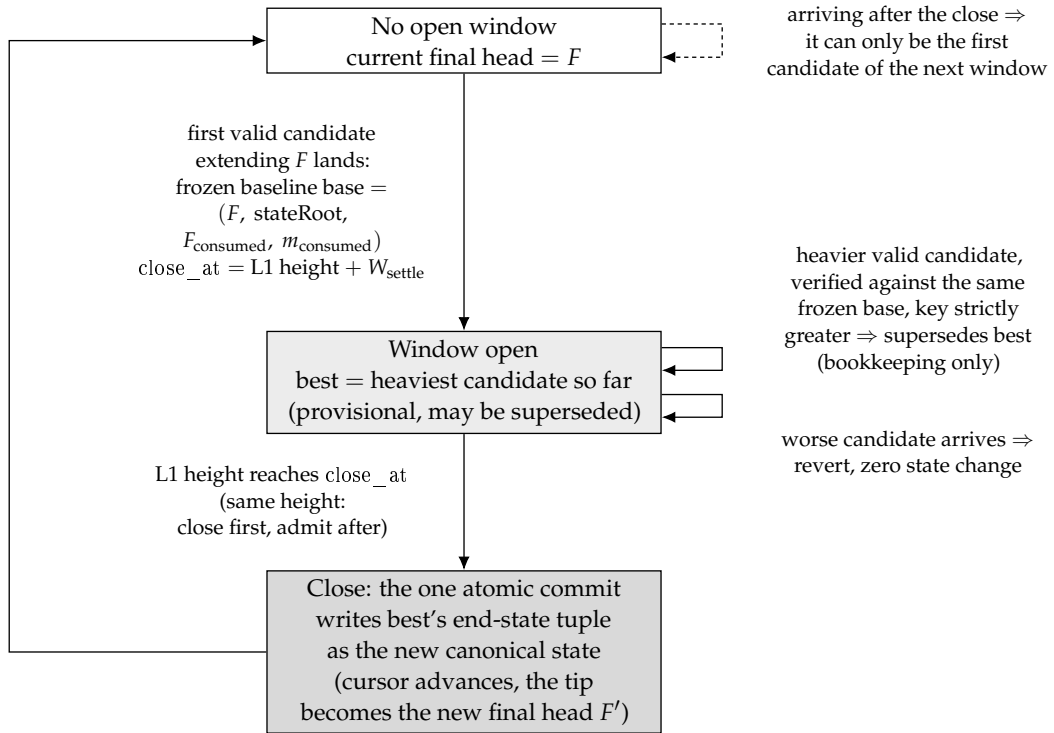


Figure 5: The life cycle of a settlement window as a state machine: opening the window freezes the baseline, every candidate inside the window is verified against that same baseline and only updates bookkeeping, and not until the close is the heaviest candidate's end-state tuple committed atomically as the new canonical state.

frozen base (the proof's starting cursors/state root = base, consuming the same queue heads), and passing verification records only that candidate's own end tuple $\text{end_i} = (\text{tip_i}, \text{stateRoot_i}, \text{F_consumed_i}, \text{m_consumed_i})$ together with its §5.2 weight key key_i — provisional acceptance changes no canonical state (it does not advance cursors, does not switch parents, and does not execute bridge effects). At window close, only the end tuple of the current heaviest candidate (the one with the largest key) is committed atomically, in one step, as the new canonical state; its tip becomes F' , and the next window starts from the new base derived from F' . Pseudocode:

```

openWindow(F): base  $\leftarrow (F, F.\text{stateRoot}, F.\text{consumed}, m.\text{consumed})$ ; best  $\leftarrow \perp$ ; close_at  $\leftarrow L1.\text{now} + W.\text{settle}$ 
acceptCandidate(c): requires best =  $\perp \vee L1.\text{now} < \text{close\_at}$ ; # close precedes acceptance (r44 normalization, W2):
# at the same L1 height, closeWindow is evaluated first, acceptance afterwards;
# a candidate arriving after that point can only open the next window
if best =  $\perp$ : openWindow(current window-final head); # the first candidate opens the window
verify(c.proof, from = base); # all candidates share one frozen baseline
requires best =  $\perp \vee \text{key}(c) > \text{key}(\text{best})$ ; # §5.2 triple lexicographic order, strictly heavier
best  $\leftarrow (c.\text{end}, \text{key}(c))$  # bookkeeping only, canonical untouched
closeWindow(): requires  $L1.\text{now} \geq \text{close\_at} \wedge \text{best} \neq \perp$ ;
canonical  $\leftarrow \text{best}.\text{end}$ ; # the single atomic commit
credit(best.beneficiary,
 $\varphi_{\text{land}} \times \Sigma \text{base\_fee}(\text{best})$ ) # the base-fee share is booked at close (§6.5, r47)

```

(What the baseline freeze covers is [the cursors and the state]; it does not include queue contents^[89]: the forced-inclusion/message queues on L1 are append-only, entries are referenced by global sequence number, and an entry's content is immutable once enqueued — so there is no need to "freeze a queue snapshot": two candidates starting from the same frozen cursor necessarily read the same content at every sequence position; new entries enqueued midway through the window are visible and determinate to any later candidate "whose blocks can reach them" (a heavier candidate legitimately consumes more), and the window state remains a pure function of L1 history — enqueue events included. In implementation this requires the queue commitment to be organized per-entry and append-only (a hash chain or an MMR), and proofs to reference entries by sequence number; a "single mutable queue root as of the moment of landing" must not be used as a public input — otherwise the public inputs of successive candidates within a window would drift. Model property P12. `closeWindow` may be called by anyone, or folded into the first transaction of the next window as a lazy close; the adjudication order at one and the same L1 height is normalized to "close precedes acceptance" — candidates arriving at or after the `close_at` height do not take part in this window and can only be the first candidate of the next window, and any implementation (including lazy close) must be consistent with that semantics^[90]; the window state and `best` are both pure functions of L1 history, and under a shallow L1 reorg they roll back together with L1 — folded into item 9 of §11.) Stated honestly: this is a small window state machine (three actions: baseline freeze / candidate versioning / close commit) — far smaller than challenge/deposit/DA or the old episode, but v1.41's claim of "no extra state machine" was inaccurate, and r42 corrects it. The atomic advance rule for the `m_consumed` cursor of §7 is correspondingly re-read: both the starting check and the advance are relative to the window baseline and the close commit, and no longer to "whoever lands first advances".

- **Why "the heaviest proven batch" is by itself a mechanical protection of the tail:** an honest lander brings the complete honest tail (necessarily the heaviest under the §5.2 total order — under an honest majority, any branch that excludes honest blocks contains fewer blocks, see §5.4) onto L1 together with its proof, and it wins outright. What if a malicious lander gets ahead by landing a worse chain? Inside the window it is overwritten by the

honest candidate; nothing is left on chain and no compensation has to be adjudicated, and the malicious party has paid L1 gas plus proving fees for nothing (a natural grieving cost that requires no design). Every "how do we prove that it should have landed at the time" difficulty of the old v1.39/v1.40 ceases to exist: the chain that ought to land lands by itself and wins, and nobody needs to prove "that it once existed".

- **"Sniping" at window close is benign (self-review rounds 1 and 2):** what if someone lands a slightly heavier candidate at the very last moment before close? Under the total order, "heavier = more genuine signed blocks" — forgery is impossible (forging would require stealing a builder's private key), so a heavier candidate necessarily contains more genuine preconfirmed blocks, and its winning is better for users (a more complete tail is finalized). The only party harmed is the earlier lander, in gas — an economic residual risk, not a safety residual risk. A fixed window therefore suffices, with no need for a chess clock or extensions (an extension would instead introduce gameable timing). Withholding a block body and releasing it late (a builder withholds the body → the fallback lands the visible tail → before close the attacker releases the body, lands the full tail and overwrites it) is the same story: the result is that a more complete tail is finalized, and the fallback lander that was "overwritten" loses gas — handled through the fallback-compensation path of §6.3 and listed as an economic residual risk.
- **The setter invariant for W_settle :** $W_settle \geq P_prove,max + T_include,max + margin$ — this guarantees that, from window opening onward, anyone who sees a heavier tail has time to prove it and land it. An honest lander need not wait for the window: it starts proving as soon as it sees the tail, so under normal conditions the heaviest candidate is already on its way when the window opens. The steady-state cost is zero: with no competition the window contains only the aggregator's own single candidate, and nobody spends an extra cent — the competition mechanism is exercised only when the system is under attack.
- **The window state is a pure function of L1 history:** candidates are all L1 transactions, window opening and close are L1 heights, and the current best is the deterministic result of transaction-by-transaction comparison — any node that replays L1 obtains the same window state; under a shallow L1 reorg it rolls back together with L1 (folded into the atomic-rollback list of item 9 of §11, with no extra state machine).
- **Liveness and anti-spam (self-review rounds 5 and 7):** superseding requires a strictly greater weight plus a valid proof, so a candidate spammer must pay a real proving fee every time and is capped by the length of the real tail — there is no livelock; a candidate arriving after window close is benignly stranded (it targets the old F , and it suffices to rebuild it onto the new F' , the same stranding semantics as in §6.4). The candidate's beneficiary (the reward/refund address) is built into the proof's public inputs, so mempool copying cannot change where it points ^[91].
- **Explicitly stated residual risks (stated honestly):** (a) temporary ordering swings during the window: the "current best candidate" inside a window may be superseded by a heavier one, so the temporary head that L2 nodes follow is mutable within the window — this corresponds precisely to the existing tier-2 preconfirmation semantics (revocable, decided by window close); it adds no new exposure and merely makes that exposure explicit; withdrawals and bridging execute only from window-final state (§6.1/§6.2). (b) **Fallback for excluded blocks:** if the heaviest candidate still excludes certain genuine blocks (for example, an attacker wins some window with a sparse branch built on enough slots of its own in the lookahead — under an honest majority this requires a coalition with dense lookahead

scheduling, bounded as in §5.4), the excluded transactions return to the mempool/forced-inclusion queue as before and are repacked in the next window — the worst-case depth is still $\leq$ the unlanded tail, the same bound as the existing residual risk of §5.4, and C does not widen it. (c) **Finality + W_{settle}** : this is the real cost of option C, and the owner accepts it explicitly (accounted for in §6.2).

6 Landing

6.1 Batches

- A batch = a contiguous segment of the signature chain $[m+1..k]$ (extending from the current window-final head m , §5.6) + block-body data (blob) + a validity proof that verifies all the rules of §5.1 together with the per-block state transitions.
- **Atomic landing + window finality** ^[92]: data, proof and state root are still submitted in a single transaction and verified on the spot by L1 — there is still no "proposed but not yet proven" intermediate state, no sealing deadline / cancellation horizon / cancellation cascade / timing-exemption net (the v15 complexity reduction is retained). But what "verification passed" now yields is the current best candidate (provisional) of that window, rather than immediate finality; finality is granted at settlement-window close to the heaviest proven candidate in the window (§5.6). Under normal conditions, with no competition, the aggregator's candidate is the only candidate and close is finality — the semantic difference appears only under attack. **Withdrawal and L2→L1 bridging effects execute only from window-final state**, never from provisional state.
- Batch size is bounded above by blob capacity and proving cost; batches need not align with epoch boundaries.
- **Future slots must not be landed** ^[93]: the slot of a batch's last block must be $\leq$ the $L2_{\text{slot}}$ corresponding to the timestamp of the L1 block in which it lands (the permitted clock skew is initially 0) — checked at the L1 entry point and proven identically in the circuit. The lookahead is published 768 slots in advance; without this constraint a builder could pre-sign blocks for future slots, and a batch could push the landed chain head ahead of wall-clock time — destroying the timestamp/anchoring/forced-expiry semantics and artificially driving the lag of §6.3 negative. Correspondingly, the arithmetic definition of lag is truncated below at 0 (safe arithmetic).

6.2 Cadence

- Target landing cadence: the aggregator should keep the lag lag defined in §6.3 (the number of slots by which the tip of the best provisional candidate trails L1 wall-clock time — observable on L1, referring to no P2P state)^[94] within Δ_{lag} . A proving latency of about 10–15 minutes means that in steady state provisional landing trails the live chain head by about 15–20 minutes; window-final finality adds one further W_{settle} ($\approx$ one round of proving plus landing, §5.6/§11)^[95] — the decoupling of the preconfirmation experience (seconds-scale) from the finality cadence (minutes-scale) is unchanged, and it is the former that users perceive. The ladder of four levels of certainty, and the two lags:
- L2→L1 bridge (withdrawal) delay = landing cadence + W_{settle} (withdrawals execute

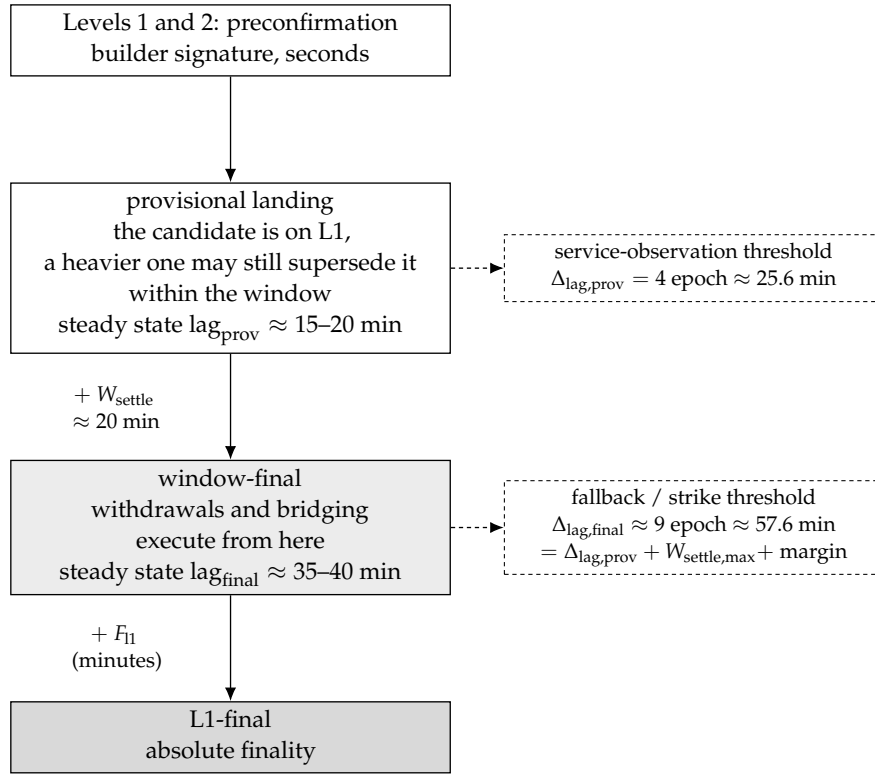


Figure 6: The four-level certainty ladder and the two lags: preconfirmation is a matter of seconds, and provisional landing is one settlement window away from window-final. lag_{prov} serves only to observe the service tempo, while it is lag_{final} that enters fallback and slashing accounting.

only from window-final state, §6.1).

- **No fast close (fast path):** even when a window contains only a single candidate, the full W_settle is awaited before finality — a test for "closing early" would reintroduce gameable timing, and v1 does not do it; it is listed in §11 as an optimization (for example, "a candidate covering all known signature-chain heads may close early" may be introduced only after it has been proven not to be manipulable by block withholding).
- **Two lags, each with its own role** ^[96]: lag_prov = the amount by which the tip of the best provisional candidate trails — used only for observing service cadence (governance signals such as whether the aggregator ought to be replaced); lag_final = the amount by which the window-final head trails — used for safety/revocable-exposure accounting and for fallback accounting (the next item and §6.3). Two thresholds, in one-to-one correspondence ^[97]: $\Delta_lag,prov$ (= the old Δ_lag , initially 4 epoch $\approx$ 25.6 min, setter invariant $\geq$ the upper bound of the normal provisional lag band)^[98] applies to lag_prov ; $\Delta_lag,final = \Delta_lag,prov + W_settle_max(\text{wall-clock conversion}) + \text{margin}$ (initially $\approx$ 9 epoch $\approx$ 57.6 min)^[99] applies to lag_final . Rationale: under window finality the steady-state $lag_final \approx lag_prov + W_settle \approx$ 35–40 min, which by itself already exceeds the old threshold of 25.6 min — if fallback/strikes were adjudicated on lag_final while the old threshold were retained, the fallback window would be permanently open in a completely healthy system, the "cost + markup" reward commitment would remain in force in perpetuity, and a diligent aggregator would accrue strikes continuously until termination; r42's switching the adjudicated quantity without raising the threshold is precisely this bug. The honest bound on revocable exposure (medium 1): although a provisional candidate has already landed on L1, it can still be superseded before close, so the worst-case revocable set of tier-2 preconfirmation = the unlanded tail + the provisional increment within the window, and the true bound = $\Delta_lag,prov + W_settle$ (wall clock) + fallback response $\approx \Delta_lag,final + \text{fallback response}$ — it can no longer be written as " $\approx \Delta_lag$ " on its own; elsewhere in this document, the Δ_lag appearing in depth bounds of the form "depth $\approx \Delta_lag + \text{fallback response}$ " is, per r44, to be read as the fallback-window-opening threshold $\Delta_lag,final$ (a new name for the same quantity; the bound is unchanged, only the naming is made honest). In order to pin this bound down, W_settle is given, besides its lower bound, a consensus upper bound W_settle_max (§11; including the margin for wall-clock conversion when L1 slots are missed), which is folded into the user-facing semantics and the wording of §5.4.

6.3 Landing Obligation and Permissionless Fallback (This Is the Aggregator's Only Obligation)

*This section was rewritten after review r1-2. The v1 draft stated the obligation as "cover up to the P2P chain head" — but the P2P chain head is a quantity that is not observable on L1, so the obligation cannot be enforced mechanically; a malicious aggregator that lands a micro-batch containing only 1 block just before every deadline could both "never time out" and let finality fall arbitrarily far behind. The fix: move every determination onto the L1-observable **lag**, and make the fallback landing itself the incriminating evidence of the aggregator's dereliction.*

- **Lag (L1-observable):** $lag_* = L2 \text{ slot number of the current L1 timestamp} - \text{slot number of the reference head}$; taking the tip of the best provisional candidate as the reference head yields lag_prov , and taking the window-final head yields lag_final (the §6.2 split)^[100]. Terminological convention ^[101]: in the remainder of this section (window opening / strikes /

late fees / threshold-riding determinations), bare $\text{lag} / \Delta_{\text{lag}}$ are always to be read as $\text{lag}_{\text{final}} / \Delta_{\text{lag,final}}$; $\text{lag}_{\text{prov}} / \Delta_{\text{lag,prov}}$ carry only the service-level observation of §6.2 and enter no penalty rule; the Δ_{lag} appearing in historical calibration notes (r10-2 and others) refers to the single threshold of that time, whose value is now inherited by $\Delta_{\text{lag,prov}}$. Normative definition ^[102]: $\text{L2_slot}(t) = \text{floor}((t - \text{GENESIS_L2}) / 1 \text{ second})$, where t is the timestamp of the L1 execution block in which the determination is made (constrained by L1 consensus, a single proposer has only second-scale jitter influence); it references no L2 or P2P data. Both quantities are pure functions of L1 state, and any contract call can compute them.

- The causal chain of the fallback window and the strikes (the precise definition of each link is given in the subsequent items of this section):
- **The fallback window is a state condition, not a timer:** whenever $\text{lag}_{\text{final}} > \Delta_{\text{lag,final}}$ (initially ≈ 9 epochs ≈ 57.6 minutes $= \Delta_{\text{lag,prov}} + W_{\text{settle_max}} + \text{margin}$)^[103], the fallback window is automatically in the open state — anyone may land any valid batch and be paid from the aggregator's bond at indexed cost (L1 fees + proving cost + markup) (a transplant of the fault-pays model of v15 §6.7). The window does not close because "the aggregator landed a batch" — it closes only because $\text{lag}_{\text{final}}$ falls back within $\Delta_{\text{lag,final}}$. Micro-batches are therefore ineffective: landing 1 block does not reduce the lag, and the window stays open regardless. Fallback eligibility and reward are determined by $\text{lag}_{\text{final}}$ and are snapshot-held until the settlement window closes ^[104]: once the fallback window has opened under $\text{lag}_{\text{final}} > \Delta_{\text{lag,final}}$, the eligibility and the "cost + markup" reward commitment remain valid for every candidate within this settlement window until close; the reward is paid to the closing winner and to every candidate that strictly improved the best chain (covering the full proving + inclusion cost) — an honest improver overtaken by a heavier candidate does not work for nothing, and a short provisional candidate cannot take the eligibility away.
- **Fault and incriminating evidence (mechanically decided on L1):** one "strike" = a batch from someone other than the aggregator was accepted while the fallback window was open. The elegance of this definition lies in its zero false positives: if there is simply no block available to land on P2P (all builders absent), then nobody can land anything and the aggregator is not struck for "having nothing to land"; whereas once someone does land, that very fact proves on L1 that "there was landable content and the aggregator did not land it" — the dereliction is self-evidencing by fact, with no need to observe P2P. Each strike slashes one tier of the aggregator's liveness bond. Strikes accumulate with the duration for which the window stays open and are rate-limited by G_{strike} ^[105]: the unit of a strike is changed to a sustained open-window interval — for each full G_{strike} (initially 1 epoch) that an open-window period (a contiguous interval in which $\text{lag} > \Delta_{\text{lag}}$) persists, one strike is counted if ≥ 1 non-aggregator batch was accepted within that interval. Two properties are preserved simultaneously: (i) rate limiting against spam: however many micro-batches an attacker splits its landings into within the same G_{strike} interval, only one strike is counted, and the attacker cannot accelerate time ^[106]; (ii) sustained dereliction necessarily accumulates: as long as the window stays open and fallback landings keep occurring, the strike count rises once per G_{strike} , so m_{agg} (2 strikes) is reached within ≈ 2 epochs and the aggregator is replaced — the drip-feed stalling of r13-1 falls precisely into this branch. The first strike still requires the window to have been continuously open for $\geq G_{\text{strike}}$ (the grace period for transient faults is unchanged). The

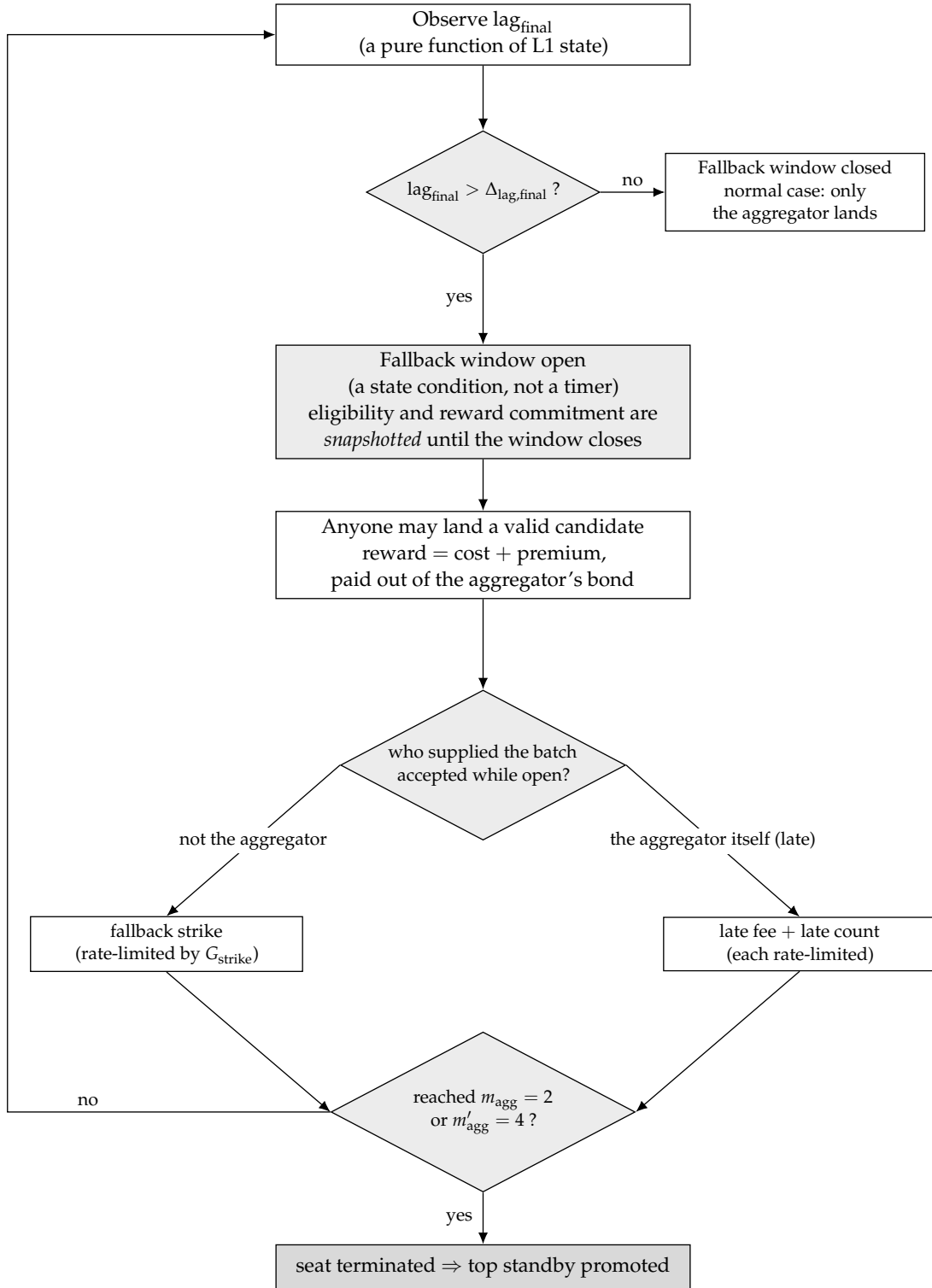


Figure 7: The causal chain of the permissionless fallback window and of seat strikes: fallback eligibility is opened by a state condition, namely $\text{lag}_{\text{final}}$ crossing its threshold as observed on L1. Who lands while the window is open decides what is recorded: a fallback strike, accrued at a rate limited by G_{strike} , or a late fee priced on the overshoot together with a separately rate-limited late count. Strikes accumulating to the threshold terminate the seat.

symmetric reverse weaponization (a malicious fallback lander drip-feeds micro-batches to invalidate the aggregator's catch-up proofs and to run up strikes on its behalf) is met as follows: the aggregator's own batches never incur a strike and it may bid arbitrarily high to get in first, and batch proofs must support incremental continuation/aggregation from an already-proved prefix (an engineering requirement, folded into §11 item 6) — once a micro-batch that advances the chain head no longer forces anyone to re-prove from scratch, the drip-feed's teeth are pulled; the residual risk is still "sustained L1-level censorship of the aggregator's landing transactions" (the most expensive censorship category, already priced in v15 §11.5), and is explicitly accepted. Bond exhaustion is itself termination: late fees and strikes draining the liveness bond through the floor (below the minimum maintenance amount, a calibration item) counts as a termination trigger, so the fees are also an independent termination path and no longer mere bleeding.

- **A late landing is itself charged — closing the "threshold-riding" strategy** ^[107]: counting strikes solely on "a fallback batch was accepted" has one hole: an online malicious aggregator can squat right at the threshold — as soon as lag just exceeds Δ_lag and the fallback window opens, it front-runs the fallback lander with its own catch-up batch, closes the window, never takes a strike, keeps finality permanently riding the Δ_lag line, and burns the fallback lander's proving cost for nothing. The fix (equally mechanically decidable on L1): any batch accepted while $lag > \Delta_lag$ — including the aggregator's own — charges the aggregator a late fee priced by the excess (deducted from the liveness bond), and the event of the aggregator's own late batch being accepted is recorded in a lighter violation counter, which upon reaching m_agg' (initially 4) likewise triggers termination. The counter is attached to "a late batch was accepted" rather than to "the window was opened" ^[108]: the window may open for reasons unrelated to the aggregator, such as all builders being absent or a proving-system failure, and in that case nobody can land at all, so any count would be unjust; whereas "a late batch was accepted" (whether the aggregator's own or a fallback lander's) itself proves on L1 that "there was landable content", so zero false positives holds. The threshold-riding front-running loop remains closed: every time the front-runner closes the window it necessarily lands a late batch of its own, and every such time is counted. Honest degradation ^[109]: "not a penny is charged during an externality-caused failure" currently holds fully only for the fallback strike; the late fee and the late counter can still hit "the honest aggregator that lands the first recovery batch after a network-wide proving-system failure is resolved" (such a recovery batch is necessarily late). Fix one: the late counter accumulates under the same G_strike rate limit as the fallback strike (for each G_strike of sustained window opening within whose interval one of the aggregator's own late batches was accepted, at most one is counted) ^[110]. Fix two: the exemption mechanism for systemic proving outages is elevated to a blocking open item (§11 item 8) — until it is fully defined and mechanically verifiable, the "zero false positives" claim is limited to the fallback strike and does not cover the late fee. The landing gas lost by a front-run fallback lander is compensated out of the late fee according to a pre-registered intent (the detailed rules are folded into §11 item 3).
- **Closing the zero-penalty reset of "self-wiping the lag + landing a worse chain"** ^[111]: charging solely on "whether lag is $> \Delta_lag$ at acceptance time" gives the aggregator a threshold-pinning loop: it can time its landing exactly at $lag = \Delta_lag$ (not $>$) and land a short chain of its own that overrides the healthy tail; acceptance then carries a zero late fee, the lag drops back after landing, and the cycle repeats every proving period — finality is suppressed indefinitely at zero penalty and the seat is never replaced. Two closures: (a)

Landing a worse chain is self-defeating (the main closure)^[112]: each worse short chain it lands is merely one candidate inside the window — anyone can land that healthy content tail with a proof into the same window, and by the total order it directly overrides the malicious candidate; the malicious candidate wastes gas + proving fees and suppresses nothing. "Zero penalty, repeatable" becomes "zero revenue, guaranteed loss", and no evidence submission or slashing adjudication is required. (b) **The lag debt is settled against the pre-acceptance state, leaving no room for integer threshold-riding**^[113]: the determination for the late fee and the late counter is changed to $\text{lag} \geq \Delta_{\text{lag}}$, and the fee is priced as $\text{late_units} = \max(0, \text{preLag} - (\Delta_{\text{lag}} - 1))$ — that is, at $\text{lag} = \Delta_{\text{lag}}$ we have $\text{late_units} = 1$ (the minimum non-zero late fee, no longer an "excess = 0 exemption point"); the m_agg' counter is changed to maintain independently, for "one of the aggregator's own late batches was accepted", a rate-limit state keyed on the L1 slot of the last count (no longer depending on the old condition "the window has been continuously open for a full G_{strike} ", which is undefined at the equality point — finding 8 pointed out that the old rate-limit definition has neither an open window nor any duration at $= \Delta_{\text{lag}}$). Timing a landing precisely at $= \Delta_{\text{lag}}$ is thereby no longer penalty-free and no longer strike-free. Rate-limit units are unified^[114]: the normative unit of G_{strike} is L2 seconds (1 epoch = 384 L2 seconds); the m_agg' rate-limit state stores "the L1 slot of the last count", and the comparison converts via $(\text{current L1 slot} - \text{last L1 slot}) \times 12 \geq G_{\text{strike_L2s}}$ — eliminating the ambiguity in which "a count of L1 slots is used directly as L2 epochs", which causes a factor-of-12 discrepancy. The two act together: the self-wiping-lag loop either lands the genuinely better tail (harmless) or is overridden inside the window at a pure loss plus a late fee, and the "Byzantine aggregator is bounded" claim of §6.3/§8 is restored (under the same fallback-feasibility condition that someone within the window is willing to land the honest tail).

- **Termination and replacement**: an accumulated m_agg fallback strikes (initially 2) or m_agg' of the aggregator's own late batches (initially 4) → the seat is terminated and the highest standby is promoted, with a promotion delay of 1 epoch (in this design the aggregator does not issue preconfirmations, so a replacement is not perceptible at the service level and the delay can be aggressive).
- **Reward anti-grinding**: the aggregator's service reward is charged by the number of slots that a landing advances, not by the number of batches — a micro-batch not only fails to suppress the fallback window, it also earns nothing.
- **Aggregator liveness is a "bound under an economic assumption", not a pure protocol bound**^[115]: strikes and m_agg' accumulate only when a batch is accepted inside the fallback window. If the aggregator silently stops landing and nobody performs a fallback landing — because the proving cost exceeds the reward, because of an L1 fee spike, or because no prover is available at all — then no strike is produced, the seat does not change hands, and finality may stall indefinitely. The bound on the "Byzantine aggregator is bounded" row of the §8 table is therefore conditional: it holds if and only if permissionless fallback is economically feasible (someone is willing and able to bear the proving and gas cost of the fallback). This is an economic/engineering assumption, not protocol-enforced liveness; stating it explicitly is an honesty clause of the same kind as §1's "honest majority is an economic assumption". The indexed-cost cap on the fallback reward and the fallback incentive under extreme congestion are listed as calibration items in §11 items 3 and 7, and the availability/cost of fallback provers is formally folded into the §1 threat model

and §11.

- **The availability precondition for fallback** ^[116]: a fallback landing requires the block bodies in hand, and before landing, block-body availability is only best-effort guaranteed by P2P (§5.5). The exact condition for "the Byzantine aggregator is bounded" is therefore: the unlanded tail, or some available fork of it, is available to at least one lander willing to perform a fallback. By the structural self-healing of §5.5, this reduces to "there exists ≥ 1 non-colluding builder in the lookahead" (which automatically routes around the segment whose bodies are withheld and produces a fork that anyone can land) — that precondition is directly implied by the §1 trust model (an honest majority of builders) with ample margin, so within the model the boundedness of a Byzantine aggregator is unconditional. If the builders and the aggregator collude as a whole (bodies kept private, nobody able to land anything), the strike mechanism indeed does not fire — which is the correct behavior of the zero-false-positive principle (the stall is then attributable to the cartel, not to the individual aggregator); by the §1 trust model that case is out-of-model, the protocol offers it no finality guarantee, and after v1.51 removed forced inclusion there is no defense in depth either — the former route "a user submits a forced entry $\rightarrow$ the deadline passes and the lag exceeds its limit $\rightarrow$ anyone advances finality with a forced-only block" no longer exists.
- **Key property**: because landing requires no authority (§0 item 3), the fallback requires no new trust. Aggregator offline: the sequencing service is entirely unaffected (builders keep producing blocks and preconfirmations keep being issued), and finality is maintained near Δ_{lag} by fallback landers. Aggregator online but maliciously stalling (Byzantine): micro-batches cannot save it — once the lag exceeds the limit the fallback opens, once a fallback landing occurs a strike is counted, and after m_{agg} strikes the aggregator is replaced; under the availability precondition of the previous item, the worst-case finality lag is pinned at Δ_{lag} + the fallback response time, and is bounded. These two cases are listed separately from the full-cartel case in the table of §8.

6.4 Stall and Gap Recovery — Ordinary Block Production Takes Over in One Hop via the Landing Head, with No Episode (v1.31, Direction B Redesign)

*This section wholly replaces the "re-anchor episode" of v1.8–v1.30 (announce / existence challenge / deposits B_{ra} , B_{ch} / freshness window / s_{base} , s_{ra} / Δ_{cont} continuation / re-pinning / state machine). The foundation of that mechanism was "pinning a block inside a freshness window of wall-clock slots", whereas proving latency (10–15 minutes) and L1 landing delay are » the slot / G_{max} (64 seconds) timing window, which makes any timing rule of that kind inherently fragile; across roughly eight rounds of review ($r23 \rightarrow r24 \rightarrow r25 \rightarrow r28$ + round 7 P1 / deposit determinism $r29$ -P2) it repeatedly produced second-order holes with the same root, and point patches only relocated them without converging. On this basis the owner decided (2026-08-25, Direction B) on a wholesale redesign: **recovery is no longer a separate sub-protocol but a direct corollary of the tier (ii) parent rule of §4.2**. This redesign has undergone six rounds of adversarial self-review by the designer (see the v1.31 entry of Appendix D).*

After a stall of hours or even days, recovery is merely "one landing + one tier (ii) block":

- **The mechanism (just one rule)**: when the gap $> G_{\text{max}}$ (consecutive absences exceed the cap), or when the aggregator withholds landing so that normal block production cannot continue on the unlanded tail — any lookahead builder of the current turn takes the current final L1 landing head as parent and produces an ordinary block under tier (ii) of

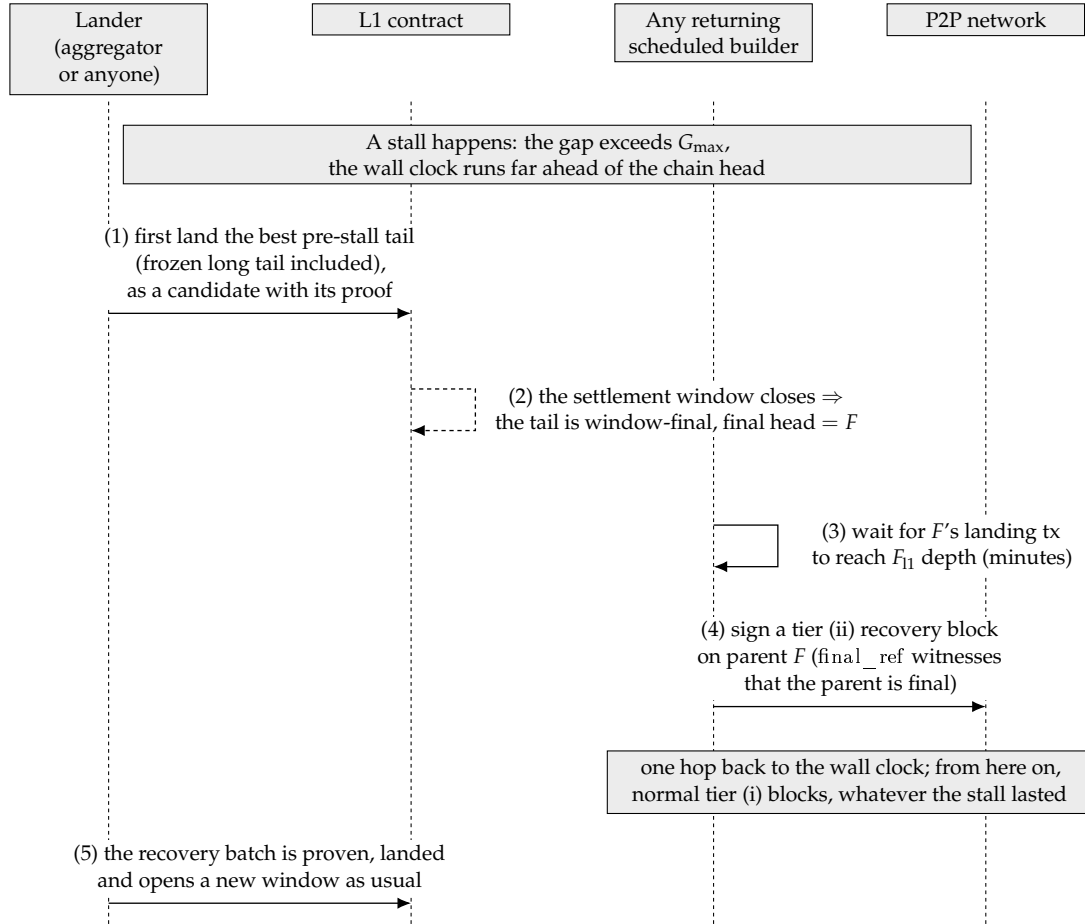


Figure 8: Recovery after a stall of hours or even days: first land the best pre-stall tail and let it close into the final head F , then let any returning builder sign a tier (ii) recovery block on parent F , which reattaches the ordering to the wall clock in a single hop. Recovery involves no separate sub-protocol.

§4.2 (a content block). That block's slot = the wall-clock slot of its turn, and the parent distance is arbitrary (tier (ii) has no cap). L2 sequencing thereby recovers: the landing head is reconnected to near the wall clock, and subsequent builders produce blocks on top of it normally (tier (i)). The production/landing of the recovery block itself has no announcement, deposit, freshness window, or state machine — it is merely an ordinary candidate in the §5.6 settlement window, competing with the other candidates by the total order. The ordering rule of landing the best chain first is given in the next item: if a frozen long tail exists from before the stall, land it first and recover from its tip, so that its preconfirmations are not lost.

- **Land the best (including frozen) chain first, then recover from its final tip — do not discard the tail** ^[117]: v1.35 once stated that "the frozen long tail from before the stall yields to a single recovery block", which is wrong — it would instruct honest landers to discard a landable healthy long tail. The correct order is: (1) by the total order of §5.2 the pre-stall long tail is still the best chain ("cannot be extended over P2P" $\neq$ "should be discarded"), so a lander lands it first (honoring all of its preconfirmations; the $G_{\max}$ gap merely prevents P2P from extending it further and does not prevent proving and posting it); (2) once it has landed and reached F_{ll} it becomes the new final tip, and the recovery block is produced with that new tip as parent under tier (ii) of §4.2, reconnecting to the wall clock in one hop from there. The "recovery vs. fork-choice deadlock" thus disappears (not by discarding the tail, but by landing the tail first), and the preconfirmations of the healthy long tail are not sacrificed. Only when there genuinely is no longer/better chain (a true total stop, with not even a frozen tail) is the recovery block itself the best chain and landed directly. Skipping a better tail in order to land a recovery block is self-defeating: anyone can land the better tail into the same §5.6 window and it wins ^[118].
- **"One hop" means re-rooting; sustained progress still requires the normal liveness floor** ^[119]: the recovery block R (tier (ii)) re-roots sequencing onto the landing head in one hop; but R is itself an unlanded P2P block, and to genuinely reconnect to the wall clock a subsequent builder must extend it within $G_{\max}$ (tier (i), exactly as for every normal block) — proving and landing R takes 10–15 minutes, and during that time sequencing continues over P2P on top of R (the normal "preconfirmation in seconds, finality in minutes" rhythm, not a new problem). "A recovery block never expires" refers precisely to the landing validity of R (tier (ii) has no gap cap, so R can land whenever it is proved), not to R being extendable without maintenance. The exact condition for recovery liveness is therefore not "one builder produces one block" but "lookahead builders keep returning at a density of ≥ 1 per $G_{\max}$ window" — that is, the $G_{\max}$ liveness floor that normal block production already requires. If only a single builder returns and its lookahead turns are sparser than once per $G_{\max}$: nobody extends R within $G_{\max} \rightarrow$ R itself freezes (§5.2) $\rightarrow$ the next builder produces another R' on the same old landing head (R is unlanded and not final, so it cannot be referenced by tier (ii)). This retry loop is benign and loses no ground (the landing head has not moved, and each attempt re-roots from the same point); once participation returns above the $G_{\max}$ floor, some R is extended, lands, and recovery completes. This floor is identical to the normal block-production floor and is not a new requirement specific to recovery; a genuine long-term outage of all participants is a matter for social recovery and is out-of-model. (After v1.51 removed forced inclusion there is no longer a parallel "any-key forced-only block" route here — the $G_{\max}$ density floor can only be met by scheduled builders.)

- **Why a stall of any duration can be recovered in one hop (this is the fundamental improvement over the old design):** tier (ii) imposes no cap in the gap dimension, so a recovery block does not expire because its parent distance is too large (there is no "expiry" in the gap dimension). Whether the stall lasted 2 hours or 2 days, recovery is always "produce one block with the landing head as parent → one round of proving → landing". The whole family of problems from the old design in which "proving latency blows through the gap freshness window" (r28 / round 7 P1) therefore disappears. Note the distinction of dimensions: what is stated here is that the gap does not expire; the other dimension, anchor freshness, still applies (see the next item + §7) — "never expires" refers precisely to the former, not the latter, and the two are not in conflict.
- **The precise boundary of "one hop" — the transient F_{l1} wait** ^[120]: the parent of a tier (ii) block must be a final landing head, whereas §5.6 lets only candidates that extend the current canonical landing tip land. The two fail to coincide in exactly one transient: the stall begins shortly after a landing and the current tip has not yet reached F_{l1} . In that case the recovery block can only wait for the current tip to become final (an extra $\leq F_{l1}$, on the order of minutes) — building on an older final head would fork off the newer tip and §5.6 would refuse the landing. The precise recovery bound is therefore $\max(0, F_{l1} + D_{\text{anchor}} - \text{the time the current tip has already existed}) + \text{one round of proving and landing}$ ^[121]: a steady-state disaster stall (in which the last landing is long past that point) degenerates to a pure one hop, while the worst transient, a stall starting immediately after a landing, adds a wait of $\leq F_{l1} + D_{\text{anchor}}$ (≈ 19.2 minutes, on the order of minutes). In either case this is independent of the total stall duration (it does not grow with 2 hours or 2 days).
- **The constraint anchor freshness places on the recovery block — "never expires" needs a qualifier** ^[122]: what tier (ii) eliminates is gap expiry (no gap cap, so R is valid however far its parent is), but anchor freshness (§7, $D_{\text{anchor_max}}$) still applies: R pins its anchor at signing time and must land within landing L1 block height — anchor $\leq D_{\text{anchor_max}}$, failing which R's anchor is too stale, does not pass §7, and must be re-signed and re-proved against a fresh anchor. The precise statement is therefore: "a recovery block never expires because of the gap" holds, but it must be proved and landed within the $D_{\text{anchor_max}}$ freshness window of its anchor. The key difference from the old re-anchor s_{ra} : the old s_{ra} freshness window was structurally pinned to G_{max} (a 64-second wall-clock window), which proving latency (10–15 minutes) necessarily blows through, whereas $D_{\text{anchor_max}}$ is a freely settable parameter, and its setter invariant (the full formula is in §11) ^[123] guarantees that a recovery block does not expire under normal L1 liveness — the old design lost because its window was structurally too small, not because of the mechanism itself. Since r41, the stale-anchor tail deadlock is resolved by the window geometry of §7 ($D_{\text{anchor_max}} \geq D_{\text{anchor}} + \Delta_{\text{lag,final}} + P_{\text{prove,max}} + T_{\text{include,max}} + \text{margin}$) ^[124]. If L1 censors R's landing transaction continuously for longer than $D_{\text{anchor_max}}$ and R has not been committed, R is rebuilt and re-proved with a fresh anchor (with no loss of sequencing) — that case is a degradation of L1 liveness and does not occur under the §1 assumption that L1 is safe and live.
- **Determinism and benign stranding (designer self-review round 1):** the parent of a tier (ii) block = a final landing block (having reached the F_{l1} finality depth, so the predicate is stable under shallow L1 reorgs; linked to the L1-reorg item of §11); determinism at landing time is backstopped by the §5.6 landing rule — a batch may only extend the current canon-

ical landing tip. If a recovery block built on an old tip finds that the tip has already been advanced $\rightarrow$ benign stranding, and it need only be rebuilt (there is no episode or deposit that can go wrong). And the tip being advanced $\Leftrightarrow$ a new landing occurred $\Leftrightarrow$ the chain has not truly stopped; during a genuine stall the tip does not move and the recovery block necessarily lands.

- **L1-sync and the forced backlog catch up under per-block caps (designer self-review rounds 4/5)**^[125]: "one-hop recovery" refers precisely to L2 sequencing recovering in one hop; the L1 $\rightarrow$ L2 messages and the forced-entry backlog accumulated during a long stall catches up under a per-block cap (v1.51 dropped the forced-entry side): the L1 $\rightarrow$ L2 message-processing cursor $m_consumed$ advances by $\leq C_anchor$ messages per block (§7 — C_anchor is bound to the message cursor; the anchor reference itself has no per-block advancement cap and may jump to a fresh height in one hop; the earlier text of this line, "the anchor's L1 advancement is $\leq C_anchor$ per block", was a leftover in conflict with §7 and has been corrected). The full accounting is therefore: sequencing = 1 hop; L1-sync and the forced backlog = backlog / per-block cap blocks (after recovery, at 1 block per second, this is usually cleared in seconds to minutes).
- **The exact condition for recovery liveness (designer self-review round 2)**^[126]: a tier (ii) content recovery block still requires $signer = lookahead(slot)$, so the exact precondition for discretionary recovery = lookahead builders keep returning at a density of ≥ 1 per G_max window (= the G_max liveness floor of normal block production, not something specific to recovery; a single sparsely returning builder can only re-root and cannot sustain progress, see the "'One hop' means re-rooting" item below). After v1.51 removed forced inclusion **there is no censorship floor left**: the former backstop by any-key forced-only blocks no longer holds, and recovery depends entirely on scheduled builders returning. An outage of all participants (for example a network-wide client bug) is a matter for social recovery and is out-of-model, as it is for any chain. A targeted DoS against near-future lookahead builders merely postpones recovery by $\approx 1/p$ slots, which is of the same order as the normal residual risk of §3.2 and is not a new hole.
- **Deletion list (relative to v1.30)**: the entire §6.4 episode — announcement/ B_ra , existence challenge/ B_ch /deposit settlement, parent pinning/re-pinning, freshness floor/ s_base/s_ra , Δ_cont continuation authorization, and the episode state machine — is deleted in full. The review to-dos that disappear with it: r28 (freshness cannot keep up with landing), round 7 P1 (Δ_cont shorter than proving), and r29-P2 (three contradictions in deposit settlement / no unique winner with multiple H_ch / a second settlement after Executed) — these mechanisms no longer exist, so the problems no longer apply. §11 is amended accordingly.
- **The recovery accounting is given in §8** (one hop + one round of proving, including a single hop for a long stall; L1-sync catches up according to the backlog). The residual risk is given in §5.4 (single-block orphaning via the landing head $\leq$ the unlanded tail $\leq \Delta_lag$, which takes effect only under a colluding lander and has the same bound as the existing worst case). The lander's tail-selection strategy is given in §5.2.

6.5 Aggregator Seat and Economics

- The seat is produced by a perpetual auction (inheriting the v15 §4 skeleton: bonded bidding, the standby queue, the q delayed transition, bonds, and re-auction upon T_max

expiry — all carried over, with the semantics of q and $T_{\max}$ unchanged).

- **The fee source must not form a circular dependency with liveness** ^[127]: if the service fee comes from the protocol fee stream, then when the chain is idle the fee stream dries up and creates the loop "underpayment $\rightarrow$ rationally going offline $\rightarrow$ dependence on the fallback"; the fallback reward is therefore paid solely from the aggregator's bond (already the case, §6.3), and the design of the service-fee source (§11 item 4) must guarantee that the aggregator still breaks even when the fee stream is insufficient (for example, a minimum service fee backstopped by the treasury). The combination of winning the seat cheaply and then slacking off is already closed off by the strike mechanism of §6.3 (slacking off = strikes = replacement), so all that needs to be handled here is the "honest but underpaid" case.
- **The direction of payment is reversed, stated explicitly**: in v15 the seat paid the treasury because it held an exclusive sequencing right; in this design the seat is a service post (batching + proving + L1 gas), and what is auctioned is the lowest service fee rate (a reverse auction, with the service fee paid out of protocol fee revenue), with the bond accounted for separately. The bidding dimensions are the pair (service fee rate, bond): the lower rate wins, and at equal rates the higher bond wins. Refinement is a calibration item of §11.
- MEV and priority fees go to the builder of each slot (coinbase). The aggregator does not touch sequencing and therefore structurally has no MEV position — this is intentional.
- **Base-fee share — a positive incentive to land a longer chain** ^[128]: a fixed proportion φ_{land} (a §11 calibration item) of the sum of the base fees of all blocks in the candidate batch serves as the landing reward and is credited, in the atomic commit at the §5.6 window close, to the beneficiary address of the closing winner (the beneficiary is part of the proof's public inputs, per the anti-plagiarism clause of §5.6, so mempool plagiarism cannot change where it points); the remaining base fee follows the current protocol's destination. Three immediate corollaries: (i) the base of the share grows monotonically with chain length — landing a longer chain yields a larger share, which closes the loop with §5.2's "landing the longest visible chain is the rational choice"; (ii) a provisional lander superseded by a heavier candidate receives nothing (winner takes all) — the opportunity cost of landing a short chain = the entire share + the gas and proving fees already paid; (iii) the share is accounted atomically at close together with the winner's final tuple, and requires no extra transaction or after-the-fact claim. The payment point is the close, not the provisional landing ^[129]: a provisional candidate can be superseded within the window, so paying on landing would require a clawback from the superseded lander, whereas paying at close is equivalent to "the last step of the landing procedure settles the accounts automatically" and needs no clawback — the cash flow of users and landers differs only by one W_{settle} . The existing §6.3 rule that "the reward is charged by the number of slots that a landing advances" (reward anti-grinding) continues to apply to the service-fee component; the base-fee share is a content-volume incentive layered on top of it, and the two point in the same anti-grinding direction (a micro-batch neither suppresses the fallback window nor has much of a share base).

7 Anchoring and the bridge (L1 $\rightarrow$ L2)

- The current per-block anchoring pattern is carried over: the first transaction of every block is the anchor transaction (anchor tx), which references an L1 block at depth $\geq D_{\text{anchor}}$

(32 L1 slot), is non-decreasing relative to the previous anchor, and whose freshness does not exceed the cap (the numerical values carry over the shape of the v15 §6.6 constraints).

- **The anchor may not be later than this block's slot time — the L1→L2 causal-ordering invariant** ^[130]: in addition to being non-decreasing and fresh, it must also hold that the anchor's L1 timestamp $\leq$ the L2 timestamp corresponding to this block's slot (equivalently, $L2_slot(anchor.L1_timestamp) \leq slot$), and this applies to all blocks alike. Why it is necessary: §6.1 rule 3 only blocks "future slots", and §7 only blocks "an anchor that is too old", and neither blocks "an old slot paired with a new anchor" — a builder could hold back an old slot, then pick after the fact a fresh anchor that has just arrived and that already contains some new L1→L2 message, and sign a tier (i) continuation that consumes that message into an L2 block whose slot timestamp is earlier than the L1 block the message originated in; a Byzantine aggregator lands it ahead of honest recovery → L1→L2 causal ordering is broken, and the bridge/message deadline semantics based on block.timestamp are distorted. With this invariant added, a block can never reference L1 state from after its own slot time. It does not affect recovery: a recovery block references an L1 block at depth $\geq D_anchor$ (≈ 6.4 minutes ago), whose timestamp is always $\leq$ the current wall-clock slot time, so an honest block satisfies it naturally; this invariant rejects only the malformed combination "old slot + new anchor". The L1 entry point and the circuit both prove it ($anchor.L1_timestamp$ is a pure function of the referenced L1 block header).
- The time geometry of the anchor's age along the worst-case path (the origin of the D_anchor_max setter invariant, Appendix C P9a):

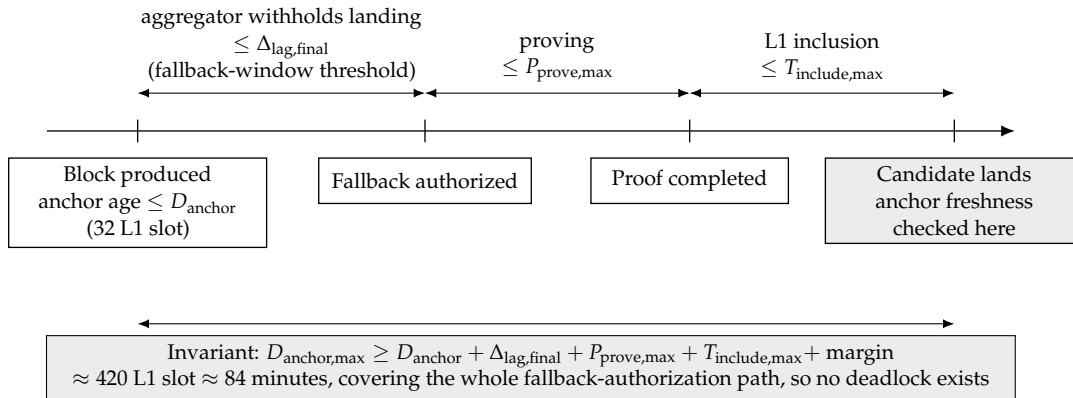


Figure 9: The time geometry of anchor age along the worst-case path: the freshness window $D_{anchor,max}$ must cover the whole route from block production, through the aggregator's withholding until the threshold authorizes fallback, and on to proving and L1 inclusion. This relation between parameters is exactly the solution to the stale-anchor tail deadlock.

- **The solution to the stale-anchor tail deadlock — D_anchor_max covers the fallback authorization period** ^[131]: when a malicious aggregator withholds landing for a long time, the anchor of an early block of an honest tail will, "at landing time", exceed D_anchor_max → the tail can then neither be landed (its first block is past the deadline) nor be skipped (parent_hash runs through it). The solution is parameter geometry, not a new mechanism: the setter invariant $D_anchor_max \geq D_anchor + \Delta_{lag,final} + P_{prove,max} + T_{include,max} + margin$ ^[132] — that is, the freshness window must cover the entire course "from block production, through lag_final crossing the threshold and opening the fallback,

to the fallback proving and landing it". Consequently, any tail that the fallback is authorized to land (produced while $\text{lag_final} \leq \Delta_{\text{lag,final}}$ and landed by the fallback after the threshold is crossed) necessarily still has a fresh anchor, and no deadlock exists. A rough calculation with the initial parameters: $32 + 288 + 75 + 10 + \text{margin} \approx 420$ L1 slot (≈ 84 minutes)^[133], which is still a deployable value. For situations beyond that window (the fallback too is persistently absent for $> D_{\text{anchor_max}}$), the stale prefix can no longer be landed and the chain is rebuilt from the window-final head — the same conditional residual risk as §6.3 fallback feasibility, stated honestly. The causal-ordering invariant (the previous item) is unaffected.

- Because there is a block every second (under normal conditions), the intake cadence of L1→L2 messages is on the order of seconds — better than the fallback path of the forced-only cadence in v15. During a gap, messages are consumed by the next honest block; after v1.51 removed forced inclusion, inbound messages have no alternative entry route under a full cartel and can only wait for scheduled builders to resume producing blocks (§8).
- **The L1→L2 message-processing cap C_{anchor} — bound to the [processing cursor], not to the anchor reference**^[134]: distinguish two independent quantities:
 - **The anchor reference** (which L1 block the block references as its freshness baseline): still non-decreasing + required to be fresh (landing L1 block height — anchor $\leq D_{\text{anchor_max}}$). A recovery block references a current, fresh anchor — even if the anchor of its parent (a landed head from days ago) is very old, this one large monotone jump is permitted (it merely references a recent L1 block and satisfies freshness), and the anchor itself carries no per-block advance cap.
 - **The L1→L2 message-processing cursor m_{consumed}** (a global message sequence number, not an L1 height)^[135]: the unique canonical algorithm = [the uniform maximum prefix]^[136]: L1→L2 messages form a public FIFO queue (the contract maintains a queue root + tail cursor, kept separate from m_{consumed}); every block must consume the longest prefix of that FIFO that satisfies both "entry count $\leq C_{\text{anchor}}$ " and "cumulative declared gas $\leq$ the L1→L2 message gas share" (restricted to those that have arrived within the height referenced by this block's anchor and are unconsumed; taking that longest prefix is legal, and only taking less than it is illegal — as in §5.1 rule 5, with no second "exactly some entry count" criterion). A recovery block references a fresh anchor (spanning days of L1), each block digests this prefix, and the remainder is continued by subsequent blocks (m_{consumed} advances monotonically). The backlog catch-up bound is correspondingly two-constraint: backlog drain time = $\max(\text{ceil}(\text{entry count} / C_{\text{anchor}}), \text{ceil}(\text{cumulative gas} / \text{the } G_{\text{llmsg}} \text{ share}))$ blocks (the bounds of §6.4/§8 are to be read this way)^[137]. Without this lower bound, m_{consumed} is monotone yet may stagnate: a builder that keeps processing 0 messages while still using a fresh anchor could censor L1→L2 messages indefinitely, invalidating the backlog / C_{anchor} catch-up bound of §6.4/§8^[138]. With the lower bound added, the catch-up bound is circuit-enforced and redeemable. Consumption is not execution — an inbound message that is illegal against the preceding state is consumed-and-discarded under the same consume-and-discard rule as §7. **The real anti-censorship bound is $D_{\text{anchor_max}}$, not C_{anchor} alone**^[139]: the maximum prefix only forces "consumption of those that have arrived within the height referenced by this block's anchor", so a builder that pins its anchor at the oldest height freshness permits (landed head — $D_{\text{anchor_max}}$) can legally consume no newer message at all. But this is not indefinite censorship: the

anchor must satisfy the freshness lower bound $\text{anchor} \geq \text{landing L1 block height} - D_{\text{anchor_max}}$ and be non-decreasing, so as the landed head advances, that lower bound rises, and a message that arrived at height h is forced into "within the anchor-referenced height" once $\text{landed head} - D_{\text{anchor_max}} > h \rightarrow$ it must then be consumed. Hence the worst-case inclusion delay of an L1→L2 message $\approx D_{\text{anchor_max}}$ (freshness forces the anchor frontier forward) + $\text{backlog} / C_{\text{anchor}}$ (the processing rate once it has entered the to-be-consumed set); the two parameters jointly bound it, not C_{anchor} alone.^[140] Why it is split this way^[141]: the §4.2 tier (ii) recovery block has to reconnect to the wall clock in one jump from a landed head days old. If C_{anchor} were imposed on the anchor reference (the error in the original v1.31 text), then the anchor of a recovery block could advance only C_{anchor} from its stale parent anchor, would still be stale, and would fail the freshness check — failing precisely on the long stall it is meant to support^[142]. Once it is rebound to the message-processing cursor: the anchor reference is fresh (satisfying §7) and L2 ordering recovers in one jump; while a huge volume of L1→L2 messages is processed block by block at C_{anchor} per block, so a full sync takes $\text{message backlog} / C_{\text{anchor}}$ blocks (after recovery, 1 block/second, which is fast). This is the same technique and the same shape as a fresh reference plus a rate-limited cursor. C_{anchor} is a §11 calibration item.

- **A shared block-level gas budget + a per-message gas cap — sealing off the combined-gas chain stall**^[143]: the original risk was that the "forced prefix" and "L1→L2 message consumption", two mandatory obligations each with its own quantity, could together burst the block gas cap and leave no legal block even with everyone honest. With the forced prefix gone that combination disappears, but **one side alone can still reproduce the same shape of deadlock**: if the head of m_{consumed} is an inbound message whose execution cost exceeds "the block cap minus the fixed anchor overhead", rule 4 requires it to be consumed and it does not fit, so no legal state transition exists and the chain dies permanently. The first two clauses below are therefore retained, and clauses 3/4 restated for two parties (fully deterministic, verifiable by both the circuit and L1):
 1. **A per-item admission gas cap**: inbound L1→L2 messages get a per-item gas cap at enqueue time, and an over-cap item is refused admission — this is the load-bearing clause against one-sided deadlock, and it is not relaxed just because the forced queue is gone;
 2. **A shared block-level budget invariant**: the fixed anchor overhead G_{anchor} + the L1→L2 message gas share $\leq \text{block_gas_limit}$, as a **setter invariant on the consensus constants**; the message side's "guaranteed capacity after deducting the anchor overhead" always fits into the block;
 3. **Deterministic priority and overflow**: the enforced execution order within a block = $\text{anchor} \rightarrow \text{L1} \rightarrow \text{L2 message prefix}$; following m_{consumed} , the block "consumes up to its guaranteed capacity, and the remainder deterministically overflows into subsequent blocks" — the cursor is advanced only by legal blocks, so a backlog buys delay and cannot buy a chain stall;
 4. **A watermark for lowering parameters**: when C_{anchor} or the message gas share is lowered, the new share must be $\geq$ the maximum gas among the unconsumed entries of the message queue, guaranteeing that in any reachable state there is no item that is "already enqueued yet unable to fit within the guaranteed capacity". The three separately rate-limited obligations are thereby coordinated by one shared budget, and can

no longer be combined into a "no legal block" deadlock. The numerical gas shares are folded into §11.

8 What happens when each role goes offline (liveness accounting, in a single table)

Who goes offline	What the user sees	Recovery mechanism	Recovery time
A single builder	Its slot becomes a gap; every other slot proceeds as usual	No recovery is needed; the next slot continues naturally	~1 second
Consecutive absences $\leq G_{\max} - 1$ (63 slots, so the next block's parent distance is $\leq G_{\max}$)	No new blocks for a few seconds	The next builder resumes under §4.2 tier (i); an absentee that supplies its signature after the fact can also bridge the interval	~number of offline builders $\times$ 1 second
Consecutive absences $\geq G_{\max}$ slots, extending to hours or days (catastrophic stall)	No new blocks for the duration of the stall	Land the best chain first, then recover from its final tip (§5.2/§6.4)^[144]: if a frozen long tail existed before the stall, the lander lands it first (thereby honoring its preconfirmations), and once it has reached F_{l1} the recovery block takes its tip as parent and is produced under §4.2 tier (ii); where there is demonstrably no better chain, the recovery block is itself the best chain and lands directly. Landing a worse candidate is superseded inside the §5.6 win-	Ordering $\approx$ one round of proving + landing ($\approx 10\text{--}15$ minutes, independent of the length of the stall) + the W_{settle} window close^[146] + in the worst case an additional wait of $\leq F_{l1} + D_{\text{anchor}}$ when the stall begins immediately after a landing^[147]; where a frozen long tail exists, this includes one round of proving to "land the tail first"; the L1-sync backlog catches up at C_{anchor} per block; sustained progress requires scheduled builders to keep returning at a window density of $\geq 1/G_{\max}$ (the normal-mode $G_{\max}$ liveness floor)^[148] — after v1.51 removed forced inclusion there is no forced-only floor to serve as an alternative

By comparison with v15: there, "one seat goes offline" = a 20–26 minute service interruption (when a standby exists); here the same class of event costs 1 second (a builder) or 0 seconds (an aggregator, which affects only the cadence of finality). This is the reason this design exists.

9 Master table of slashing and bonds

Fault	How it is adjudicated	Consequence
Builder equivocation (two block headers for the same slot)	A challenger submits the two signed block headers and the L1 contract verifies the signatures directly	Slash L_{eq} ($\geq 80\%$ burned, the remainder paid to the submitter) + removal from the registry
Builder skips the previous block / is absent	Not a fault (producing a block is an opportunity, not an obligation)	No slashing; a skip is profitable for the skipper (fee/MEV transfer) ^[151] . Orphaning depth follows the three tiers of §5.4: an adjacent skip affects a single slot; under tier (i) a single-block deep skip orphans at most $G_{max} - 1$ unlanded blocks, and a minority coalition relaying a sparse branch can reach the entire unlanded tail $\approx \Delta_{lag}$ ^[152] ; under tier (ii) a single malicious builder can, with one block built on a final landed head, orphan at most the entire unlanded tail $\approx \Delta_{lag}$ ^[153] . The worst-case depth is the same in all three tiers, namely $\leq$ the actual length of the unlanded tail ($\approx \Delta_{lag}$ when the fallback lands on schedule, conditional) ^[154] ; in every tier the attacker must beat any heavier honest candidate inside the §5.6 window before it can become final, the signature is on record for public attribution, and the frequency is constrained by the honest-majority assumption; an absentee forfeits the fee income of its own slot
Builder signs a header but withholds the body (withhold-body, §5.5)	Not adjudicable ("it has the body and is not releasing it" cannot be proven)	No slashing; the block is certain to be skipped and the withholder gains nothing at all; the receipt is itself public evidence, and the response is reputational
Aggregator dereliction (a successful landing by someone else while the fallback window is open = a strike, §6.3)	Mechanically adjudicated on L1 (the lag state plus the fact that a fallback batch was accepted)	Each strike is penalized at the liveness tier; m_{agg} accumulated strikes $\rightarrow$ termination + promotion of the standby
Aggregator takes a prefix / lands a worse chain (truncating the suffix)	Not a slashable fault ^[155]	No slashing; landing a worse chain is self-defeating : inside the §5.6 settlement window anyone can land a heavier proven candidate and supersede it, so the malicious party has wasted its gas and proving fees; and if only a prefix is taken, the suffix can still enter a subsequent window
Anyone submits an invalid batch	The proof fails verification and the transaction reverts	Wasted gas, no protocol consequence

The structural property of this table, in the honest formulation adopted after review r1-11: every slashable fault is adjudicated either by direct signature verification on L1 (equivocation) or by a mechanical judgment over the L1 clock and L1 facts (landing strikes) — there is no class of slashing that requires an L2 evidence chain plus an arbitration period (the watchtower-dependent *slashing* of the v15 §8b kind does not exist here, because "preconfirmation" and "block" are one and the same signed object, so equivocation is simply signing two headers for

one slot and is directly verifiable). There do exist two classes of misbehavior that are not slashable and are only deterred (skipping and body withholding, §5.4/§5.5) — this design does not eliminate them; the precise statement ^[156]: withholding a body yields the actor no benefit, whereas skipping is profitable (the fee/MEV transfer of a single slot), and its frequency is constrained by the honest-majority assumption of §1 — both leave public signed evidence for attribution.

10 Detailed comparison with v15 (for readers who have read v15)

- **What became of v15's nine invariants:** I1 (derivation is a pure function) is inherited and strengthened (timestamps and the lookahead are all pure functions); the spirit of I2 (a single adjudication / computed-state certificate) is inherited by the landing-timeout adjudication, but there is no longer an epoch adjudication object; I3 (the epoch always advances) is replaced by "the L1 chain head can always be advanced by anyone (fallback landing)" — after v1.51 removed forced-only blocks this replacement holds only where there is landable content; under a full cartel there is no route anyone can advance; I6 (forced content always flows) is **no longer inherited** — v1.51 removed forced inclusion, so this invariant has no counterpart here; I7 (sealing is sender-agnostic) is generalized into "landing carries no authority"; I8 (pre-committed beneficiary) is inherited (the recipients of the fallback reward and of the slashing bounty are made public inputs of the proof); I9 (safety slashing is denominated in ETH) is inherited by L_{eq} .
- **v15's review legacy is not voided:** the auction skeleton (§4), the at-fault-pays model (§6.7), the anchor freshness constraint (§6.6), the burn-dominant split of slashed funds (§8) and the discipline of invariant checks in parameter setters are all carried over directly. The parts of v15 that revolved around "one irreversible adjudication per epoch" (EBC, AC, default derivation, anarchy mode, cancellation cascade) have no counterpart in this design, so the corresponding historical review findings do not apply.
- **Response to the four criticisms in the assessment document:** the single point of failure is merely relocated → resolved (landing carries no authority, plus lag-based fallback and strikes, which are bounded for both the offline and the Byzantine aggregator — §6.3); builder faults are all of the watchtower-dependent class → the slashable class is resolved (equivocation is verified directly on L1), but two classes of non-slashable misbehavior (skipping, body withholding) remain as deterrence-grade residual risk and are stated explicitly (§5.4/§5.5) — this is "downgraded and explicitly priced", not "eliminated"; fair exchange has become load-bearing → partially resolved (intermediate deletion is eradicated by the signature chain; the residual risk is as above); the epoch mechanism needs a rewrite → conceded, and this document is that rewrite, whose result is shorter than v15.

11 Open items

1. **Builder-set size and admission rules:** N_{max} , the weight cap w_{max} on the bond-weighted sampling, the rules for excess competition, and liveness requirements (whether prolonged absence should reduce weight).
2. **Pricing of L_{eq} :** a method for estimating the worst-case extractable value of a single slot, and whether governance should adjust it as market conditions change.

3. **Landing parameters:** $\Delta_{\text{lag,prov}}/\Delta_{\text{lag,final}}$ ^[157], m_{agg} , δ_{slash} , and the exponential cost cap on the fallback reward; **recovery parameters (after the v1.31 direction-B redesign):** F_{l1} (the L1 finality depth of the final landed block in tier (ii)), C_{anchor} (the per-block cap on the number of entries consumed by the §7 message-processing cursor m_{consumed} — not a cap on anchor advancement, since anchor references have no per-block cap) ^[158], Δ_{prop} (the propagation settling amount of the §5.2 landing depth, normally $\ll \Delta_{\text{lag}}$) and $D_{\text{anchor_max}}$ (the anchor freshness cap) ^[159]. $D_{\text{anchor_max}}$ **setter invariant** ^[160]: $D_{\text{anchor_max}} \geq D_{\text{anchor}} + \Delta_{\text{lag,final}}$ (converted to L1 slots) + $P_{\text{prove,max}} + T_{\text{include,max}} + \text{queue/clock margin}$ ^[161] (r42 adds the Δ_{lag} term and requires every addend to be converted to L1 slots first — independent review round 5, high-severity finding 3: this line previously omitted Δ_{lag} , so a legal parameter set such as 130 would make an honest tail that is withheld from landing until the fallback window opens expire with certainty; the contract setter implements only this single strongest invariant. $D_{\text{anchor}}=32$ L1 slot; $P_{\text{prove,max}}$ =the worst-case proving latency; $T_{\text{include,max}}$ =the bounded-inclusion bound of §1. The other setter relation ^[162]: $T_{\text{include,max}} < \min(W_{\text{settle}} - P_{\text{prove,max}} - \text{margin}, D_{\text{anchor_max}} - D_{\text{anchor}} - \Delta_{\text{lag}} - P_{\text{prove,max}})$) — v1.38 once omitted the term expressing that an anchor is already at least D_{anchor} old at signing time; writing merely " $\geq$ proving + landing" yields a parameter set that fails permanently even with no attacker present (the reviewer's counterexample: 70 configured, 102 actually required). Otherwise the recovery block (and any block) would have its anchor expire inside the proving-plus-landing window and would have to be re-signed and re-proven (§6.4, anchor freshness clause); this is the precondition for recovery "not expiring because of a gap", and it replaces the 64-second structural window of the old s_{ra} with a free parameter that can be set large enough. Stated honestly: a finite $D_{\text{anchor_max}}$ is still a "proving + inclusion delay vs. freshness window" tradeoff, only with a window that can be configured large enough, which is why C1 is explicitly conditional on the bounded-inclusion assumption "temporary L1 congestion or censorship $\leq D_{\text{anchor_max}}$ " and claims nothing unconditional against censorship of arbitrary length. **Settlement-window parameters** ^[163]: W_{settle} (the length of the settlement window, counted in L1 blocks; setter invariants: the lower bound $W_{\text{settle}} \geq P_{\text{prove,max}} + T_{\text{include,max}} + \text{margin}$, and the consensus upper bound $W_{\text{settle_max}}$ (which includes a margin for wall-clock conversion under missed L1 slots — the upper bound pins down the revocable exposure $\Delta_{\text{lag}} + W_{\text{settle}}$ (wall clock) + fallback response of §6.2) ^[164], with an initial suggestion of ≈ 100 L1 slot ≈ 20 minutes); the L1-reorg rollback of window state is folded into item 9. The semantics of final_ref (a §4.2 tier-(ii) block-header field, attesting that the parent was already $F_{\text{l1-final}}$ at signing time, already part of the block-header tuple and of §5.1). Δ_{prop} is demoted to an off-chain strategy reference for the completeness of a lander's view (§5.2, not an on-chain attribution parameter). (The $W_{\text{chal}}/L_{\text{land}}/L_{\text{chal}}/L_{\text{commit}}/\delta_{\text{land}}$ of v1.39/v1.40 were removed together with the challenge and commitment layers and are no longer parameters; $C_{\text{force}}/C_{\text{bridge}}/F_{\text{delay}}/H_{\text{force}}$ left with the removal of forced inclusion in v1.51.) Shared gas shares (§7): G_{anchor} and the L1→L2 message gas share, subject to sum of the two $\leq \text{block_gas_limit} + \text{a per-entry admission cap} + \text{a watermark}$ ^[165]. (The $\Delta_{\text{ra}}/\Delta_{\text{ra_ext}}/B_{\text{ra}}/B_{\text{ch}}/\Delta_{\text{cont}}$ of the old re-anchor episode were removed together with the episode and are no longer parameters.) The behavior of the strike mechanism under extreme congestion (an L1 fee spike that leaves nobody willing to perform the fallback). **The concrete bounds given around these parameters in §7 and §8** (backlog/ C_{anchor} , **recovery time**, the $D_{\text{anchor_max}}$

anti-censorship bound and so on) are all [provisional values] until the parameters have been calibrated ^[166] — the form is settled, the numbers are not.

4. **Details of the aggregator reverse auction:** the funding source for the service fee rate (protocol fees or the treasury), the bid-evaluation function over rate and bond, and the bindingness of the standby queue.
5. **P2P layer specification:** the propagation protocol for signed blocks, the engineering guarantees for block-body availability (the format of preconfirmation receipts, the relaying obligations among builders, whether erasure-coded propagation is worth it) and public metrics for skipping and body withholding (published as a public dashboard, the lever that makes deterrence bite — the deterrence-grade residual risk of §5.4/§5.5 rests entirely on it).
6. **Proving-circuit cost:** an assessment of the in-circuit cost of per-block signature verification (at most 384 signatures per epoch) plus recomputation of the lookahead; whether BLS aggregate signatures are worth introducing (replacing per-block ECDSA with an aggregatable signature, lowering both the circuit cost and the potential cost of direct verification on L1 — this bears on the choice of builder key scheme); and incremental continuation and aggregation for batch proofs ^[167]: once the chain head has been advanced by a small batch, a batch already being proven against the old head must be able to reuse the proven prefix and continue from it, rather than being re-proven from scratch — this is the precondition for defanging the drip-feed stalling tactic.
7. **Concurrent multi-batch landing and reorgs:** how to handle the gas competition when several parties race to land inside the fallback window (expected to follow the v15 §11.3 conclusion that "the competitors bear their own costs and the chain advances as usual", to be re-checked).
8. **Exemption mechanism for proving-system outages (blocking)** ^[168]: the form follows v15 §10.4 level 3 (independent corroboration + a bounded exemption + an `H_toll_max`-style cap), substantially simplified in the absence of the cancellation cascade, but it requires a dedicated derivation of its own; until that is done, the "zero false positives" claim of §6.3 is limited to fallback strikes.
9. **Handling of L1 reorgs:** the rollback semantics for a landed batch that meets an L1 reorg (expected to be considerably simpler than v15 §3.1/§13-S.1, because there is no cross-transaction certificate state machine, but it still has to be written down).
10. **Migration path:** the upgrade ordering from the current deployment, and from v15, to this design.
11. **Interaction sequence diagram** ^[169]: the interaction of batch landing, equal-height fork resolution and the fallback window is currently spread across §5–§7, and deserves a single normative sequence diagram.
12. **~~Protection of forced entries against nonce preemption (blocking)** ^[170]~~ — **closed with v1.51:** forced inclusion was removed wholesale, so nonce preemption (r27-1 / DeepSeek round 6 Critical 1) disappears with it. The item number is kept so that references to "item N" elsewhere in the document are not disturbed.
13. **Availability and cost of fallback provers** ^[171]: to be folded into the threat model — the Byzantine-aggregator liveness bound of §8 is conditional on it; and the design of fallback incentives under extreme congestion or when the proving cost exceeds the reward.

14. **Upper bound on the L1 cost of the `saveForcedInclusion` enqueue validation** ^[172]: decoding a signed L2 transaction on L1, verifying its signature and checking `chainId` and the share all carry non-trivial gas costs, and they expose a grieving surface against malformed calldata. A gas cap on the validation itself and a cheap rejection path for unparseable submissions (paid for by the submitter) must be defined, or the validation must be pushed down to a cheaper domain.
15. **Worst-case calibration of the margin on the lag thresholds** ^[173]: $\Delta_{\text{lag,prov}}$ (4 epoch ≈ 25.6 min) leaves only ≈ 5.6 min of margin over the upper bound of the normal provisional lag band (20 min); for $\Delta_{\text{lag,final}}$ (≈ 9 epoch ≈ 57.6 min) ^[174], the margin over the steady-state `lag_final` band (≈ 35 – 40 min) is carried by the margin term of `W_settle_max`. A proving spike compounded by jitter in L1 finality could push a conscientious aggregator past the threshold and open the fallback window in error. Whether to widen it needs to be assessed.
16. **~~Migration of the forced-inclusion data carrier** ^[175]~~ — **closed with v1.51**: forced inclusion was removed wholesale, so there is no forced-entry carrier problem left. But **the migration side gains a new problem**, folded into item 10: the current `MainnetInbox` already holds enqueued forced entries (`LibBlobs.BlobReference`), and the upgrade must define what becomes of them (drain before switching, or void and refund), or those enqueued entries are stranded permanently at the moment of the upgrade. The original text follows: the current `MainnetInbox` stores a `LibBlobs.BlobReference`. A migration or compatibility path is needed: how already-enqueued blob-ref entries are handled at the switchover (replayed as calldata, both carriers accepted during a grace period, or the switch deferred until the old queue has drained) — otherwise entries sitting in the queue at the instant of the upgrade could be stranded.
17. **Genesis and bootstrap semantics** ^[176]: the initial lag (with no landed head, lag may be arbitrarily large and the fallback window is "open" from genesis onwards), the selection of the first aggregator, and strike accounting before the first batch has landed are all undefined. A genesis anchor (a genesis landed head) plus rules suppressing fallback and strikes during the bootstrap period must be defined.
18. **Strengthening of the formal specification — the pre-implementation gate** ^[177]: **(delivered, r43)** an executable reference model of the §5.2 total order and the §5.6 window state machine, plus property tests (P1–P7, covering every invariant of the two blocking findings of round 5) — see Appendix C and `settlement-window-model.py/settlement-window-RESULTS.md`; any change to the rules must be mirrored in the model and the model re-run. **(delivered, r44)** The second half: the model gains P8–P11 (bridge reservation, anchor geometry and causal ordering, the moment a slashing takes effect, the fallback accounting snapshot; all 19 assertions pass), plus the pre-implementation review document `settlement-window-implementation-review.md` (Solidity-level `acceptCandidate` storage and gas, the `Inbox` integration path, an item-by-item disposition of (a)/(b)/(c) below and a list of what remains open — of which (b), the precise definition of the lookahead sampling operator, is still open and must be closed before electing is implemented; the final judgment is the owner plus a human safety review, and this gate does not pass anything automatically). The original list: (a) a single set of pseudocode or state machines for block legality / parent selection / the best-chain total order (the §5.2 triple, including irreflexivity, totality and transitivity property tests) ^[178] / the settlement-window `openWindow/acceptCandidate/replaceCandidate/closeWindow` state machine (§5.6, including double-candidate supers-

ession, baseline freezing when the forced or message queue is non-empty, and L1-reorg and lazy-close tests)^[179] / landing / forced inclusion / the window challenge (§5.6) / the shared gas budget and overflow (§7) / atomic rollback of landed head + state root + F_consumed/m_consumed + lag under an L1 reorg (§11 item 9) (the current prose relies heavily on review notes and cross-references, and the P1-level parent-block / fork-choice /lander interactions are easily missed by a reader); (b) the precise definition of the deterministic weighted sampling algorithm for the lookahead — from r47 onwards §3.2 gives a complete executable candidate operator (Python code using a capped weight prefix sum plus positioning by seed modulo total weight, verifiable by running lookahead-model.py), and this item closes once the owner has confirmed that operator; (c) terminologically distinguishing L2 acceptance finality from L1 finality F_l1 with different words (the conflated use of "final" makes the two-tier parent rule harder to reason about).

Master parameter table ^[180]

Parameter	Initial value	Unit	Key invariant / source
slot	1	second	§3.1
epoch	384	slot	§3.1
H_look, lookahead horizon	768	slot (≈ 2 L1 epoch)	§3.2
D_snap, snapshot delay	5	L1 epoch	$\geq H_look + F_final + \text{margin}$, §3.2 r8-1
w_max, weight cap	20%	—	§3.2
N_max, registry capacity	64	addresses	§4.1
G_max, gap cap (tier (i): $\text{gap} \leq G_max$, independent of the parent's landing status) ^[181]	64	slot	§4.2; a depth knob, not a hard cap, r20-1
G_max_landed (tier (ii): no cap on the gap, but the parent must be an L1-final landed block)	∞ (no cap)	—	§4.2 direction B; the argument is in the v1.31 record
F_l1, final depth	TBD	L1 slot	the finality depth of a tier (ii) parent block, §4.2
C_anchor, cap on L1→L2 message processing	TBD	messages/block	bound to the m_consumed cursor in §7, r34
Δ_prop , landing propagation settling	TBD ($\ll \Delta_lag_prov$)	slot	the §5.2 landing depth, r33
δ_slash , delay before a slashing takes effect	64	slot	§4.3
Δ_lag_prov , service observation threshold	4	epoch (≈ 25.6 min)	$\geq$ upper bound of the normal provisional lag band, §6.2 r10-2/r44; does not enter the penalty rules
Δ_lag_final , fallback/strike threshold	≈ 9 ($= \Delta_lag_prov + W_settle_max + 150 + \text{margin} + 10$, counted in L1 slots = 288) ^[182]	epoch (≈ 57.6 min)	in steady state $lag_final \approx lag_prov + W_settle$ (35–40 min) $>$ old threshold, so unless it is raised the fallback window is permanently open ^[183] ; §6.2/§6.3
W_settle_max, consensus upper bound on the window	TBD ($\approx 1.5 \times W_settle$)	L1 block (including a wall-clock conversion margin)	pins down the revocable exposure and Δ_lag_final , §6.2 r42 medium-severity finding 1
G_strike, strike rate limit	1	epoch	§6.3 r13-1
m_agg / m_agg'	2 / 4	strikes	termination thresholds, §6.3
D_anchor / D_anchor_max	32 / TBD (≈ 420) ^[184]	L1 slot	anchoring depth / freshness cap; $D_anchor_max \geq D_anchor + \Delta_lag_final(\text{L1 slot}) + P_prove_max + T_include_max + \text{margin}$ ^[185]
W_settle, settlement window	TBD (≈ 100 L1 slot ≈ 20 min)	L1 block	§5.6: $\geq P_prove_max + T_include_max + \text{margin}$, r41 option C
φ_land , base-fee share ratio	TBD	—	the sum of the base fees of the candidate chain $\times \varphi_land$ is credited to the winner's beneficiary at the close, §5.6/§6.5 ^[186]
gas shares G_anchor/C_l1msggas	TBD	gas	53 §7: the sum of the two $\leq \text{block_gas_limit} + \text{a per-entry admission cap} + \text{a watermark}$, r39 (v1.51 dropped the C_force term)

(The values are initial suggestions; those marked "TBD" are calibration items under §11. Unit conversion: 1 L1 slot = 12 s = 12 L2 slot.)

Appendix A: Points of divergence from the owner's suggestions (pending the owner's confirmation)

1. **The lookahead randomness does not mix in a proof hash.** The owner suggested "L1 consensus-layer data + the hash of the most recent proof"; this document takes only the former. The reason (§3.2): a proof hash is entropy that the party producing the proof can re-grind at zero cost, so mixing it in hands partial control of the lookahead to the aggregator, which runs counter to the trust-minimization goal. The 1-bit-level bias residual of pure L1 beacon randomness is acceptable for the purposes of the lookahead.
2. **Parent linkage uses the block-header hash, not the parent's signature.** The owner put both options forward; this document makes the hash the consensus rule (it is structurally necessary and transitively commits to the entire history) and demotes carrying the parent's signature inside the child block to an optional evidence-packaging convention (§4.2).
3. **The residual risk from skipping is classified as deterrence-grade and explicitly accepted.** The signature chain eliminates intermediate deletion by the aggregator, but a skip by the adjacent builder is not falsifiable inside the protocol (§5.4). This document chooses to say so plainly and to handle it with publicity + reputation + a cap of one slot on the loss, rather than introducing an undecidable "but I did publish it" arbitration. If the owner wants elimination at the structural level, the known price is publishing a commitment to L1 every slot (fees revert to what they were before aggregation saved them) — not recommended. (This sentence is precisely the cost identity cited in §1 ^[187]: the fee for buying protocol-level commitments or DA for the pre-landing stage $\approx$ the fee this design saves by landing in aggregated batches — buy the former and the latter's savings cease to exist.)
4. **The payment point for the base-fee share: the close, not provisional landing.** The owner suggested that "the base fee should be distributed directly to the aggregator at landing time"; this document chooses to account for it in the atomic commit at settlement-window close (§6.5). The reason: a provisional landing can be superseded inside the window by a heavier candidate (§5.6), so paying at landing would require designing a clawback mechanism for the superseded party — that is a new state machine and a new attack surface. Paying at the close, by contrast, is part of the atomic commit at the close, requires zero additional machinery, and happens to implement exactly the "winner takes all" incentive (a superseded lander receives nothing, which maximizes the opportunity cost of landing a short chain). The only price is that the lander's cash flow is deferred by one W_{settle} (≈ 20 minutes). If the owner insists on paying at landing, the complexity of a clawback state machine has to be accepted back into the design first.

Appendix B: Glossary

Term	Definition
slot	The 1-second time unit of L2
lookahead	The deterministic mapping from slot to builder address
builder	The preconfir that produces and signs a block according to the lookahead
aggregator	The service seat that packs, proves and lands
signature chain	The unlanded chain linked by parent_hash and signed block by block
landing	A batch plus its proof going onto L1 as a candidate; at window close the heaviest becomes final ^[188]
batch	A stretch of contiguous signature chain + data + proof
gap	A slot with no legal block
skip	A builder's signed choice not to build on the immediately preceding block that does exist
fallback landing	Anyone landing, and claiming compensation, once the aggregator has timed out
equivocation	Signing two different block headers for the same slot; slashed by direct verification on L1
two-tier parent rule	The criterion is the gap: (i) a gap $\leq G_max$ (independent of the parent's landing status); (ii) no cap on the gap, but the parent must be an L1-final landed head ^[189]
recovery block	A block that, during a stall, is built on an L1-final landed head under tier (ii) and reconnects to wall-clock time (§6.4)
benign stranding	When the parent tip of a recovery block is advanced, that block is voided and simply has to be rebuilt; no state is damaged (§6.4)
consume-and-discard	An inbound message that is illegal against the preceding state still advances the cursor but is not executed (§7, r9-2)
best-chain total order	Lexicographic order on the candidate's own triple (count, tip_slot, tip_hash); the lane component retired in v1.51 together with forced-only blocks (§5.2)
best-chain landing strategy	The lander lands the best chain under the §5.2 total order — not an obligation: landing a worse chain is superseded inside the §5.6 window by a heavier candidate and is self-defeating ^[190]
settlement window	Opens once the first candidate extending the window-final head has landed, and closes deterministically W_settle L1 blocks later ^[191]
candidate batch	A landed batch that extends the window-final head and carries a validity proof; inside the window it can be superseded by one that is strictly heavier under the §5.2 total order (§5.6)
window-final	The heaviest proven candidate at settlement-window close; withdrawals and bridging execute only from this state (§5.6/§6.1)
challenge / head-commitment	Removed. The accountability and commitment layers of v1.39/v1.40; independent review rounds 3/4 proved they cannot be mechanized without DA, and r41 replaced them with the settlement window
re-anchor episode	Removed. The recovery sub-protocol of v1.8–v1.30, deleted in its entirety by v1.31 direction B and replaced by the two-tier parent rule (§5.6 is now settlement-window finality) ^[192]

Appendix C: Executable reference model of the settlement window (normative pseudocode + property verification)

*The delivery of §11 item 18, "the pre-implementation gate" ^[193]. The normative pseudocode is that of this appendix and the body of §5.6; the **executable version** is in `settlement-window-model.py` (dependency-free Python, runs directly), and the verification results are in `settlement-window-RESULTS.md`. **Discipline:** any change to the §5.2 total order, the §5.6 window state machine or the §7 cursor and gas rules must be mirrored in the model, the model re-run and RESULTS updated — the three being out of sync counts as a specification defect (the same practice as the v15 `model_checker`).*

Properties that have been verified ^[194]:

Property	Content	Corresponding finding
P1	The total-order key (count, tip_slot, tip_hash) is irreflexive, total and transitive ^[195] ; the round-5 $A > B > C > A$ cycle resolves to $B > C > A$	Severe 1
P2	The winner at the close is independent of the order in which candidates were submitted (verified over all permutations)	Corollary of severe 1
P3	When the message queue is non-empty: a provisional landing does not change what is canonical; a heavier candidate can be verified against the same frozen baseline and supersede it; the close commits the winner's outcome exactly once; the cursor is monotone with no double consumption	Severe 2
P5	A lazy close does not change the winner; after the close, a candidate can only open the next window	Close boundary
P6	Window state is a pure function of L1 history; after truncation by a shallow reorg, replay is consistent and reorged-out candidates disappear atomically	L1 reorg
P7	Under the shared gas budget plus per-entry admission, a legal block exists for all 300 randomly generated reachable queue states (including non-genesis cursors); the maximum prefix under both constraints respects the caps	Rounds 4/5, gas deadlock
P7b	The per-message gas cap is load-bearing (added in v1.51): with the forced queue gone it is the only thing standing between the chain and "no legal block" — an over-cap entry would stall m_consumed permanently, so enqueue must refuse it	v1.51 one-sided deadlock
P9	The setter invariants are cross-checked between independently declared deployment values ($\Delta_{\text{lag,final}} \geq \text{prov} + W_{\text{settle_max}}, D_{\text{anchor_max}} \geq \text{the worst-case}$	§7/§11; introduced in r44, made non-vacuous in r46 ^[196] 58

Boundaries of the model (stated honestly): signature, proof and execution legality are placeholders (the model models precisely only the consensus objects this design adds: the total-order key, the window state machine, cursor arithmetic, gas shares and timing geometry). From r44 onwards, anchor geometry and causal ordering, the moment a slashing takes effect, and the fallback accounting snapshot are in the model (P9–P11); the storage layout and gas cost of the Solidity-level `acceptCandidate` entry point are an analytical (non-executable) review, see `settlement-window-implementation-review.md`. The base-fee share ($\$6.5 \varphi_{\text{land}}$)^[199] is an economic action of the accounting at the close that changes neither window state nor cursor semantics, and is not yet in the model. **The final judgment is the owner plus a human safety review** — the model and the review document are a gate, not a sign-off.

Appendix D: Version History (Design Record)

A version-by-version record of the findings and fixes of every round of adversarial review (newest first, oldest last). This is the complete argumentative trail of the design; it is non-normative, and where the main text conflicts with it, the main text governs.

On the section numbers in historical entries: the entries below cite section numbers as they stood when each was written (at that time §7 = forced inclusion, and §8–§12 were anchoring and the bridge, liveness accounting, the slashing master table, the v15 comparison and the open items). After v1.51 removed forced inclusion, §8–§12 were renumbered to §7–§11. Historical entries are not retrofitted: they record the state of the document at the time.

Draft v1.51, 2026-08-28 — the owner’s decision: remove forced inclusion wholesale. The instruction was “for simplicity, let’s remove forced inclusion entirely”. This is the first time on this design line that a **safety property is given up deliberately**, so this entry states the cost rather than only the simplification.

What was removed: the whole of §7 (forced inclusion) — the `saveForcedInclusion enqueue` interface, the forced queues (the ordinary/bridge dual FIFOs with their `f_cur_ord/f_cur_br` cursors), the maturity predicate `F_delay`, the per-block capacities `C_force/C_bridge`, the overdue threshold `H_force`, and **forced-only blocks together with their validity predicate `P_forced`**. §8–§12 were renumbered to §7–§11.

The cost, stated plainly: (1) **the censorship-resistance floor is gone.** The any-key forced-only block was the only escape valve under a full cartel that did not depend on a registered identity, and it was the inheritor of v15 invariant I6 (forced content always flows) and of the current deployment’s `permissionlessInclusion-Multiplier`. Block production now belongs to scheduled builders alone; if the whole set colludes to censor an address, the protocol offers **no remedy at all**, and that address’s funds have no exit path independent of the builders. (2) Two rows of the §8 liveness table change from bounded to **unbounded**: the full-cartel case and the “builders plus aggregator colluding as a whole” case no longer have a defense-in-depth exit, and can only wait for the cartel to break up, or for new builders to enter the registry — and the latter still needs the `D_snap` (5 epoch) snapshot delay plus window alignment before they reach the lookahead. (3) §1 [G5] “the inclusion floor is retained” changes from a design goal to **withdrawn**.

Closed or simplified along the way: (a) the blocking nonce-preemption item of r27-1 (DeepSeek round 6 Critical 1) closes, because the forced path it applied to is gone (§11 item 12); (b) §11 item 16, the forced data-carrier migration, closes — but **the migration side gains a new problem**, folded into item 10: the current deployment already holds enqueued blob-ref forced entries, and the upgrade must define what becomes of them or they are stranded permanently; (c) §5.1 drops from five validity rules to four (the old rule 4, the forced prefix, is deleted, and so is the `P_forced` exception in rule 1); (d) **the §5.2 total-order key contracts from the 4-tuple** (lane, count, `tip_slot`, `tip_hash`) **to the triple** (count, `tip_slot`, `tip_hash`) — with no forced-only blocks, lane is constant and draws no distinction. The non-transitive cycle that r42 fixed does not come back: its root cause was that lane had been a pairwise criterion, whereas the three remaining components were always scalars belonging to the candidate itself. (e) the §5.6 frozen baseline tuple loses the forced cursor, becoming `base = (F, stateRoot, m_consumed)`; (f) the §7 shared block-level gas budget contracts from three parties (anchor / forced prefix / L1→L2 messages) to two.

One thing that must stay vigilant: the “combined-gas chain stall” risk of §7 did not leave with the forced prefix. **One side alone can still reproduce the same shape of deadlock** — if the head of `m_consumed` is an inbound message whose execution cost exceeds “the block cap minus the fixed anchor overhead”, rule 4 requires it to be consumed and it does not fit, so no legal state transition exists and the chain dies permanently. The per-message gas cap is therefore **promoted from one of two parallel enqueue checks to the single load-bearing clause**, and the model gains P7b specifically to verify that it is load-bearing (construct an over-cap message and confirm the cursor really does stall permanently).

Model and figures: `settlement-window-model.py` was rewritten accordingly — `~~P4~~` (lane whitewash resistance) and `~~P8~~` (bridge-queue `C_bridge` starvation resistance) are retired for want of an object, P7b is added, and the numbers are not reused so that existing references are not disturbed; **20 assertions**

now pass. Figure 9 (forced-entry consumption) was deleted and the former figure 10 (anchor-age geometry) renumbered to 9, taking the document from 10 figures to 9. lookahead-model.py (6 assertions) is unaffected.

Draft v1.50, 2026-08-27 — a stale bold lead-in that contradicted option C is corrected. The lead-in of the last bullet of §5.2 read “‘which chain ought to be landed’ is now a slashable obligation”. That is a leftover from the v1.39/v1.40 challenge-layer era: those layers were falsified in rounds 3 and 4 of the independent review and deleted wholesale by option C (v1.41), at which point §6.3 and §10 withdrew “landing the wrong chain is slashable” as well. The lead-in contradicted **the body of its own bullet** (which says L1 “does not adjudicate who ought to land what \ldots{} no DA, no ex post adjudication”), the v1.47 characterization earlier in §5.2 (landing the longest visible chain is “the rational choice under economic incentives, not a slashable obligation”), and the slashing inventory of §10 — under the current design the only mechanically adjudicable slashable faults are double-signing (§4.3) and landing-timeout strikes (§6.3). The lead-in now reads “L1 does not adjudicate ‘which chain ought to be landed’; the settlement window selects it mechanically”, in agreement with the bullet it heads. A pure consistency fix: no new rule, no parameter change, both executable models untouched.

Draft v1.49, 2026-08-27 — the paper becomes an English edition, set in an academic serif face. The owner’s instruction was that the Chinese in the PDF looked poor and that the document should be converted to English with typography closer to an academic paper. Three changes: (1) slot-chain-spec.en.md is added, a complete English translation of the Chinese specification (translated in eight parallel slices against a shared glossary derived from Appendix B; inline identifiers, §-references, list indentation and table structure were each verified against the source). The Chinese edition remains normative; the English file records the sha256 of the Chinese source in its header, and build-pdf.py --check reports when the two have fallen out of step. (2) The PDF is now generated from the English edition: the document class moves from ctexart to article, the type is Palatino (mathpazo, text and math matched) with microtype, and the CJK font dependency is gone; the title page becomes a conventional \maketitle plus abstract; 26 Unicode symbol mappings were added ($\approx \leq \rightarrow \perp \checkmark \Delta \varphi$ and the rest routed through math mode) together with a full-width punctuation normalization pass. (3) The labels and captions of all 10 TikZ figures were translated and re-tuned for width, English running 1.6–2× wider than the Chinese it replaces; both pdflatex and xelatex verify zero overfull boxes, zero missing glyphs, and a widest picture of 145 mm. Specification semantics, parameters and both executable models are unchanged. **Draft v1.48, 2026-08-27 — the typesetting pipeline switched to LaTeX: a single-column, academic-paper-style PDF, with review annotations externalized into Appendix E.** The owner instructed: “convert it to LaTeX, single column, produce the PDF directly; the figures should be black, white and grey, in academic-paper style; every comment quoted from others and from the owner goes into an appendix at the very end”. Three items landed: (1) a new build-pdf.py(Markdown → XeLaTeX + ctex → slot-chain-spec.pdf, 68 pages, A4 single column), including a title page, abstract, table of contents and headers/footers, with § and appendix references rendered as clickable cross-references; the body’s section numbering is aligned with the document’s own §0–§12, and Appendices A–E are excluded from the numbering. (2) All 10 mermaid figures were redrawn as greyscale TikZ figures(tex/figures.tex, black/white/grey only, including one genuine sequence diagram), floated with the body text and captioned. (3) Review annotations externalized: roughly 310 provenance parentheticals in the body of the form “[200]” and “[201]” were extracted into endnotes [n] collected in the newly added Appendix E; for parentheticals that put substantive content first and provenance second, only the provenance part was moved out, and normative content such as parameter values remains in the body. The HTML pipeline was deleted(slot-chain-spec.html, build-html.py); under tex/ only the hand-written figures.tex is checked in, while main.tex and the compilation intermediates are excluded by .gitignore. Normative semantics, parameters and both models(21 + 6 assertions)are unchanged. **Draft v1.47, 2026-08-27 — three instructions from the owner: characterizing the visibility of the longest chain + a base-fee revenue share as incentive, expressing the lookahead as code, and a further readability pass.** (1) Two additions to §5.2: the visibility assumption and the characterization “economic only, not compulsory” — the protocol cannot force a lander to land the longest signature chain(the builder at the chain tip can withhold its signature and thereby deprive others of visibility, which is not a fault), and landing a shorter chain carries only economic consequences; the node-local chain-selection rule is stated explicitly as the existing four-tuple

total order(including the equivocation-fork case — the equivocation is slashed, but the two chains remain comparable, so view convergence is not broken); and the trade-off “the sum of fees does not enter the total order” is recorded(fees can be self-paid to inflate volume, so letting them into the canonicity criterion would amount to letting capital buy out the honest majority — fees enter only the reward). §6.5 adds the base-fee revenue share: the sum of all base fees of the candidate chain $\times \varphi_{\text{land}}$ is credited to the winner’s beneficiary in the atomic commit at window close; the revenue-share base is monotonic in chain length and a superseded candidate receives nothing; the payment point is taken to be the close rather than provisional landing(which avoids a clawback state machine), the deviation from the owner’s proposal is recorded in Appendix A-4, and φ_{land} enters the parameter table. (2) §3.2 embeds the complete executable Python for the lookahead(with step-by-step comments), and a new lookahead-model.py(6 properties: pure function/window alignment/snapshot-finality geometry/w_max capping and proportionality to weight); the sampling operator(capped prefix sums + seed-modulo positioning)is the candidate form for §12-18(b), and that item closes once the owner confirms it. (3) Readability: about 670 mid-sentence bold spans were removed document-wide(roughly 300 line-initial label-style bold spans were kept); the HTML generator was fixed for nested-list parsing(tab_length=2 + supplying the missing blank line before a list that follows a paragraph), § and appendix references now become anchor links automatically(700+ of them), and the bold weight was reduced to 600. **Draft v1.46, 2026-08-26 — the DeepSeek-on-v1.45 batch: restoring the model’s discriminating power + recalibrating one genuine numerical contradiction. (W1) Making P9a non-vacuous — and it immediately caught the numerical slack in r44:** the model had previously defined $D_{\text{ANCHOR_MAX}}$ as “the worst-case path + 5” and then asserted “the worst-case path $\leq D_{\text{ANCHOR_MAX}}$ ”, which is trivially true and has no discriminating power. Fix: all timing parameters were changed to **independently declared deployment-value literals**(copied from the parameter table), and P9a cross-checks them with the inequality. Once non-vacuous it failed immediately — r44’s initial $\Delta_{\text{lag,final}}$ of 8 epoch(51.2 min)is below its own formula $\Delta_{\text{lag,prov}} + W_{\text{settle_max}} = 128 + 150$ L1 slot(55.6 min). Recalibration: the initial $\Delta_{\text{lag,final}} = 9$ epoch = 288 L1 slot ≈ 57.6 min; and correspondingly the initial estimate of $D_{\text{anchor_max}}$ goes $380 \rightarrow \approx 420$ L1 slot(≈ 84 minutes)(§6.2/§6.3/§8/§12/the parameter table/two figures synchronized). (W2) Clarification of the frozen-baseline semantics + P12: what is frozen is [the cursors and the state], not the queue contents — the queue is append-only, referenced by sequence number, and immutable in content, so an enqueue in mid-window is visible and determinate to any later candidate whose coverage reaches it; in implementation the queue commitment must be per-entry/append-only (a hash chain or an MMR), and a single mutable queue root taken at landing time must not be used as a public input(a new clause in §5.6 + a synchronized baseCommit section in the pre-implementation review document). The model adds P12a/P12b(a mid-window enqueue does not change the outcome for earlier candidates; the moment of enqueue does not affect the verification result), and all 21 assertions pass. (W3) **Regeneration discipline:** the HTML generator of the time gained a --check mode(regenerate and diff, exiting non-zero if dirty)^[202], and the README states the convention “edit the md $\rightarrow$ regenerate + check”; the monorepo CI workflow is out of scope for this docs branch and is listed as repository-level follow-up at merge time. (Suggestion) P9b is strengthened to check $\text{anchor.L1_timestamp} \leq \text{L2_timestamp(slot)}$ against an explicit time-base function (including the equality boundary); build-html.py should manage files with with and record its dependency versions. **Draft v1.45, 2026-08-26 — readability revision**^[203]. Round 1(structure): roughly 400 lines of version history were moved from the top of the document into this appendix, and a “how to read this document” guide(including the reading conventions for review parentheticals)plus a table of contents were added before the main text. Round 2(typography): half-width commas/semicolons/colons inside Chinese-language context were unified to full-width(about 630 of them; code blocks and inline code untouched). Rounds 3/4(figures): 10 mermaid figures were added to the body — §0 the end-to-end path and the gap timeline, §3.2 the lookahead snapshot geometry, §4.2 the decision flow of the two-tier parent rule, §5.6 the settlement-window lifecycle, §6.2 the four-level certainty ladder and the two thresholds, §6.3 the causal chain of fallback and strikes, §6.4 the stall-recovery timing, ŋFORCEDž maximum-prefix consumption over the two queues, §8 the anchor freshness geometry — all in diagram-as-code form(rendered natively by GitHub, not pasted images), and each one validated by rendering it through mermaid-cli. The §0 abstract sentences were split and polished, and references in the body to the old “change log” now point to this appendix. Round 5(final read): a full re-read of the document, residual patching, and the addition of an HTML edition slot-chain-spec.html^[204]. Normative semantics, parameters and the model(19 assertions)are all unchanged. **Draft v1.44, 2026-08-26 — fixes for the DeepSeek-on-v1.43 batch + delivery of the second**

half of §12 item 18, "the gate before implementation". (C1, high) Splitting the threshold: $\Delta_{\text{lag,prov}} / \Delta_{\text{lag,final}}$ — r42 switched the quantity used to decide fallback/strikes to $\text{lag}_{\text{final}}$ but kept the old threshold Δ_{lag} (4 epoch ≈ 25.6 min); under window finality the steady state $\text{lag}_{\text{final}} \approx \text{lag}_{\text{prov}} + W_{\text{settle}} \approx 35\text{--}40$ min is by itself already above the old threshold $\rightarrow$ in a healthy system the fallback window would be permanently open and a diligent aggregator would accumulate strikes continuously until terminated. Fix: each of the two lags gets its own threshold, $\Delta_{\text{lag,final}} = \Delta_{\text{lag,prov}} + W_{\text{settle}}_{\text{max}} + \text{margin}$ (initially ≈ 8 epoch ≈ 51.2 min), and all penalty decisions in §6.2/§6.3 now use it; §6.3 adds a terminological convention (a bare $\text{lag}/\Delta_{\text{lag}}$ reads as final, clearing W3); ripple: the lag term of the $D_{\text{anchor_max}}$ formula is likewise raised to $\Delta_{\text{lag,final}}$ (the fallback opens its window on $\text{lag}_{\text{final}}$, so the freshness window must cover the later authorization point; the initial estimate goes $250 \rightarrow 380$ L1 slot).

(C2) The bridge reservation was missing from the model — settlement-window-model.py used to give the whole of C_{force} to the bridge queue, omitting the C_{bridge} reservation of r19-1; fixed, plus a new P8 (the ordinary queue is not starved under a saturating bridge flood). (W2) Normalizing close-before-accept — the §5.6 pseudocode now explicitly requires $L1.\text{now} < \text{close}_{\text{at}}$ before a candidate can be admitted, and at the same height the close happens before admission (consistent with the model's replay semantics). (W1) **The Appendix B entry "best-chain total order" was updated to the r42 four-tuple.** Second half of the gate delivered: the model adds P9 (anchor geometry/causal ordering), P10 (slashing keyed to the candidate's landing point) and P11 (the fallback-eligibility snapshot), and all 19 assertions pass; plus a new settlement-window-implementation-review.md (a Solidity-level storage/gas analysis of acceptCandidate, the Inbox integration path, and item-by-item disposition of §12 item 18 (a)/(b)/(c); the final judgment = the owner + a human security review). **Sections touched:** §5.6, §6.2, §6.3, §8, §12 (items 3/15/18, the parameter table), Appendices B/C, the model and the new document. **Draft v1.43, 2026-08-26 — delivery of the first half of §12 item 18, "the gate before implementation": an executable reference model of the settlement window + property verification(Appendix C).** New files settlement-window-model.py (runnable with zero dependencies) and settlement-window-RESULTS.md: the §5.2 total-order key, the §5.6 window state machine (frozen baseline/candidate versioning/close-time commit/lazy close/L1 replay) and the two-constraint cursor arithmetic of $\hat{n}\text{FORCED}\hat{z}/\S8$ are written out as an executable model, and all 14 assertions P1–P7 pass — including every invariant of the two blocking findings of round 5 (dissolving the A/B/C cycle, independence from submission order, provisional does not change canonical, no double consumption), lane-whitewash resistance, lazy close, the L1-reorg pure function, and the absence of gas deadlock across 300 randomized queue states. Discipline: any change to the rules must be mirrored in the model and re-run [205]. The second half of §12 item 18 (the pre-implementation review: items the model does not cover + the Solidity-level entry points) remains open and must be completed before implementation begins. **Sections touched:** §12 item 18, Appendix C (new), and two new files. **Draft v1.42, 2026-08-26 — fixes from independent review round 5/5 (the final round): closing the mathematics and the state machine of the settlement window.** Conclusion of the final round: option C is the right direction (the fatal flaw of the two-layer committed-head/challenge design is confirmed eliminated), but v1.41's execution had two blocking problems: **(severe 1) The total order is not transitive** — the old rule "look at the first differing block of the two chains to determine the lane" is a pairwise criterion, so an $A > B > C > A$ cycle can be constructed and the winner at the close drifts with submission order. Fix: the lane becomes an invariant scalar of the candidate itself (= the class of the candidate's first block counting from F, inherited by the whole chain), and the comparison is the lexicographic order on the four-tuple (lane, count, tip_slot, tip_hash) — a total order by construction, and still whitewash-resistant (§5.2). **(severe 2) A provisional candidate advancing the canonical cursor** — the first candidate to land immediately advanced F_{consumed} /the state, so a second, heavier candidate's starting cursor no longer matched and it could not compete, reinstating "first to land wins forever". Fix: §5.6 adds the **frozen baseline + candidate versioning + close-time commit** state machine (opening the window freezes $\text{base} = (F, \text{stateRoot}, F_{\text{consumed}}, m_{\text{consumed}})$, all candidates are verified against that same base and each stores its own end tuple, and the close commits only the winner's end tuple in a single atomic write; the openWindow/acceptCandidate/closeWindow pseudocode enters the body; it is honestly acknowledged that this is a small window state machine [206], the $\hat{n}\text{FORCED}\hat{z}$ cursor rules are re-read accordingly, and the leftover text "first to land is canonical" in §5.2 is deleted. (high 1) The bounded-inclusion assumption is made explicit — a fixed $W_{\text{settle}}/D_{\text{anchor_max}}$ deadline does not entail unconditional robustness against temporary censorship; §1 explicitly adds the $T_{\text{include,max}}$ assumption + the setter relations, and the affected statements are made conditional. (high 2) Fallback accounting decides on $\text{lag}_{\text{final}}$

final and keeps a snapshot until the close — this prevents a short candidate from clearing the provisional lag and cutting off the correction reward; the reward is paid to the winner and to every strict improver. **(high 3) The D_anchor_max formula uniformly gains Δ_lag in all three places**(without it, a legal parameter set necessarily strands honest tails). **(medium 1) Splitting $lag_prov(service)/lag_final(safety)$** , with the revocable exposure honestly restated as $\Delta_lag + W_settle$ (wall clock) + fallback response, and W_settle given a consensus upper bound. **(medium 2) $submission_slot := L2_slot$** (timestamp of the enqueuing L1 block) is made unique(under a huge stall the alternative reading “take the L2 head” would fork). Consistency: a tier (ii) parent must be both window-final and L1-final(waiting bound $\max(W_settle, F_ll) + D_anchor$), the §5.1 P_forced summary gains the tier (i) continuation case, the §4.3 “no collateral damage” wording is changed to acknowledge the cascade exposure of the successors of an equivocation, and the δ_land table row/the challenge leftovers/the glossary are cleaned up. Sections touched: §1, §4.2, §4.3, §5.1, §5.2, §5.6, §6.2, §6.3, ~~§FORCEDz~~, §9, §12, Appendix B. **Draft v1.41, 2026-08-26 — the owner’s decision, option C: settlement-window finality, wholly replacing the challengecommitment layer + fixes from independent review round 4.** Round 4 proved that v1.40’s “lightweight commitment” still does not hold: a chain-head commitment cannot prove that the block body is available(commit the head then withhold the body → the honest lander lands the only chain it can land → the attacker reveals the body afterwards and challenges → the honest party is wrongly slashed / everyone stops serving), a single commitment ledger can be permanently poisoned by one deliberately-worse landing, and bounded censorship still penetrates the commitment window — repairing it = full DA + a generational commitment history + a challenge/deposit state machine = the entire complexity of the episode design flowing back in. On this basis the owner changed the root^[207]: **the entire challengecommitment layer is deleted^[208]**, and the new §5.6 = **settlement-window finality** — landing is no longer “accepted means final”; instead the heaviest proven candidate within a window W_settle becomes final at the close; a better chain wins outright by landing into the window carrying its own proof, and no after-the-fact L1 adjudication of “should have landed but did not” is required. Comparison takes place only among candidates that have all landed and are all proven, hence: the block-body availability problem does not exist(without producing the body you cannot form a candidate); there is no cross-window state and no poisoning; close-time sniping is benign(heavier = more genuine signed blocks = a more complete tail finalized); a fixed window suffices, with no chess clock. Landing a worse chain changes from “a slashable fault” to self-defeating(being overwritten wastes the cost)—§5.5/§6.3/§10 correspondingly withdraw “landing the wrong chain is slashable”. The cost (stated explicitly): finality + one W_settle (≈20 minutes, with withdrawals executed from window-final); the setter $W_settle \geq P_prove, max + T_include, max + margin$. Solved along the way: the stale-anchor tail deadlock is now resolved by parameter geometry ($D_anchor_max \geq D_anchor + \Delta_lag + P_prove, max + T_include, max + margin \approx 250$ L1 slot, §8); the grandfathering criterion for slashing = the L1 landing point of the candidate batch(unforgeable, §4.3, with δ_land deleted); and lag is measured on the best provisional candidate(so liveness is not dragged down by the window, §6.2/§6.3). **Independent bugs from round 4 fixed in the same pass:** §8 is purged of the leftover “exactly min(entry count)” text and unified on the two-constraint longest prefix, with the backlog bound changed to $\max(entry\ count / C_anchor, gas / share)$ (severe 3); G_strike units unified to L2 seconds + an L1 slot conversion(medium 3); H_force is re-verified per block and a lane continuation does not inherit maturity eligibility(medium 1); the recovery bound gains a $+D_anchor$ term(medium 2); $final_ref$ enters the block-header tuple and §5.1 (consistency 1); the three-tier depths of §10 are made conditional; and the §12 C_anchor wording is corrected(consistency 7). Six rounds of self-review have been folded into the body of §5.6(sniping is benign/fixed window/parameter invariants/L1 pure function/anti-spam/three residual risks). Sections touched: §4.2, §4.3, §5.1, §5.2, §5.4, §5.5, §5.6(rewritten as a whole), §6.1, §6.2, §6.3, §6.4, ~~§FORCEDz~~, §8, §9, §10, §12, Appendix B(§5.7 deleted). **Draft v1.40, 2026-08-25 — the owner’s decision, option B: add a lightweight [chain-head commitment layer] so that challenges become mechanically enforceable + fixes from independent review round 3.** Round 3 proved that v1.39’s challenge(pure signature verification)can neither mechanically prove that a competing chain “existed before the landing”(a block’s self-reported slot can be forged by signing after the fact)nor prove “execution validity” — so it either wrongly slashes honest landers, or it requires DA/commitments/proofs in order to stand up. On this basis the owner chose option B: a new §5.7 chain-head commitment accumulator — permissionless and bonded(L_commit), in which anyone submits the chain head + the weight of a better legal signature chain(tens of bytes, no block body, no ZK)to L1 and L1 stamps it with an unforgeable timestamp; this is partial DA(chain heads only), and its cost is explicitly accepted in exchange for “chal-

lenges being mechanically decidable". The §5.6 challenge is re-anchored on the commitment timestamp: "the competing chain existed before the landing" is proven by the L1 commitment time(closing off after-the-fact signing)^[209], the divergence point must lie within the interval B has already landed^[210], and an invalid chain is disproven by the lander's validity proof → the committer is slashed L_commit(closing off impersonation with invalid blocks)^[211]; the property that it only slashes and never changes finality is unchanged. The commitment layer simultaneously repairs: **finding 1, the stale-anchor tail deadlock** — the anchor freshness of a committed block is judged by [the commitment time] rather than the landing time, so it can still land even if landing is withheld for a long time(§8); finding 5, the F_l1 boundary pre-signing loop — a tier-(ii) block header gains final_ref to prove that the parent was already final at signing time, rejecting pre-signing(§4.2); finding 7, slashing backfill — grandfathering now uses the commitment timestamp, made precise as "did it truly exist before the effective moment" (§4.3). Independently fixed bugs: finding 3, lane whitewashing — the lane is fixed at the first block after the fork and inherited by the whole branch, so a single ordinary block cannot whitewash a forced-only scaffold(§5.2); finding 4, the gas-prefix contradiction — unified as "the longest FIFO prefix whose entry count ≤ the cap and whose cumulative gas ≤ the share", deleting "exactly min(entry count)"(§5.1/§8); finding 8, integer threshold-riding — late_units = max(0, preLag − (Δ_lag − 1)), with the minimum non-zero penalty at equality(§6.3); finding 9, H_force — folded into P_forced predicate 4(an any-key block must be overdue by ≥ H_force, nFORCEDz). Consistency: the old standalone statement "block count takes priority" in §5.2 is deleted and folded into the total order; the §5.5/§10 conclusions are taken up as the third class of slashable fault in §5.6; the §6.4 D_anchor formula is corrected; and "never expires" is scoped to the gap dimension. **Stated residual risk(§5.7):** the authenticity of HC depends on "≥1 honest committer"; a commitment proves only that the head exists, not that the block body is available(a body-withholding chain can still contest); this layer is the minimal reflux of the chain-head commitment/DA that was originally deleted to save cost, and option B accepts it explicitly. Sections touched: §4.2, §4.3, §5.1, §5.2, §5.5, §5.6, §5.7(new), §6.3, §6.4, nFORCEDz, §8, §10, §12, Appendix B. **Draft v1.39, 2026-08-25 — the owner's decision to introduce a "challenge-and-slash" layer + fixes for the severe findings of independent review rounds 1/2.** A deep independent AI review^[212] proved that under "landers are entirely untrusted + no protocol-level DA per §1", pure off-chain fork choice cannot stop a malicious lander from skipping a healthy tail and landing a worse chain, nor stop a single aggregator from resetting repeatedly at zero penalty; and that a mechanical guarantee against malicious landers is undecidable without DA. On this basis the owner decided to add an accountability layer that "only slashes, never changes the chain": **a new §5.6, the obligation to land the best chain + challenge-and-slash** — the lander shall land the "best chain"(the new total order of §5.2: content lane > forced-only lane; then block count, then higher slot, then the hash of the first differing child block)among those that have propagated and settled; if it skips a better settled chain, anyone may submit evidence to L1 within the accountability window W_chal (pure signature verification, the evidence being a chain that was signed and in existence long beforehand, with no clock race)→ the landing bond L_land is confiscated and the reporter rewarded. The key point: a challenge only slashes; it never rolls back and never changes finality (landing is final, as determined by the proving system)— which is fundamentally different from the "re-anchoring episode" deleted before v1.30(which changed canonicity, required rollback and raced the proving latency), and so avoids the old fatal flaw. On this basis the severe findings of the independent review are fixed: (1) an honest lander discarding a landable healthy long tail under the old "recovery takes priority" instruction — §5.2 is changed to "the best chain includes the frozen long tail; land it first, then recover from its final tip, without losing the tail", with §6.4 synchronized^[213]; (2) a single aggregator resetting repeatedly at zero penalty — every landing of a worse chain is slashable under §5.6, and §6.3 adds lag debt settled as of the moment before acceptance, blocking integer threshold-riding; (3) a heavier any-key forced chain overwriting a content tail — §5.2's lane priority rules it out directly, and it is slashable. Fixed in the same pass, the high-severity/consistency items of the independent review: the combined-gas chain stall(§8 adds a shared block-level gas budget + a per-message gas cap + determinism priority/overflow/watermark); the D_anchor_max formula omitting D_anchor(§12 adds ≥ D_anchor + P_prove,max + T_include,max + margin); a slashed old key backfilling historical slots(§4.3 adds a second gate, "keyed to the batch acceptance point + a δ_land grace period"); leftover C_anchor text, m_consumed changed to a global message sequence number, the "tier (i) unlanded parent" wording, a single expiry predicate for H_force/F_delay, and updates to the parameter table and the glossary. Sections touched: §4.3, §5.2, §5.6(new), §6.3, §6.4, §8, §9, §10, §12, Appendix B. **Stated residual risk(§5.6):** the challenge is economic deterrence, not a safety guarantee(submitting the evidence must be eco-

nomically viable); an honest but eclipsed lander may be wrongly slashed (mitigated by setting Δ_{prop} low); and since finality is unchanged, a skipped preconfirmation means “delay, not recovery”. All are items explicitly accepted under the no-DA model of §1. **Draft v1.38, 2026-08-25:** Codex 1×P1 + DeepSeek 1 Critical + 4 warnings, focused on anchor freshness/censorship resistance/interactions with forced-only blocks: (Codex P1) anchor freshness can make a recovery block expire — tier (ii) eliminates “expiry by gap”, but the §8 $D_{\text{anchor_max}}$ freshness rule still applies, so a recovery block must be proven and landed within the freshness window of its anchor. Fix: §6.4 scopes “never expires” to “does not expire because of the gap”, and a new setter invariant $D_{\text{anchor_max}} \geq \text{proving latency} + \text{worst-case landing delay}$ is added (§12); the crucial difference is that $D_{\text{anchor_max}}$ is a freely settable parameter, not the 64-second window that the old s_{ra} pinned down structurally. (DeepSeek Critical) The last-block-of-the-batch freshness shortcut does not hold — the anchor is only monotonically non-decreasing, so a fresh last block does not imply that earlier blocks are fresh, and checking only the last block would let earlier blocks discharge forced-inclusion obligations against a stale L1 view. Fix: !FORCED now requires freshness to be checked block by block, and the “or the last block of the batch” shortcut is deleted. (DeepSeek W1) The censorship-resistance bound is $D_{\text{anchor_max}}$, not C_{anchor} alone — correcting the over-strong claim “consuming fewer entries makes the block illegal”: the freshness lower bound forces the anchor frontier to advance as landing proceeds, so the worst-case message delay is $\approx D_{\text{anchor_max}} + \text{backlog}/C_{\text{anchor}}$ (§8). (DeepSeek W2) A tie between a forced-only block and a lookahead block at the same slot/parent — an identical fork point makes the hash tie-break ineffective. Fix: §5.2 adds “at the same slot/parent a lookahead block takes priority over a forced-only block, unless the ordinary tail is already frozen”. (DeepSeek W3) “final” is overloaded — §4.2 is annotated L1-final versus §6.1 landed-final (a document-wide unification is listed as 18(c)). (DeepSeek W4) The bounds depend on undetermined parameters — the concrete bounds in §8/§9 are marked provisional (§12). Also fixed: the nested backticks in the Markdown of P_{forced} predicate 3. Sections touched: §4.2, §5.2, §6.4, !FORCED , §8, §12. **Draft v1.37, 2026-08-25:** Codex (14:43) one P1 — a forced-only block cannot be continued after it lands, so the $F_{\text{!}}$ rate limit contradicts the per-block draining at $\text{backlog}/C_{\text{force}}$ of §7. The original P_{forced} predicate 2 required every forced-only block to use tier (ii) (a final landed parent): under a total cartel the first forced-only block becomes the canonical head once landed but is not yet final, and the second can be built neither on it ($F_{\text{!}}$ not yet reached) nor on the old final head (a landing must extend the current head) → every block must wait for $F_{\text{!}}$ finalization, and per-block draining comes to nothing. This is exactly the chaining that the v1.35 finding-5 note “the same applies to forced-only continuation” implicitly assumed but that predicate 2 forbade. Fix: P_{forced} predicate 2 is relaxed to a choice of two — (re-rooting) tier (ii) on a final landed head [the start of the chain, or re-rooting after a freeze], or (same-lane continuation) tier (i) chaining onto the previous forced-only block (the parent must satisfy P_{forced} , with $\text{gap} \leq G_{\text{max}}$). Forced-only blocks can therefore chain and drain at a density of C_{force} per block even under a cartel; restricting the parent to be a forced-only block → an any-key holder cannot inject into an honest lookahead tail via tier (i) (the two lanes are separated by “whether the parent satisfies P_{forced} ”, and a malicious forced-only block remains constrained by the §5.2 recovery priority with respect to a healthy content tail). Sections touched: !FORCED , P_{forced} . **Draft v1.36, 2026-08-25:** Codex third batch (14:38) one P1 (line 598) — the L1→L2 causal-ordering invariant that the anchor must not be later than the block’s own slot time. §6.1 rule 3 only blocks future slots and §8 only blocks an anchor that is too old; neither blocks “an old slot paired with a new anchor”: a builder can withhold an old slot, afterwards pick a fresh anchor that already contains a new L1→L2 message, and sign a tier (i) continuation that consumes that message into an L2 block whose slot timestamp precedes the L1 block the message came from, with a Byzantine aggregator racing it into a landing → breaking L1→L2 causal ordering and distorting the bridge-deadline semantics that are based on block.timestamp. Fix: §8 adds the invariant $\text{anchor.L1_timestamp} \leq \text{L2_timestamp(slot)}$ (applying uniformly to ordinary blocks and forced-only blocks, with §5.1 rule 5 and !FORCED , P_{forced} predicate 5 synchronized). Recovery is unaffected (a recovery block’s anchor is at depth $\geq D_{\text{anchor}} \approx 6.4$ minutes back and its timestamp is always $\leq$ the wall-clock slot time, so it satisfies the invariant naturally); only the malformed “old slot + new anchor” combination is rejected. **Sections touched:** §8, §5.1, §7. (The two items at line 855/1036 in the same batch = resends, against the new head)^[214] **Draft v1.35, 2026-08-25:** Codex second batch (14:31) two P1s, both refinements that tighten the claims of v1.34 (rather than overturning them): (P1 line 916) “one hop” refers to re-rooting; continued progress still requires the ordinary liveness floor — the recovery block R is an unlanded P2P block, and producing a proof and landing it takes 10–15 minutes, during which ordering continues over P2P on top of R; “a recovery block never expires” refers precisely

to the landing legality of R(tier (ii) has no gap cap), not to R being extendable without maintenance. The exact condition for recovery liveness is therefore "lookahead builders keep returning at a density of $\geq 1/G_{\max}$ " (= the ordinary $G_{\max}$ liveness floor, not something specific to recovery); a single sparse builder will cause R to freeze $\rightarrow$ a retry R' on the same old landed head, which is benign and loses no ground, and recovery follows as soon as participation returns to the floor. The wording of §6.4/§9 is tightened. (P1 line 1102) A unilateral Byzantine aggregator can isolate a branch that it sources itself — the any-key exemption lets a malicious aggregator itself (by enqueueing forced entries of its own + waiting F_{delay}) manufacture a tier-(ii) forced-only branch and land it in person, so the "collusion" in §5.4's "requires the lander to collude" can collapse to a single untrusted aggregator acting alone, with no need to suborn a lookahead builder and no slashable equivocation. The boundaries are unchanged: it is kept out of a healthy tail by the §5.2 recovery priority (it is followed only when the tail is frozen), its depth is still $\leq$ the unlanded tail, it reaches only tier-2 revocable preconfirmations, and its content is forced-only (discretionary transactions return to the mempool). Corrected wording: the frequency of the forced-only path is not constrained by "the number of malicious lookahead slots", and the harm is bounded by the three limits above; this is explicitly folded into the untrusted-aggregator residual risk of §1/§5.4. Sections touched: §5.4, §6.4, $\acute{n}$ FORCED $\acute{z}$, §9. **Draft v1.34, 2026-08-25:** reviews by Codex and DeepSeek of the new direction-B design^[215] — four genuinely new holes (not old mechanisms resurfacing) have been fixed: **(P1) Recovery/fork-choice deadlock**^[216]: after a true stall, the long tail from before the stall has the greater block count yet is frozen and cannot be extended, so §5.2's "most blocks" rule would make honest nodes cling to it and starve the one-block recovery block, deadlocking recovery against fork choice. Fix: §5.2 adds recovery priority — a branch whose head lags the wall clock by more than $G_{\max}$ is a [frozen dead branch] and yields to a tier-(ii) recovery block built on the final landed head; the yielding criterion is identical to the recovery trigger criterion, so it fires only on a true stall and never isolates a healthy tail (whose head is always within $G_{\max}$ of the wall clock). §6.4 references it accordingly. (P1) m_{consumed} has only an upper bound and no lower bound^[217]: a pure upper bound lets a builder keep processing $0 L1 \rightarrow L2$ messages while still using a fresh anchor, censoring inbound messages indefinitely and invalidating the backlog/ C_{anchor} catch-up bound. Fix: §8 adds mandatory maximum-prefix consumption $\min(C_{\text{anchor}}, \text{pending})$ (a public FIFO queue + a queue root + a new legality obligation in §5.1 rule 5, exactly the same shape as the $\acute{n}$ FORCED $\acute{z}$ forced-inclusion queue). (P1) $\leq \Delta_{\text{lag}}$ is conditional^[218]: the tail is $\leq \Delta_{\text{lag}}$ only if the fallback lands on time; if the fallback stalls or its landing is censored on L1 the tail grows without bound, and a single tier-(ii) block can isolate a tail far longer than Δ_{lag} . Fix: §4.2/§5.4 honestly note that this upper bound is conditional on the same condition as the §6.3 Byzantine liveness bound, and no unredeemable bound is claimed. (P2) The admission predicate for forced-only blocks^[219]: §5.1's reference to the deleted "three conditions" is replaced by the precise predicate P_{forced} of $\acute{n}$ FORCED $\acute{z}$ (any key + a tier-(ii) final parent + slot $\leq$ wall clock + a non-empty forced prefix up to capacity + maximum message consumption), together with an explanation that under any-key issuance slot uniqueness is resolved by fork choice + atomic landing rather than by equivocation slashing. The rest: §4.2 adds the anchor field to the block header^[220]; §12 adds forced-carrier migration (16), genesis bootstrap (17), and formal pseudocode/the lookahead sampling algorithm/terminological distinctions (18). DeepSeek's "landed but not final" Critical and its " $\acute{n}$ FORCED $\acute{z}$ three conditions" warning on v1.32 had already been fixed in v1.33^[221]. Sections touched: §4.2, §5.1, §5.2, §5.4, §6.4, $\acute{n}$ FORCED $\acute{z}$, §8, §12. **Draft v1.33, 2026-08-25:** DeepSeek's review of v1.32(2 Critical + 4 warnings + 3 suggestions, all directed at the new direction-B design rather than old mechanisms resurfacing) — adopted, and converging. Key point 1^[222]: the criterion of the two-tier parent rule was misread as "the landing status of the parent block", which made a parent that is "landed but not final" appear to fall into no tier at all and made the first block of every batch look illegal. Correction: the criterion is explicitly the gap $s - \text{parent.slot}$, evaluated on the structure of the signature chain and identical in all three places (signing/proving/landing), and it does not depend on the drifting landing status — tier (i) is a bounded gap (independent of the parent's landing status; normal block production, including the first block after a landing, always takes this tier), and tier (ii) is an unbounded gap but the parent must have landed final. "Landed but not final" therefore leaves no gap in the coverage (restated in §4.2/§5.1). Key point 2^[223]: the "one hop" of one-hop recovery elided the F_{ll} transient — if the stall begins shortly after a landing and the current landed tip has not yet reached F_{ll} , the recovery block must wait for the tip to become final ($\leq F_{\text{ll}}$) before it can land (otherwise it would fork away the newer tip and $\acute{n}$ FORCED $\acute{z}$ would reject it). Correction: the recovery bound is written out in full as $\max(0, F_{\text{ll}} - \text{how long the tip has existed}) + \text{one round of proving}$, and in steady state a catastrophic stall

degenerates to a pure single hop (§4.2/§6.4/§9/ñFORCEDž updated). Warnings: §5.1 rule 1's stale reference to "the three conditions of ñFORCEDž" is changed to "a forced-only block as defined in ñFORCEDž" (W1); the §10 slashing table is completed with the tier (ii) path of isolating a single block via the landed head, with the same bound across all three tiers (W2); the ñFORCEDž prose on cursors/snapshots gains a "net rule" sentence clarifying that the per-block anchor is the sole expiry criterion and that F consumed only records consumption progress (W3); and the tier (ii) residual risk of §5.4 gains the premise "the current tip is already final" (W4). Suggestion 3 (marking deleted parameters as historical) is partly kept: the change-history notes for Δ_stall/s_base and others are retained to preserve review provenance, and all are clearly labelled "deleted". Sections touched: §4.2, §5.1, §5.4, §6.4, ñFORCEDž, §9, §10. **Draft v1.32, 2026-08-25:** Codex's first review of v1.31, one P1^[224] — C_anchor was bound to the wrong object: v1.31 applied it to the per-block advance of the anchor reference, which conflicts with the §8 anchor freshness rule, so under a long stall ($> C_anchor + D_anchor_max$) a recovery block's anchor could only advance C_anchor from the stale parent anchor, would still be stale, and would fail the freshness check — failing precisely on the long stall it was meant to support. Fix: C_anchor is rebound to a separate L1→L2 message-processing cursor m_consumed ($\leq C_anchor$ messages per block), while the anchor reference itself remains monotonically non-decreasing and must be fresh, with no per-block cap. A recovery block therefore references a fresh anchor (one-hop recovery of ordering, preserving C1), and a flood of inbound messages is processed block by block under m_consumed (the same shape as f_consumed in ñFORCEDž). §8/§12 updated. **Draft v1.31, 2026-08-25 — the recovery subsystem redesigned as a whole (direction B), and the freeze lifted.** The entire re-anchoring episode is replaced by the two-tier parent rule: (i) an unlanded parent $\leq G_max$ (deep-reorg protection unchanged); (ii) **a final L1 landed head as parent, with no gap cap** ($G_max_landed = \infty$) — because a landed head is final, building on it reorgs no already-landed block. Recovery = a direct corollary of §4.2 tier (ii): during a stall, any lookahead builder (or any key holder on the forced-only path) produces one block with the landed head as its parent, L2 ordering recovers in one hop, and a stall of any duration (hours to days) is still one hop — because tier (ii) has no cap, a recovery block never expires, and the whole family of "the proving latency punching through the freshness window" problems of the old design (r28, round 7 P1) disappears at the root. Net deletions: the announcement/B_ra, the challenge/B_ch/deposit settlement, the pinned parent/re-pinning, the freshness floor/s_base/s_ra, the Δ_cont continuation, the episode state machine, and the two-sided lag $> \Delta_stall$ condition on forced-only blocks — and consequently open items such as r29-P2 (deposit determinism) no longer apply. The argument for $G_max_landed = \infty$: the reorg exposure through a landed head is inherently capped by the length of the unlanded tail, which is in turn capped by Δ_lag (anything beyond it is already landed and final), so "unbounded" does not increase the worst-case depth (still $\leq \Delta_lag$, §5.4) yet buys one-hop recovery from a stall of any length; a finite value loses at both ends (below the proving latency it can never catch up with a long stall, and at Δ_lag the tightening is an illusion). **Six rounds of the designer's own adversarial re-examination have been folded in:** ① the definition of "the current landed head" under an L1 reorg → bind it to the final depth $F_l1 + benign$ stranding; ② recovery liveness = ≥ 1 lookahead builder returning $\vee$ the forced-only floor; ③ rule (ii) is not gated (gating would reopen r9-1 and is redundant — an honest lander is inherently harmless via fork choice); ④/⑤ under a long stall, L1-sync and the forced-inclusion backlog catch up per block at C_anchor/C_force (ordering in one hop, synchronization amortized); ⑥^[225] the lander's tail-selection strategy = §5.2 fork choice + a landing depth $\geq \Delta_prop$ of propagation settling, and "the lander's view is complete" is added to the honesty clauses of §1. Sections touched: §4.2 the two-tier rule, §5.1 rule 2, §5.2 the lander's strategy, §5.4 the three tiers of residual risk, §6.4 replaced wholesale, ñFORCEDž forced-only blocks simplified, §8 C_anchor, §9 the recovery row, §12 the parameters, Appendix B. **Draft v1.30, 2026-08-25:** DeepSeek round 8 — r31-1 (Critical 1, outside the freeze): §4.2 still said "attach to the legal chain head with the highest slot", contradicting the "most blocks" fork choice of §5.2 per r11-2; it now references §5.2 directly (otherwise an implementation would reopen the vulnerability in which a deep skip is automatically followed). The rest: Critical 2 (forced inclusion being "unconditional" versus nonce preemption) belongs to pending decision ① (r27-1 has already added the qualification); Warnings 1/2/3 all land in the frozen §6.4/ñFORCEDž (the "exactly covers" wording for the proving lead time versus the lookahead, the uniqueness of the first forced-only block, and the Expired transition of the episode state table) — these will be dissolved together by the redesign of the recovery subsystem rather than patched point by point. **Draft v1.29, 2026-08-25:** Codex round 18, one P1^[226]: r18-2's "batch-level unique snapshot H_batch = the anchor of the first block" made the forced-inclusion obligation of an already-signed block depend on the batch boundary chosen by the lander, so landing

a prefix could strand a block that was otherwise legally signed. Changed to **a per-block snapshot, checked against the anchor already committed in the block header** — the obligation is fixed at signing time and is independent of batch boundaries. (The re-anchoring parts of §6.4/ñFORCEDž remain frozen; this item belongs to the ordinary landing path and is outside the scope of the freeze.) **Draft v1.28, 2026-08-25:** the owner's decision — the §6.4/ñFORCEDž recovery subsystem is frozen pending formalization(direction B). No more point patches to this subsystem: what the last eight rounds of review (r18/r20/r21/r23/r24/r25/r28 and round 7's P1/P2) exposed is a single family of "proving latency versus slot timing window" and deposit-settlement determinism problems, whose root-cause intersection leaves no room for a simple rule and which should be solved in one pass by a human design retrospective + state-machine model checking. A DRAFT freeze note is added at the top of §6.4 (listing the open items for the retrospective), and its mechanism description and numerical bounds are downgraded to provisional. Reviews landing on other parts (incentives, forced inclusion, parameters, trust-model wording, and so on) continue to be handled as usual. **Draft v1.27, 2026-08-25:** DeepSeek round 6^[227], 2 Critical + 5 warnings: Critical 2(the contradiction between §6.4 and ñFORCEDž over the slot of the first forced-only block)was already fixed by v1.26/r25-2; r27-1 qualifies the "unconditional" claim about forced inclusion in §0/§1/ñFORCEDž in light of the nonce-preemption gap of §12-12(Critical 1 is elevated into a correction of the wording of the claim); r27-2 specifies the different destinations of B_ra under cancellation (a)/(b)/execution(under (b), when a genuine failure heals itself, the diligent announcer is refunded in full, W3); r27-3 adds the transition "the continuation state meets a normal batch" to the state table(W2); r27-4 requires s_ra to leave enough proving lead time, with the expected wait including one round of proving(W1); and W4/W5(the L1 cost of enqueue validation, and calibration of the Δ_lag margin)are folded into §12 items 14/15. **Draft v1.26, 2026-08-25:** Codex round 16, two P1s(consistency propagation)^[228]: r25-2, the ñFORCEDž format rule for forced-only blocks still said "the first block = the execution baseline slot at window expiry", contradicting §6.4/r24-1's "the G_max window at the moment of acceptance"; these are now aligned(including item 3 of the §6.4 fourfold division); and the Cancelled row of the state table gains r24-2's conjunct for cancellation (b), "no outstanding H_ch". Pure consistency, no new mechanism. **Draft v1.25, 2026-08-25:** Codex round 15, P1^[229]: r24-1 makes the first forced-only block bring the head up to the wall clock, dropping lag to $\leq G_max$, but ñFORCEDž still requires lag $> \Delta_stall$ for a forced-only block, while Δ_cont is only a few L1 slots — so a continuation batch can never wait for lag to exceed Δ_stall again, r18-3's "one window drains the backlog" fails, and a cartel can impose a stall on every batch. Fix: a continuation batch is exempted from the three valve-opening conditions of ñFORCEDž and rides the forced-only authorization already established in this episode and carried by the proven chain state(strictly bound to "it extends this episode's forced-only head + Δ_cont has not elapsed empty + the queue still holds due entries", and lapsing as soon as the queue drains or the timeout hits); only the first batch still re-checks the three conditions. **Draft v1.24, 2026-08-25:** Codex round 14, P1^[230]+ DeepSeek round 5, 5 warnings: r24-1 changes the reference point of the re-anchoring freshness floor from "the window-expiry s_base" to "the wall clock at the moment of acceptance"(proving takes about 10–15 minutes » G_max=64s, so by acceptance time a window-expiry reference point is already hundreds of slots stale and a single re-anchoring block still cannot catch up; binding to the moment of acceptance guarantees by construction that at the instant of landing the head is within G_max of the wall clock, which the lander implements by pre-signing and pre-proving for a near-future s_ra)^[231]; r24-2 adds the conjunct "no outstanding H_ch" to cancellation condition (b)(blocking a conflicting fork that excludes H_ch from using lag $\leq \Delta_lag$ to bypass the branch commitment)^[232]; r24-3 registers multiple H_ch and settles them independently^[233]; r24-4 states explicitly that the aggregator liveness bound is conditional on permissionless fallback being economically viable^[234]; r24-5 raises nonce preemption of forced entries to a blocking open item^[235]. Also, as suggested, a master parameter was added and the episode gained an Expired state. **Draft v1.23, 2026-08-25:** Codex round 13, one P1^[236]: r17-1's "jump to the execution baseline slot s_base" was added only to forced-only re-anchoring blocks and was missing from ordinary re-anchoring blocks — the general landing rule only rejects future slots, so colluding parties could land a stale re-anchoring block whose slot exceeds the parent's by just G_max + 1 yet lags far behind the wall clock, leaving the head still not caught up and the current round's builder unable to attach, and letting the attacker keep winning announcements with a pre-signed sparse chain and drag out recovery. Fix: both re-anchoring paths uniformly gain the freshness floor slot $\geq s_base$ (§5.1 rule 2 + the §6.4 execution check), so that one re-anchoring necessarily brings the head up to the wall clock; s_base is pinned by the wall clock at window expiry and does not change across re-pinning. **Draft v1.22, 2026-08-25:** DeepSeek round 4, 7 warnings + 1 suggestion(mostly consistency

and wording corrections, plus one conceptual clarification): r22-1 clarifies that a "conflicting fork" does not reorg an already-landed pinned parent (the finality of §5.2) but extends it and isolates the unlanded tail, and that "re-pinning" only moves the pointer forward; r22-2 aligns the definition of the re-anchoring parent in §5.1 rule 2 with the pinned parent of §6.4; r22-3 changes §5.1 rule 4 from "all forced entries" to a reference to the C_force capacity rule of §FORCED; r22-4 deletes the old sentence in §5.4 about "capped at G_max" that contradicted r20-1; r22-6 clarifies the exit-delay wording of §4.1 (an address that exits after the snapshot still has valid lookahead slots); r22-7 states explicitly in §6.4 that after re-pinning a new proof must be produced against the new pinned parent; and, as suggested, an episode state-transition table was added to §6.4.^[237] **Draft v1.21, 2026-08-25:** Codex round 12, one P1^[238]: r18-1's re-anchoring cancellation criterion, "only an extension on the same branch cancels", collapses — §6.1 requires every accepted batch to extend the pinned landed head, so every batch has the pinned parent as an ancestor and passes the "same branch" test, and a malicious deep skip cancels the re-anchoring just the same. The root cause is that the episode pinned only the parent and not the branch it was meant to protect. Changed to use an existence challenge H_ch as the branch commitment: cancellation happens if and only if (a) the landed chain of the accepted batch contains some challenged H_ch (the protected tail really did land), or (b) $\text{lag} \leq \Delta_{\text{lag}}$ (full recovery); other batches that advance the landed head (including a conflicting fork that isolates H_ch) only re-pin, do not cancel and preserve the window timer, while an isolated challenge is refunded B_ch with neither reward nor penalty. **Draft v1.20, 2026-08-25:** Codex round 11, one P1^[239]: G_max's per-block gap cap bounds only the rollback depth of a single block, not the depth of a branch — §4.2 had previously claimed wrongly that it "cannot be stacked by relaying". A minority coalition can relay many blocks along a private branch (each hop's parent distance $\leq G_{\text{max}}$, each block legal, no equivocation), and once a colluding aggregator lands the whole branch it isolates the entire unlanded honest tail, at a depth far beyond $G_{\text{max}} - 1$. Throughout the document (§4.2/§5.2/§5.4/§10/§1) this is changed to two honestly stated depth bounds: a single-block deep skip $\leq G_{\text{max}} - 1$, and a relayed sparse branch $\leq$ the length of the unlanded tail ($\approx \Delta_{\text{lag}} +$ the fallback response, the same bound as the equivocation path but not slashable); G_max is repositioned as a knob for suppressing the feasibility of sparse branches (by raising the lookahead density the coalition needs), and the residual depth is filed under the explicit trade-off of having no protocol-level DA (§1/A-3). **Draft v1.19, 2026-08-25:** Codex round 10, one P1^[240]: if the head of the ordinary queue is sized against the full C_force, then when every block has due bridge entries taking up to C_bridge, it is permanently left with only $C_{\text{force}} - C_{\text{bridge}}$ available and does not fit, so a sustained bridge flow can starve ordinary forced inclusion indefinitely — the per-entry share of both queues is changed to "the guaranteed capacity after deducting the other queue's reserved quota" (ordinary = $C_{\text{force}} - C_{\text{bridge}}$, bridge = C_{bridge}), so that each queue head fits into the current block under any volume of traffic on the other queue. **Draft v1.18, 2026-08-25:** the round 10 review forwarded by the owner^[241]: r18-1 — r12-1's "any accepted batch cancels it" is too broad, since a conflicting short fork from a malicious lookahead builder (no equivocation, it only has to win the landing) also cancels the re-anchoring and voids the honest first-prover's advantage, and under wall-clock timing a slow drip suffices to bypass r13-1 without falling into the censorship residual risk; cancellation is narrowed to extensions on the same branch only, a conflicting-fork reorg merely re-pins without resetting the window, and the frequency of conflicting forks is hard-constrained by the attacker's share of the lookahead (as with the deep-skip residual risk of §5.4, with the proving-cost grieving recorded explicitly). r18-2 — "the L1 height anchored by the batch" is not unique under the per-block anchor of §8, so it becomes a batch-level unique snapshot height H_batch declared explicitly by the landing transaction (= the anchor of the first block) + a freshness cap, and the exit deadline gains a D_anchor_max staleness term. r18-3 — if a forced-only backlog reopens a window for every batch, the cost = the number of batches $\times$ the window; a continuation episode is introduced so that only the first batch pays for a window, which matches the composite bound of §7. **Draft v1.17, 2026-08-25:** Codex round 9, two P1s (r17-1: after a stall with an empty backlog, the due slot of a newly submitted entry sits at the wall-clock position, so a forced-only chain crawling slot by slot at $\text{parent.slot} + 1$ can never catch up with the due point and the escape hatch cannot bootstrap itself under a total cartel — the first forced-only block of each re-anchoring episode is changed to jump to the execution baseline slot pinned by the episode, with subsequent blocks still at +1, so bootstrapping completes in one block with no discretionary time jump; r17-2: lowering C_force/C_bridge can leave already-enqueued entries above the new share, and v1.16's borrowing rule does not cover the ordinary queue head — this becomes a setter invariant: a reduction is refused unless the new share $\geq$ the maximum gas of that queue's unconsumed entries (implemented with a watermark), and the borrowing excep-

tion is deleted). **Draft v1.16, 2026-08-25:** Codex round 8, two P1s^[242]. **Draft v1.15, 2026-08-25:** Codex round 7, one P2^[243]: an off-by-one at the boundary of the fast-recovery row of §9 — when k consecutive slots are missing the next block's parent distance is $k + 1$, so the fast path covers $k \leq G_max - 1 (= 63)$, and exactly 64 missing already forces the re-anchoring procedure; the table rows are redrawn along absences $\leq G_max - 1 / \geq G_max$. **Draft v1.14, 2026-08-25:** Codex round 6, two P2 consistency corrections(r14-1: the proving-failure row of §9 had promised an exemption under which "the landing obligation is deferred" — that exemption is the blocking open item 8 of §12 and does not yet exist, so the text is changed to state honestly the wrongful-slashing exposure of a recovery batch; r14-2: the Appendix B entry for the re-anchoring procedure still carried the old v1.11 definition "cancelled as soon as the lag catches up", contradicting §6.4's "any accepted batch cancels it"^[244], and has been corrected). **Draft v1.13, 2026-08-25:** Codex round 5, one P1^[245]: the absolute deduplication of fallback strikes, "at most one per open-window period", can be weaponized in reverse — a colluding fallback account lands a micro-batch after the window opens to consume that period's single strike, then keeps dripping micro-batches so that the open-window period never closes, m_agg is never filled, and the aggregator is never replaced. Strikes(and the m_agg ' late count)are changed to accumulate at a rate limited by the duration of the open window: every G_strike of continuously open window in which a corresponding batch was accepted counts as one strike — so that rate-limited anti-spam and the inevitability of accumulation under sustained dereliction hold simultaneously; the ability to continue a batch proof incrementally is listed as an engineering requirement(§12 item 6), and exhaustion of the bond is treated as termination. **Draft v1.12, 2026-08-25:** the round 12 deep review forwarded by the owner^[246]: the re-anchoring procedure becomes a pinned-parent scheme — the announcement pins (nonce, parent hash/slot), the proof is pre-generated only against the pinned parent, and any accepted batch cancels the announcement(which dissolves the protocol-level contradiction between "pre-generating the proof" and "taking the latest parent at execution time", and at the same time seals off the path where "a partial landing fails to catch up with the threshold while the remaining tail is still cut off by the window")^[247]; the existence challenge now carries a bond and is settled according to the outcome(a false challenge is no longer a free postpone button)^[248]; it is stated explicitly that the re-anchoring signer is a **probabilistic bound** inside the trust model, and that under a full cartel the only certain exit is a forced-only block^[249]; the deep-skip exposure gains the crucial point — with any colluding aggregator there is no race to speak of, so it is bounded as an explicitly stated safety residual risk and the effect of fork choice is downgraded to a P2P convergence assumption rather than an L1 guarantee^[250]; a fourfold division of recovery time(safety messages/queue drain/lag catch-up/discretionary service)^[251]; and the leftover sentence in §5.5, "the weak-execution class contains no profitable attack", is corrected^[252]. **Draft v1.11, 2026-08-25:** the round 11 deep review forwarded by the owner^[253]. Two items still apply to the current version: **the deep-skip exposure**^[254] — once G_max allows the parent to point back as far as $s - G_max$, the old coordination rule "attach to the chain head with the highest slot" would make the honest network automatically follow a deep-skip fork and isolate up to $G_max - 1$ honest blocks for free; §5.2 is changed to prioritize "the most blocks since the fork point"(normal block production after a genuine gap is unaffected, and a deep-skip fork is downgraded to something that must win the landing race), and the exposure statements of §5.4/§10 are correspondingly revised away from "a single slot"; **the composite time bound of the forced path under a stall**^[255] is written out explicitly as $\max(\text{the overdue point, the start of the stall} + \Delta_stall) + \text{the re-anchoring window} + \text{the proving and landing latency}$. **Draft v1.10, 2026-08-25:** DeepSeek round 3 applied on top of v1.9(2 severe + 3 warnings + 2 suggestions)^[256]: the initial value of Δ_lag , 3 epoch, is below the upper bound of the normal lag band that the specification itself declares($19.2 < 20$ minutes)and violates the r6-2 setter invariant, so it is recalibrated to 4 epoch, with Δ_stall moved in step to 5 epoch^[257]; the formalization of $f_consumed$ is reconciled into a pair of cursors, removing the implementation ambiguity introduced by the two queues of r8-2^[258]; the exit delay explicitly covers the slashing exposure of δ_slash + the evidence-submission latency^[259]; the formal definition of the fixed aligned partition of lookahead windows is $W(\text{slot}) = \text{floor}(\text{slot} / 768)$ ^[260]; and Appendix A-3 annotates the cost identity cited in §1^[261]. **Draft v1.9, 2026-08-25:** Codex round 4, three P1s, applied on top of v1.8^[262]. **Draft v1.8, 2026-08-25:** Codex round 3, two P1s, applied on top of v1.7^[263]. **Draft v1.7, 2026-08-25:** the round 7 deep review forwarded by the owner, applied on top of v1.6(2 blocking + 4 high + 1 goal-level, all adopted): the lookahead seed takes a single value per window, with the consistency formula $\text{randao_source} \leq H_snap \leq \text{the finalized L1 head} < \text{the start of the window}$ ^[264]; the forced-inclusion time bound becomes a backlog-aware formula + the bridge reserved quota C_bridge ^[265]; the $f_consumed$ cursor gains a four-step atomic state transition^[266]; the anchor is defined as a system-level

implicit transaction^[267]; the slot of a forced-only block is made unique as $\text{parent.slot} + 1$ ^[268]; the late count is deduplicated per open-window period and the "zero collateral damage" claim is downgraded to an exemption mechanism (raised to a blocking open item), completed^[269]; and the three-tier terminology for user promises + the acceptance-baseline note ^[270]. **Draft v1.6, 2026-08-25:** Codex round 2, two P1s, applied on top of v1.5^[271]. **Draft v1.5, 2026-08-25:** the first draft revised through five rounds of adversarial review (one internal review)^[272]. Round 5 (OpenAI) fixes: a batch must not land a future slot + truncation at the lower bound of lag^[273]; skip is corrected to the honest characterization "profitable but not slashable", with layered user semantics^[274]; strikes are deduplicated per open-window period and given a G_strike grace period^[275]; L_eq gains an honest upper bound on external value plus evidence idempotence^[276]; the forced-inclusion queue cursor is formalized^[277]; and the trust model states explicitly that "the honest majority is an economic assumption, not a protocol-enforced one"^[278]. The mechanism skeleton is complete, the parameters are initial suggested values, and §12 lists the open items.

Appendix E: Index of Review Notes

This appendix collects every provenance note marked ^[n] in the body. Each records which adversarial review round, which external reviewer, or which decision by the owner produced the rule it is attached to. The body therefore carries only normative statements; consult this appendix when tracing where a rule came from. The full per-version revision record is separate, in Appendix D.

- [1] **Front matter** review r39
- [2] **Front matter** independent review round 4, finding 2
- [3] **§1** r41 option C revises the original "acceptance is finality"
- [4] **§1** the owner's decision, 2026-08-25
- [5] **§1** review r12-4's refinement + r20-1's correction of the depth
- [6] **§1** r42, independent review round 5, high 1 — a fixed W_{settle} close and the $D_{\text{anchor_max}}$ freshness deadline cannot be shown robust from "it will eventually be included" alone
- [7] **§1** the refinement of review r5-1
- [8] **§1** r7-7
- [9] **§1** r5-3
- [10] **§1** the owner's decision: no in-protocol solution is required
- [11] **§1** review r33; the third non-pure-protocol dependency, standing alongside "honesty"
- [12] **§3.2** rewritten per review r7-1 — v1.6's "each slot separately takes the RANDAO D_{rand} in advance", a 768-slot lookahead horizon, and a single H_{snap} cannot all three hold at once: 768 one-second L2 slots are almost exactly 2 L1 epochs, so the per-slot RANDAO at the end of the window is not yet final and is later than H_{snap}
- [13] **§3.2** the formalization of review r10-5 — if "window" were read as a sliding horizon, the same slot would receive different lookahead assignments at different moments, destroying the determinism of G2
- [14] **§3.2** the fix of review r8-1 — v1.7's $D_{\text{snap}} = 3$ epoch omitted the horizon term: the lookahead is required to be computable H_{look} in advance, so the start of the furthest window is $\approx \text{now} + H_{\text{look}}$, and its snapshot point = $\text{now} + H_{\text{look}} - 3 \text{ epoch} = \text{now} - 1 \text{ epoch}$, which is not yet finalized and can be reorged
- [15] **§3.2** the uniqueness requirement of r1-9 is thereby satisfied automatically
- [16] **§3.2** review r1-7
- [17] **§3.2** the owner's suggestion; a deliberate divergence here — see Appendix A-1
- [18] **§4.1** the fix from review r1-5
- [19] **§4.1** review r2-3: an equivocator can still cross into the next window and keep signing before the first slashing takes effect, so the horizon cannot be counted over a single window alone
- [20] **§4.1** review r5-5
- [21] **§4.1** made explicit by review r10-4; wording clarified by r22-6
- [22] **§4.2** added to the field list by review r36, a DeepSeek warning — m_{consumed} in §7 is determined by the block header's anchor, so it must be a block-header field covered

- by the signature rather than a batch-level implicit quantity
- [23] §4.2 the owner's suggested fix, 2026-08-25; for the analysis see §5.4
 - [24] §4.2 the fix from review r1-8
 - [25] §4.2 review r6-1 → reworked in r32, direction B recovery redesign; r35 clarified the criterion and the point of evaluation, DeepSeek critical 1/2
 - [26] §4.2 the resolution of DeepSeek critical 1 — the criterion is the gap, not the landing status, so there is no class of parent for which "neither tier applies"
 - [27] §4.2 r42: F_l1 depth alone is not enough, since a provisional candidate can still be superseded before it closes, and anything built on it would be stranded along with it, independent review round 5 consistency 2; the waiting bound = $\max(W_settle, F_l1) + D_anchor$
 - [28] §4.2 r41
 - [29] §4.2 DeepSeek critical 2
 - [30] §4.2 review r40, independent review round 3 finding 5
 - [31] §4.2 the owner's decision, 2026-08-25
 - [32] §4.2 review r36, Codex line 475 P1 — must be stated explicitly
 - [33] §4.2 r24-4
 - [34] §4.2 review r20-1, restated under tiers (i)/(ii)
 - [35] §4.3 the fix from review r1-4
 - [36] §4.3 review r39, independent review round 2 finding 5
 - [37] §4.3 r41, option C
 - [38] §4.3 independent review round 4 finding 7 pointed out its dilemma: inside the grace window it lets backfills through, outside it kills in-flight blocks by mistake
 - [39] §4.3 wording corrected by r42 — independent review round 5 pointed out that "no other builders are harmed" was untrue
 - [40] §5.1 review r3-3, Codex introduced the exception; r35 updated the citation to DeepSeek warning 1; r36 fixed it as a precise predicate, Codex line 517 / DeepSeek
 - [41] §5.1 r32; the criterion is the size of the gap, not the parent's landing status, r35
 - [42] §5.1 r41 folds this into the validity rules — independent review round 4, consistency 1: v1.40 described it only in §4.2 and did not list it in this section
 - [43] §5.1 review r37
 - [44] §5.1 review r40, the sole maximum-prefix rule in a block since v1.51 (the forced prefix of the former rule 4 went with §7), eliminating finding 4's contradiction between "exactly $\min(C_anchor)$ " and "overflow as soon as gas is full"
 - [45] §5.1 r36
 - [46] §5.2 r42 switches this to window semantics and deletes "the first to land is permanently canonical"
 - [47] §5.2 introduced in review r11-2; r40 merged the ordering into the total order and deleted the conflicting old standalone "most blocks" formulation, independent review consistency 1
 - [48] §5.2 review r12-4, still holds

- [49] §5.2 introduced in review r39; r40 fixed lane whitewashing, and v1.51 retired the lane; from r41 on it serves the settlement window's mechanical comparison
- [50] §5.2 r42 fixes independent review round 5, severe 1 — the old "look at the first differing block of the two chains to determine the lane" was a pairwise criterion, and a non-transitive cycle $A > B > C > A$ can be constructed: $A=[X,a]$ vs $B=[X,b1..b3]$ diverge at $a/b1$, so content wins; B vs $C=[Y,c1,c2]$ diverge at X/Y , so block count decides; C vs A again diverge at Y/X , so block count decides — a cycle. Fix: the quantities compared must be each candidate's own invariant scalars, relative to the fixed baseline F and independent of the object of comparison
- [51] §5.2 r42 abandons "the hash of the first differing child block", which is a pairwise quantity, in favor of a scalar belonging to the candidate itself, which guarantees a total order
- [52] §5.2 r42
- [53] §5.2 review r33, raised by the owner; from r41 mechanically endorsed by the §5.6 settlement window
- [54] §5.2 r33
- [55] §5.2 resolved by the settlement window from r41
- [56] §5.2 the pre-v1.39 premise that "landing the wrong thing cannot be corrected" has been eliminated by the window
- [57] §5.2 r47, the wording confirmed by the owner
- [58] §5.2 made explicit in r47
- [59] §5.2 r47
- [60] §5.2 review r1-1, a residual risk that must be stated explicitly
- [61] §5.2 the review r16-1 correction — v1.15 and earlier claimed that it was
- [62] §5.2 option C, r41
- [63] §5.4 see the two-tier bound of r20-1 in §4.2 and below; r22-4 deleted the old sentence here that sweepingly said "capped within $G_{\max}$ ", which contradicted the text below.
- [64] §5.4 the review r5-3 correction
- [65] §5.4 clarified in review r11-2; depth corrected in r20-1
- [66] §5.4 r20-1 of §4.2
- [67] §5.4 review r12-4
- [68] §5.4 r20-1 had two tiers; v1.31 direction B added the single-block-via-landed-head path of tier (ii)
- [69] §5.4 new in v1.31, $G_{\max_landed} = \infty$
- [70] §5.4 review r35, DeepSeek warning 4
- [71] §5.4 review r36, Codex line 475 P1
- [72] §5.4 the honest version after reviews r1-1/r1-6
- [73] §5.4 the r16-1 correction — see the honest bounding in §5.2
- [74] §5.4 r5-3
- [75] §5.4 r20-1
- [76] §5.4 r11-2, §5.2

- [77] §5.4 r12-4, the explicitly priced residual risk above
- [78] §5.4 adopted in review r7-7
- [79] §5.4 r5-3
- [80] §5.5 the review r1-6 fix
- [81] §5.5 one half of review r2-1
- [82] §5.5 updated in r41 — option C withdraws "landing the wrong chain is slashable" and replaces it with mechanical competition inside the window
- [83] §5.5 this is the conclusion of independent review rounds 3/4
- [84] §5.5 the review r12-6 correction — this passage once retained the pre-r5-3 conclusion that "the weak-enforcement class contains no profitable attack", which directly contradicts §5.4
- [85] §5.6 review r41, the owner's decision, option C
- [86] §5.6 round 4 finding 1 disappears
- [87] §5.6 round 4 finding 2 disappears
- [88] §5.6 r42 fixes independent review round 5 severe 2: v1.41 had the first provisional candidate immediately advance the canonical cursors/ state, so that a second, heavier candidate's starting cursor would not match and it could not even enter the contest — a restoration of "the first to land wins permanently"
- [89] §5.6 r46 clarification, DeepSeek-on-v1.45 W2
- [90] §5.6 r44, W2; the replay of the executable model is exactly this semantics, Appendix C P5b/P6
- [91] §5.6 v15 I8
- [92] §6.1 r41 option C revision
- [93] §6.1 the fix from review r5-2 — blocking
- [94] §6.2 r41: lag is measured on the provisional candidate, so that the liveness determination is not dragged down by window delay
- [95] §6.2 the explicitly accepted cost of r41 option C
- [96] §6.2 r42, independent review round 5 high 2 / medium 1 — conflating them first switches off the error-correction reward and then understates the revocable exposure
- [97] §6.2 r44, DeepSeek-on-v1.43 C1 — splitting only the adjudicated quantity without splitting the threshold would leave the fallback window permanently open
- [98] §6.2 r10-2
- [99] §6.2 r46 recalibration: r44 once wrote 8 epoch ≈ 51.2 min, but $\Delta_{\text{lag,prov}} + W_{\text{settle_max}} = 128 + 150 \text{ L1 slot} = 55.6$ min already exceeds it, so 8 epoch violates the very formula on this same line; once model property P9a was made non-vacuous (deployment values declared independently, inequalities cross-checked) this was caught immediately, and the value was changed to 9 epoch = 288 L1 slot, with a margin of 2 min
- [100] §6.3 r42/r44
- [101] §6.3 r44 cleanup, DeepSeek W3
- [102] §6.3 review r2-4

- [103] §6.3 introduced in r44 and numerically recalibrated in r46 — the threshold must be raised together with the quantity it judges; for the derivation see §6.2; $\Delta_{\text{lag,prov}}$ is initially 4 epochs $\approx$ 25.6 minutes; the calibration of review r10-2 — in v1.9 and earlier the initial value was 3 epochs $\approx$ 19.2 minutes, below the 15–20 minute upper bound of the normal lag band that §6.2 itself declares, violating the setter invariant of r6-2, $\Delta_{\text{lag}} \geq$ upper bound of the normal lag band: the window would open during normal operation and wrongly penalize an honest aggregator, so "zero false positives" would fail. After recalibration the margin is $\approx$ 5.6 minutes
- [104] §6.3 r42, independent review round 5, high 2 — if the determination were made on the provisional lag, a malicious accomplice could first land a short candidate that zeroes out lag_{prov} and thereby switch off fallback eligibility before an honest heavier candidate lands, cutting off its proving-fee compensation; rational fallback landers then withdraw and the bad candidate closes the window unopposed
- [105] §6.3 review r5-4 introduced the de-duplication; r13-1 corrected its granularity — Codex round 5 pointed out that v1.12's "at most once per open-window period" is an absolute de-duplication: a fallback account colluding with the aggregator lands one deliberate micro-batch right after the window opens, absorbing that period's single strike, and thereafter drip-feeds micro-batches so that finality crawls forward arbitrarily slowly, the open-window period never closes, m_{agg} is never filled, and the aggregator is never replaced — the shield against strike-spamming is turned around and used as the attacker's shield; moreover, each micro-batch that advances the landing head also invalidates the proof an honest fallback lander had pre-generated against the old chain head, so it cannot win the landing right back
- [106] §6.3 the original purpose of r5-4
- [107] §6.3 review r3-1, Codex
- [108] §6.3 review r4-2, DeepSeek — v1.3 once specified counting per window opening, which contradicts this section's own zero-false-positive principle
- [109] §6.3 review r7-6
- [110] §6.3 r13-1 corrected the granularity of both counters together, away from "once per open-window period"; otherwise an aggregator riding the threshold with its own micro-batches could likewise pin m_{agg} at 1
- [111] §6.3 review r39, independent review round 2, finding 3-i
- [112] §6.3 r41 §5.6 settlement window
- [113] §6.3 review r39; r40 fixed the remaining zero penalty at equality — independent review round 3, finding 8
- [114] §6.3 r41, independent review round 4, medium 3
- [115] §6.3 review r24-4, DeepSeek round 5 W3
- [116] §6.3 the fix from review r2-1 — this item must be stated explicitly
- [117] §6.4 correction from review r39, superseding v1.35's erroneous "recovery takes priority / the tail yields"; independent review rounds 1/2, findings 1/2
- [118] §6.4 r41
- [119] §6.4 review r36, Codex line 916 P1 — a refinement of " ≥ 1 builder returns"
- [120] §6.4 review r35, DeepSeek critical 2

- [121] §6.4 r41 added the D_anchor term — independent review round 4, medium 2: the $final_ref$ of a tier (ii) block header must be $\leq$ the anchor, and the anchor is at least D_anchor deep, so the L1 block that witnesses the parent's finality must age a further D_anchor before it can be referenced; the earliest signable point = parent landing + F_ll + D_anchor
- [122] §6.4 review r38, Codex P1
- [123] §6.4 r40 corrected it to $D_anchor_max \geq D_anchor + P_prove,max + T_include,max + margin$ — independent review round 3 pointed out that the old formulation omitted the term for the anchor already being D_anchor old at signing time
- [124] §6.4 with r44 raising the lag term to the final threshold, so that it covers the entire span of fallback authorization — any tail a fallback lander is authorized to land necessarily still has a fresh anchor
- [125] §6.4 r39 corrected the wording of the C_anchor binding — independent review, consistency 2
- [126] §6.4 r39 unified it with "sustained progress requires the liveness floor" below — independent review, consistency 6: the conflicting old wording " ≥ 1 return is enough" was deleted
- [127] §6.5 review r1-10
- [128] §6.5 r47, raised by the owner
- [129] §6.5 one point of deviation from the owner's suggestion of "sharing directly at landing time"; see Appendix A-4
- [130] §7 review r37, Codex line 598 P1
- [131] §7 r41, replacing the commitment-time waiver of v1.40 — the commitment layer has been deleted; independent review round 3, finding 1 and round 4, scenario A
- [132] §7 r44: the lag term is $\Delta_lag,final$ — the fallback opens its window on lag_final , so the authorization point is later than under the old Δ_lag , and the freshness window must cover it, or else the C1 fix would in turn break this geometric solution
- [133] §7 r46 raises this in step with the recalibration of $\Delta_lag,final$
- [134] §7 introduced in review r32, self-review round 5; r34 corrected what it is bound to — Codex pointed out that binding it to the anchor would conflict with freshness
- [135] §7 review r39, independent review consistency 3: a single L1 height may contain several messages, so a cursor that records only the height would consume messages twice or skip them within that height, and could not uniquely prove a "maximum prefix"; hence $m_consumed$ = the global sequence number in the inbound message queue, or the equivalent (L1 height, tx index, log index) triple
- [136] §7 r41 cleaned out the residual text "exactly min(entry count)" — independent review round 4, severe 3 pointed out that this residual text and the two-constraint prefix of §5.1 negate each other, which could cause a consensus split or leave no legal block
- [137] §7 r41 fixed "dividing by entry count alone underestimates the gas bottleneck"
- [138] §7 Codex line 1031
- [139] §7 review r38, DeepSeek warning 1 — correcting the overly strong statement "consuming less makes the block illegal"
- [140] §7 Together with the anchor $\leq$ slot time upper bound of Codex line 598, the anchor

- is squeezed into $[\text{landed head} - D_anchor_max, \text{the height corresponding to the slot time}]$.
- [141] §7 r34
 - [142] §7 Codex r34-1
 - [143] §7 review r39, independent review rounds 1/2, findings 1/5 — severe; contracted from three parties to two in v1.51 along with the removal of forced inclusion
 - [144] §8 r39 correction — the frozen long tail is not lost
 - [145] §8 r41
 - [146] §8 r41
 - [147] §8 r35/r41; this term is absent in steady state
 - [148] §8 r36
 - [149] §8 r24-4: if nobody performs the fallback, no strike is recorded and the stall can continue indefinitely
 - [150] §8 r14-1, stated honestly; the exemption is expected to take the form of v15 §10.4 level 3
 - [151] §9 r5-3
 - [152] §9 r11-2/r20-1
 - [153] §9 v1.31/r35, provided the current landed tip is already final, see §5.4/§4.2
 - [154] §9 r41, harmonized with §4.2/§5.4
 - [155] §9 r41 option C — "what ought to be landed" was shown not to be mechanically adjudicable, independent review rounds 3/4
 - [156] §9 r5-3 correction
 - [157] §11 split in r44; defined in §6.2
 - [158] §11 r41 corrects the stale text on this line, independent review round 4, consistency item 7
 - [159] §11 r18-2/r30-1
 - [160] §11 introduced in review r38, with r39 correcting a term missing from the formula — independent review round 2, finding 4
 - [161] §11 r44: the lag term is Δ_lag_final , in sync with §7
 - [162] §11 §1, r42
 - [163] §11 r41 option C, §5.6; r42 adds the upper bound
 - [164] §11 r42 medium-severity finding 1
 - [165] §11 the "lower it vs. keep it $O(1)$ " tension in the watermark is discussed in independent review consistency item 7 and is listed as an implementation refinement
 - [166] §11 review r38, DeepSeek warning 4
 - [167] §11 an engineering requirement from r13-1
 - [168] §11 r7-6
 - [169] §11 review r2-8
 - [170] §11 r24-5
 - [171] §11 review r24-4, DeepSeek round 5

- [172] §11 review r27-5, DeepSeek round 6, W4
- [173] §11 review r27-5, DeepSeek round 6, W5; from r44 onwards this covers both thresholds
- [174] §11 recalibrated in r46
- [175] §11 review r36, DeepSeek warning
- [176] §11 review r36, DeepSeek warning
- [177] §11 raised in review r36; designated blocking in r42; first half delivered in r43, second half in r44
- [178] §11 r42/v1.51
- [179] §11 r42 blocking, must be completed before implementation
- [180] §11 review r24 suggestion — units and invariants gathered in one place, since scattered numbers are easy to misread
- [181] §11 r35
- [182] §11 recalibrated in r46, since 8 epoch violates this formula
- [183] §11 r44, DeepSeek C1
- [184] §11 raised in r46 in step with the recalibration of $\Delta_{\text{lag,final}}$
- [185] §11 r42 unified the whole document on the strongest form, r44 raised the lag term to final
- [186] §11 r47
- [187] **Appendix A** annotation added in r10-6
- [188] **Appendix B** §5.6, r41
- [189] **Appendix B** §4.2, v1.31/r35
- [190] **Appendix B** r41/r42
- [191] **Appendix B** §5.6, r41 option C
- [192] **Appendix B** r41
- [193] **Appendix C** first half r43, second half r44
- [194] **Appendix C** against the blocking findings and requirements of independent review round 5
- [195] **Appendix C** v1.51 dropped lane, so the ordering is by block count first
- [196] **Appendix C** DeepSeek W1 / suggestion 3
- [197] **Appendix C** r44
- [198] **Appendix C** DeepSeek W2
- [199] **Appendix C** r47
- [200] **Appendix D** review r39
- [201] **Appendix D** independent review round 5, high 1 — \ldots{}
- [202] **Appendix D** after the typesetting pipeline migrated to LaTeX in v1.48, that guard is carried by build-pdf.py --check
- [203] **Appendix D** the owner’s instruction: five rounds of self-review, zero changes to normative semantics
- [204] **Appendix D** that HTML pipeline was deleted in v1.48 and replaced by LaTeX/PDF;

what follows describes the practice at the time: generated from this markdown, with mermaid rendered client-side, for convenient reading in a browser

- [205] **Appendix D** following the v15 model_checker practice
- [206] **Appendix D** correcting v1.41's "no extra state machine" wording
- [207] **Appendix D** option C
- [208] **Appendix D** v1.39 §5.6 + v1.40 §5.7, with $W_chal/L_land/L_chal/L_commit/\delta_land$ all retired
- [209] **Appendix D** finding 2a
- [210] **Appendix D** the high-slot extension challenge of finding 2a, consistency 3
- [211] **Appendix D** finding 2b
- [212] **Appendix D** of v1.34/v1.38
- [213] **Appendix D** superseding the erroneous v1.35 rule
- [214] **Appendix D** of cHR4b/cHR4c already fixed in v1.35; replied and resolved.
- [215] **Appendix D** Codex $3 \times P1 + 1 \times P2$; DeepSeek 1 Critical + 6 warnings + 4 suggestions
- [216] **Appendix D** Codex line 822
- [217] **Appendix D** Codex line 1031 / DeepSeek
- [218] **Appendix D** Codex line 475
- [219] **Appendix D** Codex line 517 / DeepSeek
- [220] **Appendix D** DeepSeek
- [221] **Appendix D** its review target was the old v1.32 head
- [222] **Appendix D** r35-1
- [223] **Appendix D** r35-2
- [224] **Appendix D** r34-1
- [225] **Appendix D** the owner's question in r33
- [226] **Appendix D** r30-1, outside the scope of the freeze — the determinism of the ordinary forced-inclusion path
- [227] **Appendix D** reviewing v1.25
- [228] **Appendix D** my r24 fix to §6.4 was not synchronized to the remaining normative locations
- [229] **Appendix D** r25-1
- [230] **Appendix D** r24-1
- [231] **Appendix D** this ends the re-anchoring slot-timing family of r18/r20/r21/r23, and covers DeepSeek W2
- [232] **Appendix D** DeepSeek W1
- [233] **Appendix D** DeepSeek W4
- [234] **Appendix D** DeepSeek W3
- [235] **Appendix D** DeepSeek W5
- [236] **Appendix D** r23-1
- [237] **Appendix D** The stale Appendix B entry DeepSeek pointed out had already been up-

- dated in v1.21/r21-1.
- [238] **Appendix D** r21-1
 - [239] **Appendix D** r20-1, important
 - [240] **Appendix D** r19-1
 - [241] **Appendix D** target head v1.17, 2 P1 + 1 P2, all valid and adopted
 - [242] **Appendix D** r16-1: the isolatable depth under equivocation collusion had been described as "hard-capped by Δ_{lag} " — opening the window is only an authorization and does not force acceptance, so it is honestly bounded as a soft upper bound of Δ_{lag} + the fallback response time, with the assumption stated explicitly; r16-2: bridge entries have their gas validated against C_{force} but can only be consumed against C_{bridge} , so an entry in the interval $(C_{\text{bridge}}, C_{\text{force}}]$ deadlocks the moment it reaches the queue head — enqueue validation is changed to use the quota of the queue the entry belongs to, plus a deterministic borrow-and-drain rule for legacy oversized entries
 - [243] **Appendix D** r15-1
 - [244] **Appendix D** r12-1
 - [245] **Appendix D** r13-1
 - [246] **Appendix D** target v1.11, 1 blocking + 5 high + 2 medium, all valid and adopted
 - [247] **Appendix D** r12-1
 - [248] **Appendix D** r12-3
 - [249] **Appendix D** r12-2
 - [250] **Appendix D** r12-4
 - [251] **Appendix D** r12-5
 - [252] **Appendix D** r12-6
 - [253] **Appendix D** the review target was v1.6; 8 of its 10 items had already been fixed by r7–r10, and the item-by-item comparison is in the PR reply
 - [254] **Appendix D** r11-2
 - [255] **Appendix D** r11-4
 - [256] **Appendix D** its severe 1 duplicates Codex r9-3 and had already been fixed in v1.9
 - [257] **Appendix D** r10-2
 - [258] **Appendix D** r10-3
 - [259] **Appendix D** r10-4
 - [260] **Appendix D** r10-5
 - [261] **Appendix D** r10-6
 - [262] **Appendix D** r9-1/r9-3: delete the immediate-recovery exception "if $\text{lag} > \Delta_{\text{stall}}$ at landing time then G_{max} is waived" — that condition looks only at the L1 landed head and cannot see the P2P tail, so an aggregator withholding landings + one malicious lookahead builder + a colluding lander could use it to isolate an unbounded honest tail without equivocating; and honest recovery from a genuinely large gap was forced to wait out the full Δ_{stall} — all of it is replaced by the §6.4 **re-anchoring procedure**: announcement + an existence-challenge window + cancellation as soon as the lag catches up, with the same rules applying to the landing of forced-only blocks, which also nar-

rows the stall residual race of r2-2; r9-2: forced entries gain two deterministic lines of defence — validation at enqueue and consume-and-discard at consumption — sealing off "a single malformed paid entry putting the inclusion obligation and execution legality in conflict and stalling the chain permanently"

- [263] **Appendix D** r8-1: D_{snap} gains the lookahead-horizon term, $= H_{\text{look}} + F_{\text{final}} + \text{margin} = 5 \text{ L1 epoch}$, and this is established as a setter invariant; r8-2: the forced-inclusion queue is split into ordinary/bridge dual FIFOs with dual cursors, fixing the problem that the C_{bridge} reserved quota was purely nominal under a single FIFO
- [264] **Appendix D** r7-1, fixing the three-way contradiction around 768 slot $\approx 2 \text{ L1 epoch}$
- [265] **Appendix D** r7-2
- [266] **Appendix D** r7-3
- [267] **Appendix D** r7-4
- [268] **Appendix D** r7-5
- [269] **Appendix D** r7-6
- [270] **Appendix D** r7-7
- [271] **Appendix D** r6-1: the parent-gap cap G_{max} + the stall-recovery exception, sealing off the deep-reorg path that requires no equivocation; r6-2: the on/off threshold of the forced-only valve is changed from H_{force} to $\Delta_{\text{stall}} > \Delta_{\text{lag}}$, fixing its numerical conflict with the normal finality lag band
- [272] **Appendix D** two rounds by DeepSeek, one by Codex, and one round of OpenAI review forwarded by the owner; all 31 findings were fixed or explicitly characterized; the body marks each fix location as "review rN-M", and the review originals are in the PR record
- [273] **Appendix D** r5-2, blocking
- [274] **Appendix D** r5-3
- [275] **Appendix D** r5-4
- [276] **Appendix D** r5-5
- [277] **Appendix D** r5-6
- [278] **Appendix D** r5-1, a refinement decided by the owner